TFPT — Research Contracts for the Remaining Interfaces:

v_{geo}, G_{net}, F_{transfer}

(U_{wall}): the parabolic flavor wall-selection and (G_{metric}): the full quantum-gravity measure

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: **R.** `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\rm car} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1,1,2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
$v_{\rm geo}$	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
$G_{\rm net}$	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$	[C]
$F_{\rm transfer}$	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit $v_{\rm geo}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\rm net}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit $v_{\rm geo}$ plus the typed $F_{\rm transfer}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\rm wall}, G_{\rm metric}, F_{\rm frontier}$) are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus F_{\rm transfer}$, with the seam $G_{\rm net}$ closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven **[C]** conditional **[O]** open / axiom **[X]** kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1, 1, 2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8, h = 30, \varphi_0$ *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this document is about

After the compiler closure the entire residual is

$$\boxed{\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})}$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. (F_{frontier}) (Koide, η_B, axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** (U_{wall}) *first* (finite, algebraic, falsifiable), then (G_{metric}) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\boxed{\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}}$$

— $(U_{\text{wall}}) \to v_{\text{geo}}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), U_{point} collapses to the *one* dimensionful unit v_{geo} shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where U_{point} still appears); $(G_{\text{metric}}) \to G_{\text{net}}$: holographically reduced to the single boundary-net statement *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/`GATE.METRIC.05-07`); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\to SO(16)_1 \to (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank $= c$; Wick/Pfaffian exact) is the certified Calderón datum — closing G_{net} therefore also closes the carrier choice **[E]** ([`verification/v113_quasifree_kernel.py`]); $(F_{\text{frontier}}) \to F_{\text{transfer}}$: named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the $K \to C \to F$ sheet axis $M(1,t)$, with the Koide source→pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([`verification/v224_diamond_ftransfer_path.py`]).

The residual is certification, not construction ([`verification/v384_residual_certification.py`]). A ledger-CI audit classifies every remaining item into exactly three *external* kinds, with *zero* open *internal* TFPT mechanisms: **[E]** (A) an external math proof — `SEAM.EQUIV.01` continuum existence (the cited MMST theorem plus the crossed-product extension package, v336/v469, with AGT/AMT as second witness), `ALPHA.QUILLEN.EXACT.01` (v382, its level and normalisation sharpened by v470, the determinant-line/moduli bridge lemma exhibited at the finite level by v472 — the continuum ζ-det identification stays the open part) and the constructive `QG.AMB.01` measure (a **[C]** redundancy, v369); **[E]** (B) theorem-forbidden — v_{geo}, the one unit, by the No-Unit theorem (v153/v364); **[E]** (C) external physics — F_{transfer} (v371–v374, firewalled v187). The two load-bearing facts are checked (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), so the only hard things left are theorems and external physics, not missing dynamics (`RESIDUAL.CERTIFICATION.01`).

The Coxeter-clock facet symmetry, resolved as a structural lemma (`RES.COXETER.SYMMETRY.01`, [`verification/v394_coxeter_atoms.py`]/[`verification/v409_coxeter_prime2_lemma.py`]). The order-30 clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ has all three primes facet-typed, and they are exactly the three anchor atoms ($2{=}|Z_2|$, $3{=}N_{\rm fam}$, $5{=}g_{\rm car}$) wearing number-field hats ($\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$, v390). The table is *not* symmetric: the seam (prime-2) appears as the *common factor* of both dynamic rates ($2/3 = |Z_2|/N_{\rm fam}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$, $|\mu_4|{=}2^2$), not as an independent attractor. *The structural lemma (v409) closes the falsifiable corollary:* the sheet involution $\sigma{=}\operatorname{diag}(1,-1,-1)$ has spectrum $\{+1,-1\}$ and projector $(I{+}\sigma)/2$ spectrum $\{0,1\}$ — none in the open interval $(0,1)$, so prime-2 supplies a *parity/projection*, never a nonzero contraction rate; the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a *factor*. Hence prime-2 is *necessary* (admissible boundary transport is sheet-paired) but *not autonomous*: **no prime-2-only attractor exists** — dynamics begins only after coupling to family prime-3 or carrier prime-5 ($|\mu_4|{=}2^2$ the Galois bridge). This closes the corollary [E] and types the necessity [C], sharpening v383's "every sector is the same gapped operator" towards "the same *seam-driven* operator"; the full "appears in every gap" universality stays the v383 reading. The corollary is **machine-proved in Lean 4** (`TfptCarrier/CoxeterPrime2.lean`, `FORM.COXETER.PRIME2.01`: axiom-pure, `lake build` clean). [E]/[C].

QFT layer (this round). The boundary QFT is now *one relative object* $\mathsf{TFPT}_{\rm QFT} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\rm rel})$, closed modulo the *single* keystone `SEAM.EQUIV.01` (whose downstream conformal-deck face is `QGEO.SYM.01`): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and E_8 live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure ($G_{\rm metric}$) is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

Research Contract 1 — ($U_{\rm wall}$)

Goal. Prove

$$\nabla_F^\star = \operatorname{Sel}_{D_4,\,\mathrm{cusp},\,\det R=8,\,\mathrm{Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\rm par}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4. The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4-symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation)**.** *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\rm cusp} = \{(M_1, \ldots, M_4) \in C_{\rm cusp}^4 : \ M_1M_2M_3M_4 = \mathbf{1}\}/SU(3)$ with $C_{\rm cusp} = \mathrm{Ad}_{SU(3)} \operatorname{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \ldots, \gamma_4 \mid \prod \gamma_i = 1\rangle$. To show: $\mathcal{M}_{\rm par}^{D_4} \simeq \mathcal{X}_{\rm cusp}^{D_4}$ as polystable Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and D_4-action are finite and codable (Sage/Wolfram).

Finite rigidity — now proven [E]. The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz,\ z \mapsto 1/z\rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\rm fam} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$, $k = 1, 2, 3$, are μ_4-eigenforms ($z \mapsto iz : \ \omega_k \mapsto i^k\omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1, 2, 3) = \mathrm{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1, 2, 3)$ grading (`QGEO.RIGID.01`, [E], [`verification/v168_qgeo_rigidity.py`]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` ([C]).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O] ([`verification/v176_seam_collar_realisation.py`])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4-character decomposition; (6) an admissible c=8 boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1, 2, 3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

1–2 *Geometry / uniformisation* [E] (v168): μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$.

3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx-|k|y}$), the free chiral dispersion — bulk-detail-independent.

4 *H^1 character = generation grading* [E] (v137/v168): $\omega_k = z^{k-1}dz/(z^4 - 1)$ carry μ_4 characters i^k with weights $(1, 2, 3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$ — natural, not merely isomorphic.

5 *One-particle contraction* [E] (v162): a μ_4-equivariant operator is diagonal in the cusp basis $(\frac{1}{|\mu_4|}\sum_j U^j X U^{-j} = \operatorname{diag} X$ for $U = \operatorname{diag}(1, i, -1, -i))$; its spectrum is the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, so the transfer eigenvalues $(1 - \alpha)^{p_2}$ (p_2=6) are $\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 *AQFT closure* [E] (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5+3 = 8$, μ-index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120+128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the *two* genuine open doors [O] ([`verification/v177_seam_marking_kernel.py`]). Taking the marks/D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\textsc{Realize} = \textsc{Marks} \cdot \textsc{Uniform} \cdot \textsc{Cohom} \cdot \textsc{Module} \cdot \textsc{Kernel} \cdot \textsc{Net},$$

of which *four nodes are now closed constructively*: Uniform [E] (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as `FORM.QGEO.02`: ρ^4=id, $\rho^2 \neq$id, σ^2=id, $\sigma\rho\sigma$=ρ^{-1}, the orbit scaling to μ_4, the μ_4 permutation, and the multiplier classification "order-4 $\Leftrightarrow \zeta = \pm i$" are machine-proved in `TfptCarrier/MobiusUniformisation.lean`, audit: pass); Cohom [E] ($\rho^* \omega_k = i^k \omega_k$, grading $(1, 2, 3)$ — also **Lean-formalised** as `FORM.QGEO.03`: the pullback characters i^1, i^2, i^3 and the reflection parity $\sigma^*\omega_1$=ω_3, $\sigma^*\omega_2$=ω_2, $\sigma^*\omega_3$=ω_1 are machine-proved in `TfptCarrier/CohomologyGrading.lean`, audit: pass); Module [E] (a *unique* μ_4-equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed; the parity is the Lean `FORM.QGEO.03` above, the *uniqueness* stays symbolic); and Net [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: **`QGEO.MARKS.01`** — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a)$=4=$|\mu_4|$, collisions excluded, $\tau\rho\tau$=ρ^{-1}) — and **`QGEO.KERNEL.01`** — the *raw* seam Calderón operator equals the μ_4-equivariant free c=8 gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1} C_{\mu_4} U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

How far each door opens by computation ([`verification/v178_marks_kernel_attempt.py`]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For Marks: given* a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving faithful D_4 — so Marks reduces to the *single* topological premise "the seam double is genus-0 with an order-4 clock whose marks form one orbit". *For Kernel:* on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the "spectrum vs operator" distinction *vanishes* on the finite block. So Kernel reduces to the *single* premise "the full DtN on L^2 reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation" (principal symbol $|k|$, Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

Marks without a geometric posit: a Lefschetz/character argument ([`verification/v195_marks_lefschetz_character.py`]). The marks need not be *assumed* as D_4

boundary points: they follow from the H^1 character alone. The order-4 clock $\rho{:}z \mapsto iz$ fixes exactly $\{0,\infty\}$, so a puncture at a fixed point would contribute a *trivial* character to H^1; but the carrier datum is $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents $(1,2,3)$, with $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$), which has *no* trivial component — so no puncture sits at a fixed point. With rank $H^1 = n-1 = 3 \Rightarrow n = 4$ punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free μ_4 orbit (cross-ratio 2), and the closed-surface Lefschetz number $L(\rho) = 1+1 = 2 = |\{0,\infty\}|$ is consistent. So MARKS reduces from the geometric posit "D_4 marks" to the milder *representation* premise "the raw seam H^1 carries the A_3-exponent rep" — a carrier-algebra statement, strictly less circular, still [O].

And the two doors are in fact ONE (`[verification/v179_conformal_realisation.py]`). MARKS's residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\,2\pi\,\chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$, and $\chi = 2-2g \Rightarrow g = 0$, so "genus-0" is the *same* $\chi = 2$ that supplies the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points $\{0,\infty\}$, and a free order-4 orbit has exactly $4 = e_1(a)$ points. So MARKS reduces to "the carrier clock acts *conformally* on the genus-0 double", and KERNEL then *follows* from that same geometry (Lee–Uhlmann $|k|$ + the $c = 8$ rigidity v158 + the cusped-surface spectral split, residual $= H^1 = \mathbb{C}^3 = N_{\mathrm{fam}}$). **Hence the entire structural residual of the theory is a single geometric premise `QGEO.CONF.01`:** *the raw RP seam normal double is conformally* $(\mathbb{P}^1, \mu_4)$ *with the carrier clock as the order-*4 *Möbius automorphism* $z \mapsto iz$. Given it, everything — genus-0, the four D_4 marks, the $(1,2,3)$ grading, the DtN block, the rank-8 polarisation and the $(E_8)_1$ net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

And even that conformal premise reduces to a milder isometry premise (`[verification/v180_clock_is_mobius.py]`). `QGEO.CONF.01` need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is* $\mathbb{P}^1$ — the target is *produced*, not assumed — and by *Kerékjártó* (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3){=}4$) is topologically $z \mapsto iz$. An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of $\mathbb{P}^1$, and an order-4 Möbius map is $z \mapsto iz$ (elliptic, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the strictly milder, non-circular premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does *not* name $\mathbb{P}^1/\mu_4$ and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly [O].

And one step further reaches bedrock (`[verification/v181_clock_is_conformal_symmetry.py]`). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), i.e. the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification `QGEO.SYM.01`: *the seam's conformal structure is the* μ_4*-deck structure of the carrier clock* (the carrier μ_4 glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam's conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain `REALIZE` → theorem → MARKS+KERNEL → `CONF` → `ISO` → `SYM` (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

A sharper, operator-checkable restatement: `QGEO.ENERGY.01` (`[verification/v192_qgeo_energy_clock.py]`). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock* ρ *preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double*, $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$. In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality → Möbius → $z \mapsto iz$ → μ_4). Its appeal is checkability: $\Lambda_\Sigma = |k|$ is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** "ρ preserves Λ_Σ" is plausibly *equivalent* to "ρ is the conformal deck" (`QGEO.SYM.01`), so `QGEO.ENERGY.01` *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if ρ is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays [O]; the gain is a sharper operator-analytic target, not a closure.

The non-circular sharpening: `QGEO.ENERGY.02` ([`verification/v193_qgeo_energy_commutator.py`]). The way to make "ρ defined independently" precise is the *energy commutator* $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims: (1) ρ is the Coxeter element of $W(A_3) = S_4$ (order $4 = |\mu_4|$, v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* — Λ_Σ must be the DtN of the *raw* RP seam (definable from the RP data alone), not the μ_4 normal-form operator; (3) $[\rho, \Lambda_\Sigma] = 0$ forces the μ_4 character grading $(1, 2, 3)$ on H^1, already established (`QGEO.COHOM.01`). An exact consistency rider pins the direction: the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ *iff* $\ln m = q(A_3) = \frac{3}{4}$ (the *family* glue norm), while $q(D_5) = \frac{5}{4}$ (the carrier norm) misses by $g_{\mathrm{car}}/N_{\mathrm{fam}} = \frac{5}{3}$ — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds, $[\rho, \Lambda_\Sigma] = 0$ turns `QGEO.SYM.01` from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, [O].

Subclaim (2) tackled — one notch closer, not closed ([`verification/v194_raw_seam_dtn_rp.py`]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator $T = e^{-H}$ from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$, a contraction). So Λ_Σ is RP-definable without ever naming $\mathbb{P}^1 \setminus \mu_4$. The commutator $[\rho, \Lambda_\Sigma] = 0$ then *closes on the finite cohomology* H^1 [E]: $\rho : z \mapsto iz$ acts on $\langle w_1, w_2, w_3 \rangle$ ($w_k = z^{k-1}dz/(z^4 - 1)$) as $\rho^* w_k = i^k w_k$, distinct characters $i, -1, -i$, so any ρ-equivariant operator is diagonal there (v177). What does *not* close is the *full-L^2* commutation beyond H^1 — it is exactly the statement that the quasi-free seam state's modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to "intrinsic BW geometric modular covariance of the quasi-free seam state" — a named, standard (hard) AQFT property, one notch sharper, but still [O] (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

A concrete variational handle on the residual ([`verification/v196_seam_energy_functional.py`]). The open BW statement can be made a discretisable extremum problem via the failure functional $E_{\mathrm{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, whose zero locus is exactly the `QGEO.SYM.01` conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite H^1 block $E_{\mathrm{fail}} = 0$ at the μ_4 configuration ($\rho = \mathrm{diag}(i, -1, -i)$, Λ μ_4-equivariant, Θ = conjugation) and > 0 for any μ_4-breaking perturbation, so μ_4 is a genuine minimiser [E]; minimising E_{fail} over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

The decisive crack: principal symbol exact + Tomita–Takesaki ([`verification/v198_modular_commutator_reduction.py`]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is $|k| = \sqrt{-\partial_\theta^2}$ (Lee–Uhlmann, v156); in the Fourier basis $e^{in\theta}$ it is $\mathrm{diag}(|n|)$ and the clock $\rho : z \mapsto iz$ (rotation by $\pi/2$) is $\mathrm{diag}(i^n)$, both *diagonal*, so $[\rho, |k|] = 0$ *exactly* on all of L^2 — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator Δ_ω is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with Δ and hence with $\log \Delta \sim \Lambda_\Sigma$: thus $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$ — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite H^1 block is already μ_4-invariant (v177), the entire residual is now the single *state*-invariance "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega$, equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of `QGEO.SYM.01`. A foundational symmetry postulate cannot be derived from nothing (the role $c = \mathrm{const}$ plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

The residual, fully localised and numerically probed ([`verification/v199_seam_state_invariance.py`], [`verification/v200_seam_variational_scan.py`]). Attacking $\omega \circ \rho = \omega$ on the *raw* seam (not the μ_4 normal form) confirms it is non-circular — ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance $C_\Sigma = \frac{1}{2}(1 + \mathrm{sgn}\, H_1)$, to $[\rho, H_1] = 0$, i.e. H_1 *is block-diagonal in the four μ_4-character classes* $\{n \equiv r \bmod 4\}$. The principal symbol $|k| = \mathrm{diag}(|n|)$ passes automatically, so the entire residual is the single bounded statement "the sub-principal symbol of the raw DtN has no off-character matrix elements". The companion variational scan minimises

$E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 +$ (faithfulness) over a free clock angle and finds its *unique* minimum at $\theta = \pi/2$ — the μ_4 clock $z \mapsto iz$ (the only faithful order-4 grading $i, -1, -i$) — a numerical sign that the variational principle selects μ_4. This is the "next executable theorem" a working mathematician can take up; it does not close the postulate, [O].

The bounded sub-principal residual is the image of the μ_4 marks, not a free postulate ([`verification/v201_seam_subprincipal_marks.py`]). One genuine step further: the seam DtN is $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature function $f(\theta)$ on the seam circle (the standard Steklov/DtN structure), so $\langle n|M_f|n'\rangle = f_{n-n'}$. Then M_f is μ_4-character-block-diagonal *iff* f has Fourier support only on modes $\equiv 0 \pmod 4$, i.e. iff f is $\mathbb{Z}_4$-invariant (the principal $|k|$ already commutes, v198). But a curvature sourced by the four marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ for *any* profile g, has $f_m = 4\, g_m\, [m \equiv 0 \bmod 4]$ — automatically $\mathbb{Z}_4$-invariant. So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 equally-spaced marks ($\mathbb{Z}_3$) or 4 generic marks both break it (negative controls). The residual of $\omega \circ \rho = \omega$ thus collapses to "the DtN sub-principal symbol is *mark-local* (the seam is flat away from the μ_4 marks = the conformal-deck structure, v177)" — a locality/flatness property, structurally definitional, the narrowest form yet of `QGEO.SYM.01`, [O].

Certified on realistic Steklov profiles, and at the state level ([`verification/v210_mark_local_dtn.py`]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump $g(\theta)$ localised at a puncture, summed over the four marks $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, has *every* off-(mod 4) Fourier coefficient vanishing to $< 10^{-16}$ (exact 4-fold cancellation), the assembled DtN $\Lambda = |D_\theta| + M_f$ has $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the *quasi-free state* it defines, with covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, is clock-invariant: $\rho\, C\, \rho^\dagger = C$ to $< 10^{-13}$, i.e. the actual $\omega \circ \rho = \omega$, not merely $[\rho, \Lambda] = 0$. A single off-centre bump, a $\mathbb{Z}_3$ (three-mark) source and four unequally-spaced marks each break both by $O(1)$, and the mark-local commutator stays $< 10^{-10}$ as the truncation grows ($N \in \{16, 32, 64\}$) — not an artefact. This *numerically certifies the implication* "mark-local DtN $\Rightarrow \omega \circ \rho = \omega$" for real profiles; the premise that the physical seam *is* mark-local stays [O] (it does not close `QGEO.SYM.01`).

The same residual, read dynamically: modular flow + a recovery Lindblad generator ([`verification/v238_modular_lindblad_dynamics.py`]). The whole `QGEO.SYM.01` discussion above is *static* — a commutator condition $[\rho, \Lambda_\Sigma] = 0$. Read as a *dynamics* it is the statement that the modular flow $\sigma_t = \Delta^{it}$ (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian $\Lambda_\Sigma = \log \Delta_\omega$) *conserves* the μ_4 clock: $\rho\, \sigma_t\, \rho^{-1} = \sigma_t\ \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ — so the open premise is exactly "modular time conserves the carrier charge". The dissipative half is the recovery channel T of v221 read as a generator: $L = \log T$ is a doubly-stochastic (CPTP-classical) Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time), and its dissipative gap is $-\log((2/3)^6) = 6\log(3/2) = \Delta$, the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together $G = i\Lambda_\Sigma + L$ is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real≤ 0) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that σ_t is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra — is the same `QGEO.SYM.01` bedrock, [O].

From the dynamics to a QFT skeleton on the seam ([`verification/v239_kms_thermal_time.py`], [`verification/v240_gns_os_reconstruction.py`], [`verification/v241_dhr_sectors.py`], [`verification/v242_spin_statistics_modular.py`], [`verification/v243_haag_ruelle_braiding.py`], [`verification/v244_spectral_action_contract.py`]). With the static→dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at $\beta = 1$, and the seam unit $2\pi = 1/(4c_3)$ fixes the modular temperature as the horizon temperature $T_H = c_3/M$. *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from $(\mathcal{M}, \omega)$, and the OS Hamiltonian $H_{\text{OS}} = -\log T = -L$ is positive with gap Δ — the v64 "$H \geq 0$ + mass gap" is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (`FORM.GLUE.01`); fusion is recovered from the modular S by Verlinde, the Gauss–Milgram sum returns $c = 8$, and the condensation tower $16 \to 4 \to 1$ (v235/v237) selects the holomorphic $(E_8)_1$. *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed $(E_8)_1$. The one genuinely 4d step is typed as a contract, `CONTRACT.QFT4D.01` (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open [O] target. Its representation-theory core is now discharged ([`verification/v245_spectral_matter_content.py`]): the

16 is exactly one *anomaly-free* SM generation ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$) with the forced SM hypercharges and the spectral-action value $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\, g_1$), so only the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one contract* (`CONTRACT.QFT4D.01`), not a diffuse list.

The perturbative 4d S-matrix is constructible ([`verification/v269_spert_paqft_skeleton.py`]). The 4d scattering layer is typed precisely as three *distinct* objects: $S_{\rm top}$ (the 2d DHR braiding monodromy, $|M|$=1 statistics, v243), $S_{\rm pert}$ (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and $S_{\rm phys}$ (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. [E] the spectral-action a_4 interaction is power-counting renormalizable (every operator mass dimension ≤ 4: 7 marginal +3 relevant, a *finite* counterterm basis), so [C] by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix $S(g) = T\exp(i\int g\, L_{\rm int})$ is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit. [C] this is *perturbative* only: the nonperturbative ambient measure `QG.AMB.01` is now a [C] redundancy (v369), a certification object rather than missing dynamics, and the canonical 4d reading remains boundary-only (v265); $S_{\rm pert}$ is the perturbative cross-section layer on top (`QFT4D.SPERT.01`).

A concrete one-loop EG number ([`verification/v271_eg_oneloop_quartic.py`]). The skeleton is taken from "exists" to an actual Epstein–Glaser renormalization: the order-1 product $T_1 = {:}L_{\rm int}{:}$ needs no extension, and the first one is the order-2 T_2, whose UV behaviour is the Steinmann *scaling degree*. [E] the massless propagator has sd $= 2$ in $d = 4$, the one-loop bubble (two propagators) has sd $= 4 = d$, so the singular order $\omega = \text{sd} - d = 0 \Rightarrow$ *exactly one* logarithmic local counterterm ($C(\omega{+}d, d) = 1$ — finite, renormalizable, no infinite tower); the one-loop factor is $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly, the *same* $1/(16\pi^2)$ that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop running is then $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s, t, u channels), with the EG initial condition fixed by the spectral action, $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$. [O] this is order-2/one-loop, one diagram class; the all-order perturbative construction stays open (`QFT4D.SPERT.02`), while the nonperturbative measure is discharged as a [C] redundancy (below).

Red-team caveat, now discharged ([`verification/redteam/rt_F_qft4d.py`]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the local R^2/Weyl^2 *truncation* gives a four-derivative propagator $1/(p^2(p^2 + M^2))$ whose massive spin-2 mode has a negative residue (the Stelle ghost). But that ghost is a Seeley–DeWitt *truncation artefact*: the entire seam KMS form factor $a(p^2){=}e^{p^2/M^2}$ keeps its only pole at $p^2{=}0$ (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established. The matter+gauge $S_{\rm pert}$ is unitary outright; the *nonperturbative* `QG.AMB.01` (asymptotic safety v207 / the boundary $(E_8)_1$ net) is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap.

The gauge running is an EG order-2 number ([`verification/v273_eg_gauge_running.py`]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content: $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$, i.e. [E] $b_3 = -7$, $b_2 = -\frac{19}{6}$, $b_1 = \frac{41}{10}$ (GUT-normalised; the same 41 as the carrier algebra $10\, b_1 = g_{\rm car}\, 2^{g_{\rm car}-2} + 1$, v159). [C] the physical RG directions ($b_3 < 0$ asymptotic freedom, $b_1 > 0$ the $U(1)$ Landau pole) are EG-derived and feed the unification cross-check (v246/v249); [O] the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (`QFT4D.SPERT.03`).

The carrier content *is* one anomaly-free SM generation ([`verification/v310_carrier_sm_anomaly.py`]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor $16 = \Lambda^{\rm even}(\mathbb{C}^5) = 1{+}10{+}5$ decomposes under $SU(5) \subset SO(10)$ as $10{+}\bar{5}{+}1$ = exactly one Standard-Model generation ($Q, u^c, d^c, L, e^c, \nu^c$), and that content is [E] gauge-anomaly-free ($\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$, exact) while reproducing $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family would shift b_1 off $41/10$, so $N_{\rm fam}{=}3$ *pins* the RG. [E] for the spectrum, anomalies and one-loop β; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay [C]/[O].

The bridge to the physical S-matrix and one-loop unitarity ([`verification/v278_lsz_bridge_unitarity.py`]). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of $S_{\rm pert}$ are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives $S_{\rm pert} \xrightarrow{\text{LSZ}} S_{\rm phys}$ (the three-layer S-matrix $S_{\rm top}\,|\,S_{\rm pert} \xrightarrow{\text{LSZ}} S_{\rm phys}$), valid because the gap $\Delta{=}6\log(3/2)$ gives Haag–Ruelle asymptotic states. **[E]** *one-loop unitarity* is the optical theorem: the s-channel bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ is *exactly* the two-body phase space (turning on at the physical threshold $s=4m^2$), so $2\operatorname{Im} M = \sum\int d\Pi\,|M|^2$ holds and the matter+gauge sector is unitary order by order. **[C]** the gravity sector's Stelle ghost is a truncation artefact (perturbative spin-2 unitarity established, v304/v370/v380); the nonperturbative `QG.AMB.01` is a **[C]** redundancy (v369) (`QFT4D.SPERT.04`).

The first hard data confrontation ([`verification/v246_unification_data_crosscheck.py`]). TFPT's gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ "Dark matter = no new state"); the E_8 cascade is an arithmetic even-integer spine ($D_n=60-2n$, v5), *not* a particle/threshold tower. So a "TFPT-adapted" RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over $\sim 10^{13}$–10^{17} GeV with residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV; at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $\mu \approx 1.7\times10^{14}$ GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, **[X]**: with "no new state" there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

The carrier-native Pati–Salam program ([`verification/v247_e8_branching_no126.py`], [`verification/v248_ps_algebra.py`], [`verification/v249_ps_unification.py`], [`verification/v250_dirac_triple_contract.py`]). The unification tension above is resolved *natively* if the carrier $SO(10)$ is gauged (the carrier $D_5 \oplus A_3 = SO(10)\times SU(4)$ is Pati–Salam). *First*, the E_8 audit hull *forbids* the **126**: the **248** branches as $(45,1)+(1,15)+(10,6)+(16,4)+(\overline{16},\overline{4})$, so the available $SO(10)$ reps are exactly $\{1,10,16,45\}$ — the renormalisable 126_H is algebraically absent, making the "16_H over 126_H" selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ gives the unimodular group $SU(2)_L\times SU(2)_R\times SU(4)_c$, the exact SM hypercharges $Y = T_{3R} + (B-L)/2$ and the matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale, $M_{PS} = \kappa M_s$, $\kappa \in [1.0, 1.7]$ at one loop ($\to [1.0, 1.2]$ at two loops), and proton decay selecting the content (minimal excluded, the E_8-allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual "carrier is gauged" reduces to ONE constructive object, the spectral triple (A, H, D_F, J, γ) with A_F Pati–Salam and $H_F = 3\times16$, whose NCG axioms (the first-order condition above all) and the σ =scalaron identification are the open contract `CONTRACT.QFT4D.DIRAC.01` (v250). *Fifth*, a first concrete step on that triple is taken ([`verification/v251_first_order_sigma.py`]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/σ term — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ (honest correction: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([`verification/v252_full_finite_triple.py`]) — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — and the NCG axioms are verified by direct computation: reality J, grading γ, KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on $\{\nu_R, e_R\}$), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}} = m_D^2/M_R$ — thereby discharging five of the seven obligations of `CONTRACT.QFT4D.DIRAC.01` to **[E]** (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio κ in $M_{PS} = \kappa\, M_s$ is pinned to a tight $O(1)$ interval (≈ 1.3) by the RG and shown to be a scale-free ratio of heat-kernel coefficients, $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, over the same 96-dim H_F ([`verification/v253_kappa_scalaron_ps.py`]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations Ω_D^1 give exactly the one SM Higgs doublet $(1,2)_{1/2}$ with no junk and no **126**, while σ (the ν_R Majorana scalar, $B-L=2$) appears only as the beyond-first-order Pati–Salam field ([`verification/v254_inner_fluctuations.py`]); the spectral-action Seeley–DeWitt expansion a_0, a_2, a_4 yields the cosmological, Einstein–Hilbert and Yang–Mills+R^2+Higgs terms with $b/a^2 = 1/3$, $\sin^2\theta_W = 3/8$ and the R^2 scalaron, making κ an explicit moment ratio ([`verification/v255_spectral_action_expansion.py`]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three ν_R forces a skew intersection form, corank

1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([`verification/v256_orientability_poincare.py`], [`verification/v257_quasiorient_modpd.py`]). *Ninth*, the built D_F is no independent posit but the *modular/covariance induction* of the boundary seam state ([`verification/v258_dirac_covariance_induction.py`]): a quasi-free (KMS, β=1) state is fixed by its covariance C (a positive contraction), with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta) inverse to the KMS occupation $C = (1+e^H)^{-1}$; inducing the carrier covariance $C_F = P_F C_\Sigma P_F$ returns the v252 operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw m_D^2/M_R recovered — so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ of the boundary geometry and the v18 Yukawas are the *readout* of C_Σ, **[E]** modulo the one **[O]** seam realisation. *Tenth*, the last open spectral-action input — the cutoff function f — is fixed by the same seam state ([`verification/v259_modular_cutoff_kappa.py`]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (v239) and the seam unit 2π=$1/(4c_3)$ (v58) fixes that β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*, not chosen, with moments $f_2/f_0 = f_4/f_2 = 1$ *exactly* (a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$, so the choice is non-vacuous); hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\mathrm{grav}}} = \sqrt{c_{PS}/c_{\mathrm{grav}}}$ loses its scheme factor and the v253/v255 open **[O]** cutoff freedom becomes a **[C]** finite trace ratio over the 96-dim H_F (the RG κ≈1.3 pinning c_{PS}/c_{grav}≈1.7). So unification is now *one* theory decision (gauge $SO(10)$?) plus *one* constructive object, both falsifiable — not a fog. The colour $SU(4)_c$ here sits inside $SO(10)$=D_5 and is distinct from the carrier family factor A_3.

The implication is now Lean-formalised (`FORM.QGEO.01`). The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ($f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$, i.e. $[\rho, M_f] = 0$) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext`, `Classical.choice`, `Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the document-1 analytic input — so the conditional theorem "`SeamDeckPremise` $\Rightarrow \omega \circ \rho = \omega$" is now **[E]** while `QGEO.SYM.01` stays the one open **[O]** postulate.

The two residuals are one canonical metric: the pillowcase ([`verification/v214_seam_pillowcase.py`]). The isometry residual `QGEO.ISO.01` (v180) and the mark-locality residual `QGEO.SUBPRIN.01` (v201) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class.* The four marks make the orbifold sphere $S^2(2,2,2,2)$, whose orbifold Euler characteristic $\chi_{\mathrm{orb}} = 2 - 4(1-\frac{1}{2}) = 0$ is *Euclidean*: by Troyanov's cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits π sum to $4\pi = 2\pi\chi(S^2)$) — this *is* the v201 "flat away from the marks". And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that v168 already proved: $j(\lambda) = 256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$ gives $j(2) = 1728$, so the modulus is the *square* torus $\tau = i$, whose complex multiplication by $\mathbb{Z}[i]$ supplies the order-4 isometry $z \mapsto iz$ (explicitly $(x,y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$, order 4); a generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$, so the clock is specific to μ_4. Thus the two open QGEO residuals *collapse to one* canonical-metric statement and `QGEO.ISO.01` ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise "the RP seam metric *is* the uniformising constant-curvature representative of its conformal class" stays **[O]** — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (`QGEO.PILLOW.01`).

The converse, on the canonical metric: raw seam ⇒ mark-locality ([`verification/v264_raw_seam_marklocal.py`]). Composing the chain explicitly closes the gap between v216, v201 and v198 on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (v216), so the DtN sub-principal symbol M_f is a μ_4-orbit sum, hence $\mathbb{Z}_4$-invariant (Fourier support on $4\mathbb{Z}$, verified by FFT), hence block-diagonal in the clock-character basis, hence $[\rho, M_f] = 0$; with the principal symbol $|k|$ already commuting (v198) this gives $[\rho, \Lambda] = 0$ and $\omega \circ \rho = \omega$ *on this metric*. A negative control (curvature *off* the μ_4 orbit) breaks $\mathbb{Z}_4$-invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement "the raw RP seam carries the uniformising flat pillowcase metric" — which *is* `QGEO.SYM.01`. A genuine reduction/sharpening (`QGEO.ENERGY.03`), **[O]** — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

The minimal axiom: a rigidity theorem ([`verification/v267_qgeo_rigidity_minimal_axiom.py`]). One can sharpen the bedrock to its irreducible kernel. **[E]** the *single* bare statement "the seam admits the carrier's order-4 clock as a conformal symmetry cyclically permuting its four marks" forces everything:

an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 *independent of* a; cross-ratio 2 ⇔ $j{=}1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) ⇔ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2{=}x^3{-}x$; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality, $[\rho, \Lambda]{=}0$ and $\omega \circ \rho{=}\omega$ all follow (v214/v201/v264/v198). **[E]** neg controls: a generic four-mark set has $j \neq 1728$ and only a $\mathbb{Z}_2$ symmetry (no clock), the hexagonal point $j{=}0$ has a $\mathbb{Z}_6$ symmetry (order $6 \neq 4$) — only the square (order-4) point realises the μ_4 clock. **[C]** a second, independent reason: $\{1, i, -1, -i\}$ is defined over $\mathbb{Q}$ with $j{=}1728$ (CM by $\mathbb{Z}[i]$), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. **[O]** So `QGEO.SYM.01` is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like c=const in relativity) cannot be derived from nothing. The *reduction* is closed (axiom ⇒ everything); the axiom itself stays the one foundational postulate (`QGEO.SYM.02`).

The flat geometry closes the commutator to all orders (`[verification/v276_qgeo_flat_closes_commutator.py]`). The state-level residual $[\rho, H]{=}0$ — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat $\tau{=}i$ pillowcase is granted. **[E]** if $H = f(\Delta_{\text{flat}})$ is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck ρ $(z \mapsto iz)$ is a flat-$\tau{=}i$ isometry, then $[\rho, \Delta]{=}0 \Rightarrow [\rho, H]{=}0$ *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full H — flatness removes the lower-order terms. **[O]** so `QGEO.SYM.01` collapses to a *single geometric* statement, "the raw RP seam state is the flat $\tau{=}i$ pillowcase quasi-free state ($H{=}\sqrt{-\Delta_{\text{flat}}}$)"; everything downstream is **[E]**, and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (`QGEO.SYM.03`).

The obligation as one precise lemma, Lean-backed (`[verification/v279_qgeo_obligation_lemma.py]`). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise "the raw seam carries the flat $\tau{=}i$ pillowcase metric g_{flat} (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{g_{\text{flat}}}}$", the conclusion $\omega_\Sigma \circ \rho = \omega_\Sigma$ (and hence mark-locality, the metric inclusion, E_8) follows. **[E]** the reduction DAG from "free RP seam" to this premise has *exactly one* un-discharged leaf — "the raw seam state *is* the flat state" — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes (**[C]**): RP-state uniqueness on the flat $\tau{=}i$ orbifold, or Troyanov + OS reconstruction. **[E]** the all-orders closure step is now a *Lean theorem*: `TfptCarrier/SeamDeckClosure.lean` (`flat_all_orders_clock`) proves that any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — upgrading v198/v201 to the full operator with only the standard axioms, no `sorry` (`FORM.QGEO.03`). **[O]** the one premise stays the honest axiom (`QGEO.OBLIG.01`).

Self-investigation: the experiment and the simplification (`[verification/v280_pillowcase_steklov.py]`, `[verification/v282_e8_tau_i_unification.py]`). Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat $\tau{=}i$ pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and $[\rho, \Delta]{=}0 \Rightarrow [\rho, H]{=}0 \Rightarrow [\rho, C]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still **[O]**. (ii) A candidate *simplification*: the $(E_8)_1$ character $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$ gives $\chi_{E_8}(i) = 1728^{1/3} = 12$, so $\tau{=}i$ is *both* the order-4 CM point (`QGEO.SYM.01`, flat geometry) *and* the $(E_8)_1$ partition-function modulus (`QG.AMB.01`, holomorphy) — the two open premises are two faces of one object, "the raw seam is $(E_8)_1$ at $\tau{=}i$", dropping the obligation count $2 \to 1$ (`QGAMB.UNIFY.01`).

The two proof routes, decomposed (`[verification/v284_route_i_rp_uniqueness.py]`, `[verification/v285_route_ii_seam_condensation.py]`). The keystone — now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` via the lattice/S_3 stack (below; the parent stays **[O]**) — can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — **[E]** five are already **[E]**/**[O]** (RP→DtN v194; marks→pillowcase v195/v216; the flat Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$, principal symbol $|\xi|$ with no curvature term; covariance from the DtN v258; μ_4-invariance v276), and **[C]** Troyanov ($\chi_{\text{orb}}{=}0$) reduces the residual to "the raw seam is the constant-curvature geometric state". *Route (ii)* (seam-net condensation) is a four-lemma chain — **[E]** three are discharged (no abelian sector ⇔ $\det K{=}1$ ⇔ holomorphic ⇔ E_8 v234/v277/v281; the tower 16→4→1 v92), leaving "the free RP seam condenses the μ_4-Lagrangian glue" =`QGEO.SYM.01`. **[E]** **the two open lemmas coincide** — both say "the raw seam is the $(E_8)_1$ at $\tau{=}i$" (v282), so the canonical equivalence between the raw seam

KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$ is the *single* [O] statement that closes both (`QGEO.ROUTEI.01`/`QGAMB.ROUTEII.01`).

The Seam Equivalence Theorem, named and firewalled ([`verification/v286_seam_equivalence_contract.py`]). That single statement is now the named target `SEAM.EQUIV.01`: *the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau{=}i$ pillowcase* (respecting the DtN, the KMS flow, the μ_4 deck, the Q-system extension, $\det K{=}1$ and the $(E_8)_1$ character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. [E] the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through `SEAM.EQUIV.01` (this kills the "E_8 smuggled into the geometry and pulled back out" attack). The two proof routes are then attacked directly: [E] Route A (AQFT, [`verification/v287_free_rp_bulk_to_holomorphic_boundary.py`]) reduces the Gral to *one* standard theorem, "invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary" (4/5 lemmas discharged); and [E] Route B (DtN, [`verification/v288_full_l2_subprincipal_z4.py`]) *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod 4$), so $[\rho,\Lambda]{=}0$ on all of L^2), shrinking the residual to the single sharper question "why is the raw seam sub-principal term mark-local?" (`SEAM.EQUIV.A01`/`SEAM.EQUIV.B01`).

The route split (2026-07-22; ID restructuring, no status change in substance). The target statement now carries *two named routes* as ledger rows of their own: `SEAM.EQUIV.MMST.01`, the conditional scaling-limit route (the lattice/S_3/Lean stack documented in this section — *closed modulo cited theorems*, its only [O] residual the abstract continuum existence v336), and `SEAM.EQUIV.TWISTOR.01`, the open Costello–Li construction (prepared by `CELEST.SEAM.01`, bridged to physics only by `WOIT.OS.TWISTOR.01`; [O]). The parent claim `SEAM.EQUIV.01` remains, with exactly this logic as its claim text: *closed if `MMST.01` or `TWISTOR.01` closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open $\Rightarrow$ the parent stays [O] as an unconditional claim.* Every "closed modulo a cited theorem" statement in the papers and on the website is henceforth attached to the MMST route ID; the assertions themselves are unchanged.

Route B sharpened to one analytic lemma; both routes precisely scoped. That last question is decomposed ([`verification/v289_marklocal_raw.py`], `MARKLOCAL.RAW.01`) into a five-lemma chain whose only open link is [O] *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the μ_4-orbit source, the $4\mathbb{Z}$ Fourier support (v264/v288) and the full-L^2 commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert "full-L^2 QGEO" — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A's "one standard theorem" is itself a composition of citable results (Kitaev/Freed–Hopkins, Müger/Kawahigashi–Longo–Müger, Kawahigashi–Longo, [`verification/v287_free_rp_bulk_to_holomorphic_boundary.py`]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B's Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

Red-team: $\mathbb{Z}_4$-invariance is not mark-locality. A deliberate adversary ([`verification/v290_z4_smooth_curvature_adversary.py`]) shows the honest limit of Route B's lift: a *smooth* $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$ *passes* the v288 commutator ($\|[\rho,\Lambda]\| = 0$, since $i^n - i^{n\pm4} = 0$) yet is not mark-local — so $[\rho,\Lambda]{=}0$ is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ($\max|\Delta\lambda| = 0.155$, $\Delta\mathrm{Tr} = 0.019$ at $\varepsilon = 0.3$), i.e. the KMS weight and $(E_8)_1$ character that `SEAM.EQUIV.01` must match. Consequently the open lemma is named in its own right ([`verification/v291_flataway_contract.py`], `FLATAWAY.RP.01`): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

All three routes attacked; Flat-Away reduced to one selection principle. Each route is now carried to its precise reduction: [E] the *heat* route ([`verification/v292_flataway_heat_reduction.py`]) shows the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature (leading invariant $\|f\|_{L^2}$), so Flat-Away ⇐ "the seam fixes the a_2 heat coefficient"; **[E]** the *spectral* route ([`verification/v293_flataway_spectral_hessian.py`]) lifts this to the full smooth-$\mathbb{Z}_4$ space — mode $4k$ splits the degenerate Steklov level $|2k|$, the mismatch Hessian over $\{4,8,12\}$ is positive-definite ($\{0.497, 0.500, 0.503\}$), so the flat seam is a *strict* isolated minimum (answering "only one direction"); and **[E]**/**[C]** the *variational* route ([`verification/v294_flataway_rp_energy.py`]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix a_2 / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A's holomorphy half, `SEAM.EQUIV.01`.

The heat route made exact and Lean-formalised. The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([`verification/v295_flataway_a2_exact.py`]): the first variation of $\mathrm{Tr}\, e^{-t\Lambda}$ vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\, e^{-t\Lambda}$ is convex (Klein/Peierls), ε=0 is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for every deformation, $= 0$ iff $f = 0$; the exact second-order coefficient is the divided-difference kernel of e^{-tx} (validated to 10^{-6}). The algebraic Δ=0 ⇔ flat step is machine-formalised in Lean (`FORM.FLATAWAY.01`, `SeamDeckClosure.lean`: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away's heat route is reduced to the single external invariant "the seam fixes a_2", with its positivity both proved and formalised. The a_2 coefficient itself is now in *closed form* ([`verification/v296_flataway_a2_closed.py`]): the operator tails telescope geometrically to $W_k^{\mathrm{tail}} = t(1-e^{-4kt})/(8k(1-e^{-t}))$, leaving a finite $(4k-1)$-term middle, e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ (manifestly positive; small-t $W_k = t/2 + O(t^3)$). Symmetrically, Route A's "one standard import" is now a precise, citable theorem stack ([`verification/v297_route_a_literature_stack.py`]): LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk ⇒ $(E_8)_1$", all three legs established, the one shared hypothesis (seam-bulk invertibility) coinciding with Route B's Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input, which the lattice/S_3 stack (below) now pins at every computable level, making the keystone's MMST route (`SEAM.EQUIV.MMST.01`) *closed modulo cited theorems.*

Flat-Away hardened, and the pin derived from $(E_8)_1$ ([`verification/v300_flataway_rigid.py`], `FLATAWAY.RIGID.01`). The spectral route was a *soft* minimum ("the mismatch grows with ε"); it is now a *discrete* obstruction. **[E]** the flat fingerprint is the integer Steklov ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (the exact 2×2 block $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, validated against the full Toeplitz operator), so **[E]** any $g_k \neq 0$ lifts the degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every $g_k = 0$, a degeneracy obstruction robust to rescaling and strictly stronger than the v293 "deviation grows" scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: **[C]** the $(E_8)_1$ vacuum character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (E_8 even unimodular ⇒ integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder ⇔ flat seam — i.e. "the seam fixes its spectrum" *is* the $(E_8)_1$ rationality of Route A. **[O]** Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two `SEAM.EQUIV.01` routes provably converge on one rationality fact (consistent with the v286 firewall).

Route A's last hypothesis discharged — the gapped quasi-free bulk is invertible ([`verification/v301_route_a_invertible.py`], `SEAM.EQUIV.A.INV.01`). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A's open input (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev's periodic table, class D, the only invariant being the chiral central charge). **[E]** the carrier is 16 Majoranas, so $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8 = 16/2$ (v148); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+ 128$ spinor $= 248$, v154); **[C]** hence a gapped quasi-free bulk is invertible automatically, and **[E]** the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) against 4 for the gauged D_8=$SO(16)$ (anyons), which free fermions cannot produce. **[O]** So LIT-A's 'invertible bulk' *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line:* together with v300, the entire open content of `SEAM.EQUIV.01` reduces to the single spectral statement "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

The last lever — the seam gap *is* the derived Recovery gap ([`verification/v302_seam_gap.py`],

`SEAM.EQUIV.GAP.01`). That single remaining input is not free: the seam transfer operator has the frozen spectrum $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap* $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ — a number forced by the carrier ($(2/3)^6$ the μ_4-deck transfer eigenvalue), with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) making it robust. [C] By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian (correlation length $1/\Delta$). *Bottom line:* with v300/v301, the whole residual of `SEAM.EQUIV.01` is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$): *no undischarged TFPT-internal assumption remains.* It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

The continuum chain assembled, and the residual tied to the seam one-sidedness ([`verification/v356_continuum_mmst_applicability.py`]**,** `SEAM.EQUIV.CONTINUUM.03`**).** Writing the chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar $\to$ **S2** invertible (Kitaev free fermions, v301) $\to$ **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) $\to$ **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk–edge) $\to$ **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, i.e. $8 \leq 8 \leq 16$) $\to$ **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). Every link is a theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3 = 1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase", [O] but theorem-bounded on both sides.

MMST in-class verified for the collar, so the scaling limit is $(E_8)_1$ ([`verification/v366_mmst_seam_collar.py`]**,** `SEAM.MMST.INCLASS.01`**).** The continuum leg is now checked *hypothesis-by-hypothesis* for the explicit seam collar: it is a $D{=}\dim S^+{=}16$ Majorana quasi-free (CAR) state (v155/v160), gapped with $\Delta{=}6\log\frac{3}{2}{>}0$ *derived* (v302), and chiral with $c_-{=}D/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, so it sits squarely in the MMST free-lattice-fermion class with rank $\leq c \leq D$ reading $8 \leq 8 \leq 16$ (*in* range, saturating rank $=c{=}8$). The MMST massless-scaling-limit theorem then applies as a cited result, and the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$, one primary) over the same-c rival $SO(16)_1$ ($\det K{=}4$, v351). [E] the class arithmetic, the derived gap, the chirality and the $(E_8)_1$ pinning; [C] the CAR-class typing and the cited MMST theorem; [O] the lone residual is exactly S3 above (the lattice realization). The continuum side of `SEAM.EQUIV.01` is thereby reduced to a single, named, theorem-bounded input. The lattice$\to$conformal-net step itself is now a *rigorous AQFT construction*: Adamo–Giorgetti–Tanimoto (`arXiv:2506.01008`, 2025) build the two-dimensional conformal net $\mathcal{A}_Q$ of an even lattice Q and *classify* every two-dimensional extension of the Heisenberg net (under a discreteness assumption) as such a lattice net — so "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage the continuum leg invokes, not a hope; and since (xxxii) ([`verification/v469_seam_crossedproduct_route.py`]) the extension step no longer even *needs* the preprint lineage — the local $\mathbb{Z}_2$ simple-current crossed product ($h_s{=}1{\in}\mathbb{Z}$, Longo–Rehren 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM 2001) certifies it on peer-reviewed ground, with AGT/AMT as the independent second witness.

S3 made concrete: an explicit gapped chiral-Majorana lattice model ([`verification/v367_seam_s3_lattice.py`]**,** [`verification/v368_seam_s3_inflow.py`]**).** The S3 input is no longer abstract. The $p{+}ip$ Bloch model $h(k){=}\sin k_x\,\sigma_x{+}\sin k_y\,\sigma_y{+}(M{-}\cos k_x{-}\cos k_y)\sigma_z$ is, at $M{=}1$, a gapped free-fermion phase with numerically-computed Chern number $|C|{=}1$ ($C{=}0$ trivially at $M{=}3$), so $N_{\mathrm{Maj}}{=}\dim S^+{=}16$ copies give a chiral edge of $c_-{=}16/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) condensed to $(E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). A finite STRIP of the same model carries an edge-localised in-gap chiral branch (absent in the trivial phase) – so by anomaly inflow the chiral edge *exists* as a consequence of the invertible bulk (the S4 leg), discharging the "does the edge exist?" question on the lattice. [E] the gapped Chern model, the $c_-{=}8$ counting and the edge existence; [C] the μ_4 condensation to $(E_8)_1$; [O] only the genuine *continuum scaling limit* (MMST, v336) stays open — S3 is a runnable lattice model that pins the target at every computable level, so the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is *closed modulo cited theorems.*

The S3 closure stack: $c = 8$, the $(E_8)_1$ character, the torus count and reflection positivity — all computed on the lattice ([`verification/v376_seam_s3_centralcharge.py`]**,** [`verification/v377_seam_s3_e8character.py`]**,** [`verification/v378_seam_s3_modular.py`]**,** [`verification/v379_seam_s3_rp.py`]**).** On that explicit model the target of the scaling limit is now pinned at every level a computation can reach. *Central charge* (v376): the Calabrese–Cardy finite-size entanglement scaling gives $c = 1.0000$ per critical complex mode, so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\text{car}}+N_{\text{fam}}$, and it is chiral ($c_- = 8$). *Convergence* ([`verification/v392_seam_s3_scalinglimit.py`]): across five sizes $L = 66 \dots 514$ the slope $c_1(L)$ is a monotone Cauchy-like sequence approaching 1 and $1/L$-Richardson-extrapolates to $c_1(\infty) \approx 1.000$ ($c \approx 8.00$ in the limit) — the existence-relevant *convergence* the two-size value did not test (evidence, not a proof of existence; the residual stays **[O]**). *The net* (v377): the exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary), and the μ_4/GSO promotion 248=240+8=120+128 (240=16·5·3) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, four primaries). *Genus-1* (v378): the KLM condensation $\mu(\text{carrier})$=16 → $\mu(E_8)$=16/4^2=1 and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 — the holomorphy / $\det K$=1 statement that the genus-0 arguments could not reach (v344). *Reflection positivity* (v379): the gapped collar's Euclidean two-point function has a positive spectral measure, so the OS reflection Gram matrix is positive semidefinite (a ghost mode breaks it). **[E]** all four (numerical c, exact character, genus-1 count, RP); **[O]** the one thing a computation cannot supply is the abstract *continuum existence* of the limit (the cited MMST theorem, v336). With RP in hand the ambient measure C7 is then discharged by the redundancy theorem (v369), so S3 is the master key: the target is pinned exactly, only the existence proof is external.

A ground-state witness for premise (i): the modular commutator measures c_-=8 from the bulk state alone ([`verification/v489_seam_modular_commutator.py`], `SEAM.CCC.01`). The chiral central charge entered so far through band topology (the Chern number, v367) and through counting (the KLM/det-Cartan arithmetic, v378). The Kim–Shehab–Kim modular commutator $J(A, B, C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022)) reads c_- off the *entanglement data of the exact BdG ground state* of the same collar — no band bundle, no representation theory enters — via the Peschel covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$) on a disk cut into three 120° wedges: per layer $c_- = 0.499998$ at L=32 (converged 0.500000 at L=48, deviation -1.8×10^{-7}), the trivial M=3 control gives 0.000000, the orientation flip M=−1 gives −0.499986 (sign flip), layer additivity is exact (10^{-14}), and the 16-layer carrier lands on $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$ (direct run 7.9967 at L=16, additivity route 8.0000). **[E]** the third, mutually independent lattice witness of premise (i) of `SEAM.EQUIV.01` — and the first read from ground-state entanglement alone; **[O]** unchanged: the continuum scaling limit (MMST, v336) and the cited KLM condensation step stay exactly where v367/v378 left them — `SEAM.EQUIV.01` stays **[O]**.

The spin-structure parity census: μ_{pre}=4 witnessed on the lattice ([`verification/v490_seam_parity_census.py`], `SEAM.MU.01`). Step 1 of the KLM chain (v378) — “the carrier sits in the $SO(16)_1$ class, μ_{pre}=4, *before* the μ_4 condensation” — was arithmetic; it is now a lattice ground-state measurement. The T^2 fermion parity per spin structure (AA, AP, PA, PP) is determined *twice* (Read–Green TRIM product and real-space Majorana Pfaffian, spectra matching to 9×10^{-15}): per layer $(+, +, +, -)$ — the ν=1 Ising signature — and for the 16-layer carrier parity^{16} = + in *all four* sectors, so the gauged torus GSD is $4 = \mu(SO(16)_1) = |\det \mathrm{Cartan}(D_8)|$. The 16-fold-way bridge then runs on *measured* numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (v489) — the gauged vortex is a boson, condensable, while the rivals $\nu = 1, 2, 8$ are not holomorphically condensable. Honest fences: the fermionic Kitaev–Preskill TEE is identically zero for *both* M=1 and M=3 ($\gamma_f = 0.000001/0.000000$) — an invertible fermionic phase has no TEE, so γ_f does *not* discriminate $(E_8)_1$ vs $SO(16)_1$; and the KLM step $\mu : 4 \to 4/2^2 = 1$ stays the cited theorem (**[C]**, the condensed model is interacting). The numerics pin exactly the premise package $\{c_-$=8, μ_{pre}=4, θ_v=1$\}$ of the condensation; `SEAM.EQUIV.01` stays **[O]**.

And that external step is now Lean-formalised as “closed modulo a cited theorem” (`FORM.SEAM.MMST.01`, `TfptCarrier/SeamScalingLimit.lean`). The continuum leg is machine-formalised in the audit-contract style of `SeamEquivChain.lean`: the collar's MMST applicability hypotheses are *provable* arithmetic facts by Lean kernel `decide` (no axioms) — D=16 Majoranas, rank =c=8, the range $8 \le 8 \le 16$, the chirality c_-=8 ≠ 0, and the holomorphy discriminator $\det K$=1 vs $SO(16)$'s 4; the MMST scaling-limit theorem and the Adamo–Moriwaki–Tanimoto OS reconstruction are declared as *named cited axioms*; and the conclusion $(E_8)_1$ follows by a well-typed composition whose `#print axioms` pins the residual to exactly those two published theorems (plus the carrier gap), with no hidden assumption and no `sorry`. Every TFPT-internal hypothesis is thus kernel-checked; only the two cited theorems are assumed — the honest closure ceiling for `SEAM.EQUIV.01`. *Two of the seven convergent routes below are now kernel-hardened the same way:* `FORM.SEAM.BW.HSMI.01` (`TfptCarrier/SeamStandardPair.lean`) proves the four Borchers/Wiesbrock *standard-pair relations* exactly over $\mathbb{Z}$ on the centred ladder ($[K, P]$=P, J^2=1, JKJ=$-K$, JPJ=P_+, all by kernel `decide`, no domain axiom), with the cited Borchers step and the one continuum input as named axioms; and `FORM.SEAM.APPLICABILITY.01` (`TfptCarrier/SeamApplicabilityLedger.lean`) kernel-checks the audit *count* itself — of the 12 cited-theorem hypotheses exactly 11 are internal and 1 external (and $8 \le 8 \le 16$, $\det K{=}1 \neq 4$), all axiom-free — so the very bookkeeping the keystone's honesty rests on is machine-verified, leaving only the one named continuum fact open.

TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure ([`verification/v308_seam_equiv_chain.py`], [`verification/v309_modular_bedrock.py`], [`verification/v311_gap_decoupled_measure.py`]**).** Three machine-checked steps push the keystone arc further. *(i) Composition certificate* ([`verification/v308_seam_equiv_chain.py`]): the closing implication is one well-typed logical chain gap>0 → invertible → holomorphic $c{=}8 \to (E_8)_1$ whose arithmetic discriminators are exact — $|\det\mathrm{Cartan}(E_8)|{=}1$ against $|\det\mathrm{Cartan}(D_8)|{=}4$, the carrier tower $4{\cdot}4{=}16 \to 4 \to 1$, $c{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}8$ — so the only undischarged premise is the cited OS step; **[E]** the discriminators, **[O]** the keystone. *(ii) Modular bedrock* ([`verification/v309_modular_bedrock.py`]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ is built explicitly and $[\rho, K]{=}0 \Leftrightarrow \omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicality of the raw collar state; **[E]** the model, **[O]** the intrinsicality. *(iii) Gap-decoupled measure* ([`verification/v311_gap_decoupled_measure.py`]): the admissible sector clusters exponentially at rate $(2/3)^6$ (correlation length $1/\Delta$, finite susceptibility 729/665) and the v76 margin $\Delta - 31/(4\pi^2) > 0$ decouples it from the ambient, so the physical sector is a convergent object while `QG.AMB.01` is a **[C]** redundancy (v369). Each pins the residual to a single named step; with the lattice/S_3 stack the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is now *closed modulo cited theorems*. *(iv) The intrinsic-BW residual as a cited reduction* ([`verification/v424_seam_lto_rp_reduction.py`]): the open bedrock lemma of (ii) — the raw collar's intrinsic Bisognano–Wichmann property — is reduced to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`), whose reflection-positivity axiom (LTO-RP) is exactly $u_\Theta{=}J$ (the reflection *is* the modular conjugation), i.e. the BW condition $\theta K\theta{=}-K$ — *not* $\omega{\circ}\rho{=}\omega$. **[E]** the μ_4 clock is a modular *symmetry* ($[\rho, K]{=}0$), not the flow ($\sigma^\psi{=}\rho$ is type-wrong, a one-parameter group versus a finite order-4 element); a positive character-block-diagonal K has $[\rho, K]{=}0$ yet, being positive definite, never $\theta K\theta{=}-K$, so clock-invariance does *not* imply (LTO-RP). **[O]** make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* commuting-projector models (Toric Code via a trace, Levin–Wen anyonic), whereas the seam is the opposite corner — invertible ($|\det\mathrm{Cartan}(E_8)|{=}1$, no anyons) and a $\beta{=}1$ KMS state (not a trace) — so two sub-steps stay open: realising the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO, and extending (LTO-RP) to the invertible KMS case. An MMST-style reduction (cf. v308/v336), *not* a closure. *Sub-step (ii) is now constructively exhibited* ([`verification/v426_seam_lto_rp_kms.py`]): a gapped reflection-symmetric μ_4-even collar at $\beta{=}1$ KMS satisfies $\Theta K\Theta^{-1}{=}{-}K$ ($u_\Theta{=}J$) and $[\rho, K]{=}0$, so (LTO-RP) holds there by direct Tomita–Takesaki — the corner NPW26 leave open — with $|\det\mathrm{Cartan}(E_8)|{=}1$ (invertible, no anyons) and the state a genuine KMS, not a trace; only the continuum sub-step (i) remains **[O]**. On the covariance side, the recent theorem of Adamo–Giorgetti–Tanimoto (`arXiv:2508.07109`, 2025) — every DHR representation of a conformal net extending a loop-group/Virasoro net is *automatically* positive-energy and diffeomorphism-covariant — defuses the old "Bisognano–Wichmann presupposes the very covariance it should produce" circularity (v215): once the seam net is in hand, geometric covariance is a *consequence*, not an input.

Twenty-six convergent attacks on the one continuum residual — intrinsic BW, a bulk continuum motor and its chiral-edge counterpart, the β-homotopy, the applicability audit, uniform rigidity with its forcing converse and the derivation of clock-invariance, three independent reads of the edge central charge, the four-point and uniform-in-N pinning of the external fact, the Ising operator tower, the $(E_8)_1$ torus modular data, the μ_4-from-marks derivation, the kill tests, the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, the chirality-from-P_1 forcing, an $(E_8)_1$ character kill test, the exact MMST citation audit, the complementary lattice-VOA route, the explicit flat-band commuting-projector LTO realisation, the strict-locality (Kapustin–Fidkowski) obstruction, the character-level 128-spinor extension, the $c{=}8$ holomorphic-uniqueness pin and the one-particle realisation rigidity ([`verification/v438_seam_hsmi_borchers.py`], [`verification/v439_seam_lieb_robinson.py`], [`verification/v440_seam_lto_rp_beta.py`], [`verification/v441_seam_applicability_ledger.py`], [`verification/v442_seam_rigidity_uniform.py`], [`verification/v443_seam_e8_discriminator.py`], [`verification/v444_seam_edge_scaling.py`], [`verification/v445_seam_rigidity_forcing.py`], [`verification/v446_seam_clock_invariance.py`], [`verification/v447_seam_edge_chern.py`], [`verification/v448_seam_edge_fourpoint.py`], [`verification/v449_seam_mmst_uniform.py`], [`verification/v450_seam_edge_entanglement.py`], [`verification/v451_seam_edge_cardy_tower.py`], [`verification/v452_seam_e8_modular.py`], [`verification/v453_seam_mu4_from_marks.py`], [`verification/v454_seam_edge_virasoro.py`], [`verification/v455_seam_bw_uniform.py`], [`verification/v456_seam_chirality_from_c3.py`], [`verification/v457_seam_character_killtest.py`], [`verification/v458_seam_mmst_citation_audit.py`], [`verification/v459_seam_lattice_voa_route.py`], [`verification/v460_seam_s3_lto.py`], [`verification/v461_seam_strict_locality.py`], [`verification/v462_seam_spinor_continuum.py`], [`verification/v463_seam_c8_holomorphic_uniqueness.py`], [`verification/v464_seam_oneparticle_rigidity.py`], [`verification/v469_seam_crossedproduct_route.py`]). None closes `SEAM.EQUIV.01`; together they ground the one continuum input on both its halves, reduce it to a single analytic fact and make the claim falsifiable. *(v) The intrinsic Bisognano–Wichmann route* ([`verification/v438_seam_hsmi_borchers.py`]): the seam supplies a Longo *standard pair* / +half-sided modular inclusion — a positive lightlike translation $P \geq 0$, the boost K with $[K, P]{=}P$ (the Borchers scaling $\Delta^{it}P\Delta^{-it}{=}e^{-t}P$), and a reflection J with $JKJ{=}{-}K$, $JPJ{=}P_{-}$ — so by Borchers' theorem the modular flow is geometric *intrinsically*, *without* presupposing the rotation-covariance of the vacuum; **[E]** the ladder algebra and the $sl(2,\mathbb{R})$ positivity (min eig > 0, with the indefinite boost as control), **[C]** the synthesis, **[O]** assembling the finite algebra and the positive-energy rep into one continuum standard pair (the MMST/NPW26 wall). *(vi) The continuum-existence motor* ([`verification/v439_seam_lieb_robinson.py`]): the explicit $p{+}ip$ collar (v367) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral massless *edge* scaling limit; **[E]** gap/clustering/light-cone, **[C]** the lift, **[O]** the edge CFT. *(vii) The β-homotopy* ([`verification/v440_seam_lto_rp_beta.py`]): the (LTO-RP)/BW data ($\Theta K\Theta{=}{-}K$, $[\rho, K]{=}0$) are β-independent, and along the whole path $C_\beta{=}1/(1{+}e^{\beta K})$ from the NPW26-proven *trace* state ($\beta{=}0$) to the seam KMS state ($\beta{=}1$) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii). *(viii) The applicability audit* ([`verification/v441_seam_applicability_ledger.py`]): a hypothesis-by-hypothesis ledger of MMST/NPW26/Adamo finds 11 of 12 hypotheses are TFPT-internal computed facts (**[E]**) and exactly one is external — the continuum existence of the chiral scaling limit — covariance being an Adamo consequence, not an input; the keystone's external dependence is pinned to that one analytic fact. *(ix) Uniform rigidity* ([`verification/v442_seam_rigidity_uniform.py`]): the band-limited μ_4-character rigidity of v398 is uniform in N — exact 4-sector commutation to $N{=}256$, a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho, V]\|{=}2$ — so it is not a small-N artifact (the full-L^2 forcing stays **[O]**). *(x) The kill tests* ([`verification/v443_seam_e8_discriminator.py`]): mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest $c{=}8$ rival $SO(16)_1$ — torus GSD 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots — with $c{=}8$ alone unable to decide (v277) and only $\det K{=}1$ (the genus-1 statement, v90) selecting E_8; a seam with GSD>1 would falsify $(E_8)_1$. **[X]** the tests, **[O]** the realisation. *(xi) The chiral-edge scaling motor* ([`verification/v444_seam_edge_scaling.py`]): the counterpart of (vi) on the *edge*, attacking exactly the chiral massless scaling limit that the bulk motor left open. On the same $p{+}ip$ collar (v367/v368) the lattice edge already carries the conformal data of a chiral $c{=}\frac{1}{2}$ Majorana CFT — a linear, chiral in-gap dispersion (velocity $v{\approx}0.98$, $R^2{=}0.9999$, with *opposite* per-edge velocities), an *algebraic* edge two-point $\sim r^{-\eta}$ with $\eta{\approx}1$ (versus the *exponential* bulk row and the exponential trivial-phase control), and a Cauchy-convergent scaling-collapse under refinement ($\eta{=}1.001{\pm}0.007$, successive spread 0.028→0.013) — so the 16-copy edge is a chiral $c_{-}{=}8$ $(E_8)_1$ CFT; **[E]** the kinematics, correlator and collapse, **[C]** the central-charge reading, **[O]** the rigorous uniform-convergence theorem (cited MMST). With (vi) for the bulk and (xi) for the edge, *both halves* of the one continuum residual are numerically grounded and pinned to a single cited theorem. *(xii) The rigidity forcing converse* ([`verification/v445_seam_rigidity_forcing.py`]): the converse of (ix), upgrading rigidity from "block-diagonal *permits* commuting" to "commuting with the order-4 clock *forces* block-diagonal". For $\rho{=}\,\mathrm{diag}(i^n)$, $[\rho, K]{=}0$ *iff* $K_{ij}{=}0$ whenever $i{\not\equiv}j$ mod 4 (verified entrywise, swept $N{=}4..128$), so the commutant is *exactly* the 4-sector block algebra: $\dim\{K : [\rho, K]{=}0\}{=}\sum_s n_s^2$, a proper subspace of the full N^2 algebra ($N{=}16$: 64<256); every off-block unit has $\|[\rho, E_{ij}]\|{\geq}\sqrt{2}$ (zero slack), and the order-2 clock leaves a strictly larger commutant (64 vs 128) — the *four* Gauss–Bonnet marks (order $4{=}|\mu_4|{=}h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det{=}1$, v281/v154/v351). **[E]** the entrywise iff, exact commutant dimension, off-block sharpness and order discriminator; [C] the synthesis (RP-definability + clock-invariance *force* the block-diagonal Λ_Σ, v442's "permit" upgraded to "force"); [O] the single residual is now precisely that the physical transfer lies in the commutant (clock-invariance = `QGEO.SYM.01`, v323/v335). *(xiii) Clock-invariance derived* ([`verification/v446_seam_clock_invariance.py`]): closing (xii)'s residual one notch further — the operator condition $[\rho,\Lambda_\Sigma]{=}0$ is itself a *consequence* of a structural symmetry, not an extra assumption. On explicit finite OS data, the canonical transfer $\Lambda{=}C(1)C(0)^{-1}{=}e^{-h}$ (the transfer fixed by RP+reflection, v54) of a μ_4-invariant covariance satisfies $[\rho,\Lambda]{=}0$ (swept $N{=}4..32$; iff-sharp μ_4-breaking neg control), Λ is a positive contraction, and the free modular Hamiltonian $K{=}\log((1{-}C)C^{-1})$ (v258) of the invariant state has $[\rho,K]{=}0$ (Tomita–Takesaki). [E] the inheritance, contraction and modular commutation; [C] so with (xii), RP-definability *plus* the four marks force the block-diagonal Λ_Σ; [O] the residual reduces from an operator commutation to the structural "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`, v216/v323). *(xiv) The independent topological edge reading* ([`verification/v447_seam_edge_chern.py`]): a second, fit-free route to the edge central charge of (xi). On the same $p{+}ip$ collar (v367/v368) the bulk Chern number is the integer $C{=}{+}1$ (Fukui–Hatsugai–Suzuki link variables; $C{=}0$ in the trivial phase, $C{=}{-}1$ at the opposite mass, matching (xi)'s opposite per-edge velocities), the open strip carries exactly $|C|{=}1$ chiral edge branch per boundary, and the Li–Haldane entanglement spectrum has in-gap modes at $\frac{1}{2}$ only in the topological phase; [E] the Chern integer, channel count and entanglement signature, [C] the bulk–edge reading $c_-{=}N_{\rm Maj}\cdot\frac{1}{2}\cdot|C|{=}16\cdot\frac{1}{2}{=}8{=}g_{\rm car}{+}N_{\rm fam}$ — the *same* $c_-{=}8$ as (xi) from a logically independent (topological) observable. *(xv) The four-point pinning of the external fact* ([`verification/v448_seam_edge_fourpoint.py`]): extending (xi)'s two-point scaling limit to the next order to support exactly the one external fact (viii) isolates. The edge state is quasi-free (the ground-state correlation matrix is an exact projector $G^2{=}G\sim 10^{-13}$, the MMST CAR hypothesis), and the edge four-point cross-ratio $Q{=}[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement to the free-fermion conformal value $\sin(5\pi/12)/\sin(\pi/12){=}2{+}\sqrt{3}\approx 3.732$ (the bulk control off-scale, exponential); [E] the quasi-free property, four-point convergence and conformal form, [C] so the one external MMST fact is empirically supported at *both* two- and four-point order. *(xvi) The uniform-in-N pinning* ([`verification/v449_seam_mmst_uniform.py`]): strengthening (xv) from convergence at a single cross-ratio to the *uniformity* the cited theorem asserts — two *independent* edge four-point cross-ratios converge to *distinct* conformal values ($Q(1,2,4,7)/12\to 2{+}\sqrt{3}$ and $Q(1,3,5,9)/12\to 2$) at one N-independent $1/N$ rate ($|Q(N){-}Q_\infty|\cdot N\le 4$ for both, $N_x\in\{36,48,60\}$), the bulk control off-scale; [E] the two limits and the uniform rate, [C] so the MMST uniform-convergence *hypothesis* is empirically met. *(xvii) The entanglement reading of c_-* ([`verification/v450_seam_edge_entanglement.py`]): a *third*, correlator-independent route to the edge central charge — the Calabrese–Cardy block-entropy law gives $c_{\rm Dirac}{=}1.000$ for the critical free-fermion chain (vs a saturating, ℓ-independent gapped control), so c per Majorana $=\frac{1}{2}$ and $c_-{=}16\cdot\frac{1}{2}{=}8{=}g_{\rm car}{+}N_{\rm fam}$; [E] the law and the control, [C] the assembly. *(xviii) The Ising operator tower* ([`verification/v451_seam_edge_cardy_tower.py`]): naming the edge CFT by its *operator content* — the Cardy finite-size spectrum reproduces the exact Ising chiral primaries $h_\sigma{=}\frac{1}{16}$ (the periodic–antiperiodic ground split) and $h_\epsilon{=}\frac{1}{2}$ (the lowest NS pair gap), both L-independent, so $\{c,h_\sigma,h_\epsilon\}{=}\{\frac{1}{2},\frac{1}{16},\frac{1}{2}\}$ is the free-Majorana minimal model $M(3,4)$; [E] the dimensions and conformal scaling, [C] the identification. *(xix) The $(E_8)_1$ torus modular data* ([`verification/v452_seam_e8_modular.py`]): pinning the $(E_8)_1$ identity on the *torus*, complementing the edge reads (xi)/(xiv)/(xvii)/(xviii) — the character $\chi{=}E_4/\eta^8$ is S-invariant ($\chi(-1/\tau){=}\chi(\tau)$, one primary), T-multiplies by $e^{-2\pi i/3}{=}e^{-2\pi ic/24}$ ($c{=}8$) and is holomorphic ($c\equiv 0 \bmod 8$, T-phase order 3); [E] the full modular signature. *(xx) The μ_4-from-marks derivation* ([`verification/v453_seam_mu4_from_marks.py`]): folding the last structural premise into the existing one — the four Gauss–Bonnet marks *are* μ_4 (roots of $z^4{-}1$), the form basis ω_k is μ_4-graded ($\to i^k\omega_k$, weights $(1,2,3){=}A_3$ exponents) and the order-4 clock $z\to iz$ fixes their cross-ratio $\lambda{=}2$ (the D_4 stabiliser, v168), so the residual `QGEO.SYM.01` of (xiii) follows from marks$=\mu_4$ plus the existing `QGEO.REALIZE.01` — [E] the three mark facts, [C] no new premise: the chain (xx)$\to$(xiii)$\to$(xii) closes modulo realisation. The integer backbone of (xiv)–(xix) ($c_-{=}8$, the Chern integers, the order-3 T-phase) is kernel-hardened in Lean axiom-free as `FORM.SEAM.EDGE.CHERN.01`. *(xxi) The lattice Sugawara current algebra* ([`verification/v454_seam_edge_virasoro.py`]): testing the cited MMST "Koo–Saleur generators $\to$ continuum Virasoro with the correct c" fact on the edge *algebra*, not just its central charge. The critical free-fermion ring has the Affleck–Cardy Casimir energy $E_0(L){=}e_\infty L{-}\pi vc/(6L)$ fitting $c_{\rm Dirac}{=}1.000$ ($\frac{1}{2}$ per Majorana, $L{=}64..1024$), the lattice $U(1)$ current closes its affine *level* exactly ($\langle 0|J_nJ_{-n}|0\rangle{=}n$, $n{=}1..8$; vanishing for a gapped control), and the level-1 Sugawara central charge $c{=}\dim G/(1{+}h^\vee){=}8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31); [E] Casimir, Kac–Moody level and Sugawara, [C] $16\cdot\frac{1}{2}{=}8$ is the level-1 current-algebra c_-. *(xxii) The* *uniform-in-N Tomita–Takesaki tower* ([`verification/v455_seam_bw_uniform.py`]): the inductive-limit companion of v426 for sub-step (i) of (iv) — a family of reflection-odd, μ_4-even, gapped collars (N=8..128) on which the intrinsic Bisognano–Wichmann condition $u_\Theta=J$ ($\|\Theta K\Theta+K\|=0$) holds *exactly at every N*, with an N-independent gap ($6\log\frac{3}{2}$) and an N-independent neg-control margin ($\|\Theta K\Theta+K\|=2\|K\|$ for a side-even reflection); **[E]** the uniform BW, gap, clock symmetry and teeth, **[C][O]** so the inductive limit inherits $u_\Theta=J$ (the abstract continuum theorem stays the cited backbone). *(xxiii) The chirality-from-P_1 forcing* ([`verification/v456_seam_chirality_from_c3.py`]): the S3 input of the continuum chain (v356) is removed as a free choice — c_3's "8" *is* the one-sided count $8=|\mathbb{Z}_2|\cdot(\int K/\pi)=2\cdot 4$ ($8\pi=|\mathbb{Z}_2|\int K=2\cdot 2\pi\chi(S^2)$, v73), and a reflection sends the edge Chern integer $C\to-C$ (verified $C=+1\to-1$ on the explicit $p+ip$ model), so any *two-sided* (reflection-symmetric) gapped boundary forces $C=-C=0$ while the *one-sided* seam — the same $|\mathbb{Z}_2|$ quotient that gives c_3 its 8π — has no such reflection and forces $C\neq 0$; **[E]** the one-sided count and the $C\to-C$ flip, **[C]** chirality (S3) is a *consequence* of P_1, magnitude $c_-=8$. *(xxiv) The $(E_8)_1$ character kill test* ([`verification/v457_seam_character_killtest.py`]): a *falsifiable* sharpening of the (x)/(x") discriminators — the $(E_8)_1$ tower $\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\cdots)$, the weight-1 count 248=120+128 (vs the free 120) and the sector count $|\det\mathrm{Cartan}\,E_8|=1$ (vs $|\det\mathrm{Cartan}\,D_8|=4$); a measured 120, four sectors, or a wrong tower coefficient would *falsify* $(E_8)_1$, and the data give $248/1/\{1,248,4124\}$ (passed). **[X]** the kill criterion, **[E]** the exact discriminators. *(xxv) The exact MMST citation audit* ([`verification/v458_seam_mmst_citation_audit.py`]): a hypothesis-by-hypothesis reading of the cited MMST theorem (`arXiv:2107.13834`) against the collar — H1 CAR class, H2 massless limit, H3 the per-copy $c=\frac{1}{2}$ Thm 4.11, H4 decoupled additivity to c=8, and H5 the WZW range $8\leq 8\leq 16$ with its 120 bilinear currents all *match*, isolating the single residual H6 to the 128-spinor extension $SO(16)_1\to(E_8)_1$ (the $\det K$ 4→1 holomorphy step), *outside* MMST's bilinear method; **[E]** the per-copy theorem, **[C]** the additivity and range, **[O]** the one open spinor extension — since (xxxii) certified at net level by the peer-reviewed crossed product. *(xxvi) The complementary lattice-VOA route* ([`verification/v459_seam_lattice_voa_route.py`]): the second, independent construction that supplies exactly H6 — for the even unimodular E_8 the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) has weight-1 space 248=8+240 with the 240 roots splitting 112 integer + 128 *half-integer* (the spinor, 248=120+128), so the lattice route builds the 128-spinor currents MMST (xxv) leaves open; **[E]** the even unimodularity and root split, **[C]** AGT constructs the holomorphic extension — since (xxxii) an *independent second witness*, the primary net-level certification being the peer-reviewed crossed product — **[O]** the routes leave one shared realisation input (the collar's scaling limit = A_{E_8}), tied to P_1 by (xxiii). The G-block arithmetic (the one-sided $8=|\mathbb{Z}_2|\int K/\pi$, the reflection $C\to-C$, $c_-=8$, the range $8\leq 8\leq 16$, 248=8+240=120+128, 240=112+128, $\det K$ 1 vs 4) is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`), reducing the residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus the two cited existence theorems. *(xxvii) The explicit flat-band commuting-projector (LTO) realisation* ([`verification/v460_seam_s3_lto.py`]): NPW26 prove the intrinsic Bisognano–Wichmann lemma of (iv) for *commuting-projector* (local-topological-order) models, so sub-step (i) asks for the seam realised as such a $\mathbb{Z}_2$-reflection LTO. On the same v367 $p+ip$ collar the flattened Bloch $Q=h/|h|$ is a frustration-free flat-band parent — $Q^2=I$ and the occupied $P=(I-Q)/2$ is an exact projector EQUAL to the ground-state projector — whose real-space kernel is gap-protected quasi-local ($\xi\approx 1.14$ at M=1, $|P(20,0)|<10^{-6}$; a clean power law $|R|^{-2.1}$ at the gapless point, so locality needs the gap), and whose flattened modular Hamiltonian $K_{\mathrm{flat}}=\mathrm{sign}(K)$ inherits the $\mathbb{Z}_2$ reflection $\Theta K_{\mathrm{flat}}\Theta=-K_{\mathrm{flat}}$ and the μ_4 clock $[\rho,K_{\mathrm{flat}}]=0$ EXACTLY; **[E]** the flat-band/projector and reflection algebra, **[C]** the quasi-locality, **[O]** the chiral $c_-=8\neq 0$ obstructs *strictly* finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`), so the realisation is exactly the quasi-local LTO net NPW26 use — advancing sub-step (i) of (iv) from abstract existence to an explicit gap-protected net, strict locality the cited residual. *(xxviii) The strict-locality obstruction made explicit* ([`verification/v461_seam_strict_locality.py`]): upgrading (xxvii)'s "strict locality is the cited residual" from a citation to an EXHIBITED topological obstruction. On the same v367 collar the FHS Chern is $C=+1$ (chiral) / 0 (trivial) / -1 (opposite mass), and the hybrid Wannier-centre (Wilson-loop) phase $\theta(k_y)$ winds by $|C|=1$ (chiral) and 0 (trivial) over the BZ — exponentially-localised Wannier functions exist *iff* that winding is 0 (Brouder–Panati et al.; Thouless), so the chiral collar has none; both phases are gapped, yet only the trivial one (winding 0) admits a strict finite-range commuting projector, so it is the *chirality* $c_-=8\neq 0$, not the gap, that obstructs. **[E]** the Chern integer, the winding= $|C|$ identity and the gapped neg-control, **[C][O]** Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0 hence $C=0$, so for $C=1\neq 0$ none exists — sub-step (i)'s realisation is *provably* the quasi-local NPW26 LTO net, strict locality topologically forbidden rather than a missing premise. *(xxix) The character-level 128-spinor extension* ([`verification/v462_seam_spinor_continuum.py`]): the constructive companion of (xxv)/(xxvi), exhibiting the 128-spinor extension three ways — the exact Jacobi/E_8 identity $\theta_2^8{+}\theta_3^8{+}\theta_4^8{=}2E_4$ makes the holomorphic character the sum of the $SO(16)_1$ vacuum and spinor sectors, $\chi_{(E_8)_1}{=}E_4/\eta^8{=}\chi_o^{SO16}{+}\chi_s^{SO16}$ (weight-1 counts 120+128=248); the finite 16-Majorana Fock space already carries them ($C(16,2){=}120$ currents and $2^{16/2}{=}256{=}128{+}128$ Ramond states); and a critical free-fermion ring converges to the chiral CFT (Casimir $c_-{\to}8$, lowest scaled gap $x_1{\to}1$ with $|x_1{-}1|{\sim}1/N^2$). **[E]** the θ identity, the 120+128=248 split and the finite-L convergence, **[C][O]** the rigorous holomorphic extension is certified at net level by the crossed product (xxxii), with AGT/AMT (xxvi) as the independent second witness. *(xxx) The $c{=}8$ holomorphic-uniqueness pin* ([`verification/v463_seam_c8_holomorphic_uniqueness.py`]): the positive lever that makes the *identification* half classification-forced rather than assumed. At level 1 the central charge $c{=}\dim\mathfrak{g}/(1{+}h^\vee){=}8$ has *three* simple solutions — A_8 ($\dim 80$), $D_8{=}\mathfrak{so}(16)$ ($\dim 120$) and E_8 ($\dim 248$) — so $c{=}8$ is necessary but *not* sufficient, and the $\det K{=}4$ $SO(16)$ rival of (xxiii) is a genuine same-c competitor, not a strawman. But a holomorphic (single-sector) $c{=}8$ theory has an $SL(2,\mathbb{Z})$-covariant character, and the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}{=}E_4/\eta^8$, whose q^1 coefficient (the one-dimensional candidate space, vacuum $a_0{=}1$) is 248 — so $\dim V_1{=}248$ is *forced*, selecting E_8 uniquely and excluding A_8/D_8. **[E]** the three candidates, the forced 248, the tower $\{1,248,4124,34752,213126\}{=}E_4/\eta^8{=}j^{1/3}$; **[C]** Dong–Mason/ Schellekens give the holomorphic $c{=}8$ VOA unique $=V_{E_8}$ (the even unimodular rank-8 lattice unique $=E_8$, Mordell–Witt), so "$c{=}8$ holomorphic limit $=(E_8)_1$" is classification-forced. *(xxxi) The one-particle realisation rigidity* ([`verification/v464_seam_oneparticle_rigidity.py`]): attacking the *realisation* input R_1 itself. Since the seam is quasi-free (v113/v160/v161), by Araki's self-dual CAR the whole net is the second quantisation of its one-particle symbol P, so R_1 collapses to a finite-per-mode statement about P: the gapped ground state is the *unique* quasi-free state with $P{=}$ the spectral projection onto $K{<}0$ ($P^2{=}P$ to $<10^{-12}$, K fixed by gap + chirality + reflection v426); its real-space kernel is Cauchy on every fixed window ($\max|P_{2N}{-}P_N|{\sim}1/N{\to}0$, so P_∞ exists), and the limit carries $c{=}1$ per Dirac (entanglement $S(L){=}(c/3)\ln L$, 16 copies $\Rightarrow$ $c_-{=}8$, tying v450). **[E]** idempotency, Cauchy convergence and the c-fit; **[C]** Araki/Shale–Stinespring give the state$\leftrightarrow$symbol bijection and implementable Bogoliubov map, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, its one-particle limit exhibited, modulo the named CAR functor". *(xxxii) The net-level crossed-product route + the invariant-level realisation* ([`verification/v469_seam_crossedproduct_route.py`]): both halves of the residual re-founded on harder-to-reject ground. The 128-spinor extension needs no preprint: the $SO(16)_1$ spinor has $h_s{=}N/16{=}16/16{=}1{\in}\mathbb{Z}$, so its statistics phase is $+1$ — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1{\rtimes}\{1,s\}$ is a *local* index-2 extension by 1995–2001 *peer-reviewed* subfactor theory (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B){=}4/2^2{=}1$ $\Rightarrow$ holomorphic $\Rightarrow$ $(E_8)_1$ by the (xxx) pin — so the AGT/AMT route (xxvi) is *demoted to an independent second witness*. The index-4 μ_4 glue runs on the same integer, $h(J^k){=}\{1,1,1\}$ for $k{=}1,2,3$ (the v125 isotropy $q(k(1,1)){=}k^2$ at net level, $\mu{=}16/4^2{=}1$). And the realisation input R_1 is reduced from model fiat to *invariants* (R1$'$): quasi-free **[C]** (v155/v160) + gap **[E]** (v302) + class D + $c_-{=}8$ **[E]** ((xxiii), from P_1) — computed on the collar as FHS Chern $|C|{=}1$ (control 0), $\nu{=}16{\times}|C|{=}16$, $c_-{=}\nu/2{=}8$; and $\nu{\equiv}0$ (mod 16) is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (the sixteen-fold way; c_- a *phase* invariant, Kapustin–Spodyneiko; Kim et al.), so any realisation with the R1$'$ invariants is phase-equivalent to the v367 stack. **[E]** the locality integers, the KLM μ-arithmetic, the computed invariants and the composed chain; **[C]** the re-founded citation package; **[O]** the cited theorems stay cited. The full G-block arithmetic — now including the (xxviii) winding obstruction, the (xxix) finite-Fock spinor counts, the (xxx) holomorphy selector and the (xxxii) locality/μ/sixteen-fold joints — is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`), the two formerly separate cited theorems are *merged* into one combined external axiom `seam_realisation_theorem`, so the residual reduces to ONE realisation axiom plus ONE cited theorem (`#print axioms` drops from six to four); and the (xxxii) route is Lean-carried as the *parallel* derivation `seamResidualClosed′` (the invariant-level axiom `collar_invariants` plus the single combined package `crossedproduct_route_theorem`, its joints — `lr_locality_integer`, `glue_locality_integers`, `klm_holomorphic`, `sixteenfold_pin` — kernel facts with no axioms). Across (v)–(xxxii), `SEAM.EQUIV.01` stays **[O]**: with (xxx) pinning the identification, (xxxi) reducing the realisation to its one-particle data and (xxxii) re-founding the extension leg on peer-reviewed ground, the residual is now *entirely certification* — a named, hypothesis-audited package of published theorems (MMST, Longo–Rehren/Böckenhauer/Böckenhauer–Evans/KLM, Dong–Mason, Araki, Shale–Stinespring; AGT/AMT as second witness) with no open internal mechanism — yet still rests on cited continuum-existence theorems we do not re-prove.

The TOE/QFT closure contracts as three citable certificates ([`verification/v398_seam_state_rigidity.py`], [`verification/v399_gravity_complete.py`], [`verification/v401_metrology_closure.py`]). Three companion contracts package the closure status as named targets without fabricating any closure. *(i) Seam State Rigidity* (`SEAM.RIGIDITY.01`): the scattered state-invariance reductions (v177/v199/v308/v309/v335) as ONE theorem target — [E] the RP-definability hinge (OS canonical transfer from RP + reflection alone, v54, quasi-free v155), [E] the band-limited μ_4-character-block-diagonal core swept N=4..64 (beyond the 3-dim H^1, made uniform in N to N=256 by v442), [E] holomorphy $\Rightarrow$ $(E_8)_1$; and [E] the forcing converse (v445) that commuting with the order-4 clock is *exactly* μ_4-block-diagonality, so the residual is no longer "does block-diagonal permit" but only $[\rho, \Lambda_\Sigma]$=0 on the full L^2 — the physical transfer lying in that commutant (intrinsic Bisognano–Wichmann). [E] Moreover (v446) that operator commutation is itself *derived*: the OS canonical transfer of μ_4-symmetric RP data inherits the clock (with the modular Hamiltonian commuting, v258), so the residual reduces further to the *structural* statement that the seam RP data is μ_4-symmetric (the four marks, `QGEO.SYM.01`), leaving [O] only that structural symmetry on the raw continuum collar — for which the four marks now have an *established* continuum frame: the four-interval multilocal modular geometry (v480, keybox below), where the invariance is manifest at the state level in the exactly solvable class. *(ii) Gravity-complete = Boundary-complete + Ambient-redundancy* (`GRAVITY.COMPLETE.01`): the local equation parameter-free (v358/v359) plus the redundancy discriminators (v369), with $\mathcal{O}_{\rm phys} \subset \mathcal{A}_\Sigma$ checked over five gravitational readout classes. *(iii) Metrology closure* (`METROLOGY.CLOSURE.01`): the final input set $\{a, \pi, v_{\rm geo}\}$ — 0 dimensionless dials + 1 unit, the dimensional bookkeeping that every dimensionful observable is a number $\times v_{\rm geo}^d$ (No-Unit, v153), and $v_{\rm geo}$ theorem-forbidden (TOE$_{\rm strict}$=TOE$_{\rm dimensionless}$+$[v_{\rm geo}]$, v384).

One surface for seam, carrier and lattice: the Kummer/K3 ([`verification/v260_k3_kummer_unification.py`]). The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z\mapsto -z)$, its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface* $\mathrm{Km}(T^2\times T^2)$, a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion $|A[2]|$=2^4=16 of the abelian surface) resolve to 16 exceptional $\mathbb{P}^1$'s $= \dim S^+ = \Lambda^{\rm even}(\mathbb{C}^5) = 16$ (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on $H^2(\mathrm{K3}, \mathbb{Z})$ is the even unimodular lattice $U^3 \oplus E_8(-1)^2$ of signature $(3, 19)$, so the same E_8 the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the $E_8(-1)^2$ summand — both exact, not identified.) Thus the v1 "glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$" is the lattice shadow of a single smooth object that already carries the seam and the carrier; E_8 appears both *locally* as the du Val $\mathbb{C}^2/2I$ singularity (the Poincaré-sphere link, v232/v236; by Kronheimer that local model *is* the hyper-Kähler quotient of the marks quiver, v479) and *globally* as this H^2 summand. A unification (`ARCH.K3.01`), *not* a closure: the lattice arithmetic is [E], the carrier$\leftrightarrow$node identification [C].

Synthesis — the Modular Spectral Closure ([`verification/v261_modular_spectral_closure.py`]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\boxed{\mathsf{TFPT}_{\rm QFT} = \big(\mathcal{A}_\Sigma,\ \omega_\Sigma,\ \Delta_\Sigma,\ \rho,\ A_F,\ H_F,\ D_F,\ J,\ \gamma,\ S_{\rm rel}\big)}$$

(seam algebra, KMS state, modular generator, μ_4 clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-*4 *KMS seam state* ω_Σ *minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the* 96*-dim KO-*6 *finite triple* D_F *(v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259)* — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 ($\dim S^+ = \Lambda^{\rm even}(\mathbb{C}^5)$ = Kummer nodes, $H_F = 96$), and one gap $\Delta = 6\log(3/2)$ (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:

- [E]/[C] **QFT-complete (modulo one premise).** The admissible OS sector, the boundary $(E_8)_1$ net, the relative spectral action, D_F (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once `QGEO.SYM.01` (the seam realisation: *the raw quasi-free seam state is μ_4-invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.

- [C] **Ambient QG — a redundancy, separate by design.** The full covariant metric-sector / ambient quantum-gravity measure (`QG.AMB.01`) is gap-decoupled ($\Delta_{\text{eff}} = 1.648 > 0$) and is now a [C] *redundancy* (v369), *not* a blocker for the QFT layer: a certification object, not a missing piece of the boundary QFT.

So "solving the QFT" here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo `QGEO.SYM.01`; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([`verification/v265_qft4d_fork_freeze.py`])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* (E_8 is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the $\alpha_1{=}\alpha_2$ scale α_3^{-1} is off by $\sim$5.6; spread $\sim$8.8 at $2{\times}10^{16}$ GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W{=}3/8$ (v248), $\kappa{=}M_{PS}/M_s$ in the frozen $O(1)$ band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied ([`verification/v266_ps_threshold_proton.py`]).** [X] the explicit two-step thresholds give $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the *minimal* 16-Higgs content ($M_{GUT}{\sim}2.4{\times}10^{15}$) gives $\tau_p{\sim}4.4{\times}10^{33}$ yr $<$ Super-K ($2.4{\times}10^{34}$) and is *killed*; the E_8-allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT}{\sim}5.9{\times}10^{15} \Rightarrow \tau_p{\sim}1.6{\times}10^{35}$ yr, proton-safe *now* but sitting at the Hyper-K reach ($\sim1.4{\times}10^{35}$ yr) — so branch B is *content-selected* (must be $+(15, 1, 1)$) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal; τ_p carries an $O(3)$ hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

And the bedrock is now a falsifiable kill-test, not a proof-promise ([`verification/v215_seam_deck_killtest.py`]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega{\circ}\rho{=}\omega \Leftrightarrow [\rho, C]{=}0$ for the quasi-free covariance, with the principal symbol $|k|$ commuting exactly, so the whole residual sits in the bounded sub-principal M_f (equivalently: the raw Gaussian bulk form A is μ_4-deck-invariant). The four levers are **K1** exactly four marks ($b_1{=}\#\text{marks}{-}1{=}3{=}N_{\text{fam}} \Rightarrow \#\text{marks}{=}4{=}|\mu_4|$); **K2** the *square* configuration (cross-ratio $2 \Rightarrow j{=}1728$, $\text{Aut}{=}\mathbb{Z}/4$; the clock order $h(A_3){=}4$ selects $\mathbb{Z}/4$, not the $j{=}0$ $\mathbb{Z}/6$, and a generic mark set has only $\mathbb{Z}/2$); **K3** the mod-4 sub-principal support ($[\rho, M_f]{=}0 \Leftrightarrow f$ is $\mathbb{Z}_4$-invariant, automatic for a μ_4-mark sum); and **K4** the transfer gap $(2/3)^6{=}64/729$. These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. The *light* version of that deferred test is now *executed* ([`verification/v503_qgeo_emergence_light.py`], `QGEO.EMERGE.LIGHT.01`), with the result *rigidity, not dynamical selection*: the free field does *not* select the square — the global det$'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, Osgood–Phillips–Sarnak), $\tau = i$ is a saddle of the full moduli space, the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$), and criticality at the square is Palais symmetric criticality — while the square *is* pinned sharply by the clock: the raw free-field DtN is mod-4 block-diagonal *iff* $\tau = i$ (an isolated zero of the K3 indicator as a modulus function, 8×10^{-15} at $\tau = i$, strictly monotone away), and $\text{Aut}(E_\rho) = \mathbb{Z}_6$ carries no order-4 element. The clock input itself is now *reduced to one bit* ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`): the deck has exactly two Möbius square roots, both automatically of order 4, and a marks-preserving root exists *iff* the deck is the central involution of the mark-stabiliser D_4 — given that alignment bit, order, uniqueness, the free 4-cycle and $\tau = i$ are all theorems (harmonicity alone is insufficient on bare mark data, exact counterexample — on free-deck circles it becomes sufficient, v510); on the Fock side the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$; the R sector splits), so the fermionic $\mathbb{Z}_4$ explains the *order* and only the spatial alignment remains the carrier input. That bit has now survived its *tautology attack* ([`verification/v507_seam_bit_origin.py`], `SEAM.BIT.ORIGIN.01`): the marks are the four half-periods of the seam double and *every* mark-free collar deck is a half-period translation — the deck's *class* is derived from the common torus origin — yet the origin does not force the alignment (the marks-preserving root exists iff the deck pairs separate harmonically, $\lambda \in \{-1, 2, \frac{1}{2}\}$, and the silver counterexample sits in-family as "$\tau = i$ with a non-CM-fixed half-period"); the bit is an equivalence tetrad (clock $\iff$ $\tau{=}i$ with the CM-fixed half-period $\iff$ deck central in the mark-D_4 $\iff$ harmonic deck pairs $\iff$ nonsplit NS Fock lift) whose physical face is the Fidkowski–Kitaev-type extension class (central $U^2 = (-1)^F$ vs edge $U_e^2 = +1$, split) — distinguished, but not derived: only the *position* (which half-period) remains the carrier input. That position half is now *topology* ([`verification/v510_seam_bit_freedom.py`], `SEAM.BIT.FREEDOM.01`): the collar deck is a *covering* deck, hence free on the seam circle; the unique free involution is the antipode ($\mathrm{Aut}(C_{16}) = D_{16}$, complete census), nonsplit $\iff$ free over all 17 involutions in exact Cl(16), and the edge/silver class — whose mark circle passes through the deck poles, the restriction of the *branched* pillowcase involution — is excluded; harmonic + free solves exactly to $b = \pm i$, so the bit reduces to the square-modulus datum $\tau = i$ alone. That datum now carries an equivalent *discrete* form ([`verification/v512_seam_tau_flag.py`], `SEAM.TAU.FLAG.01`): on the free seam circle bare mark-transitivity is automatic for every deck-invariant configuration $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ (the pair-exchanging V_4; no local jet sees the side bit), and what selects $\delta = \pi/2 \iff \tau = i$ is exactly *flag* transitivity ($V_4 \to D_4$ symmetry lift), welded into a 12-fold exact equivalence web — the remaining input is *one discrete symmetry-lift bit*, its negation concretely measurable (odd-doublet split $(2/\pi) \ln \cot(\delta/2) \neq 0$), and the search space for a future **[E]** closure is narrowed to the RP positivity of the odd sector. The full `QGEO.REALIZE.01` stays **[O]**; no marker moves. **[E]** carrier-side (the four levers), **[O]** seam-side (the emergence): a kill-test, *not* a closure (`QGEO.KILL.01`).

And the mark count is now *derived*, not assumed: four marks from Gauss–Bonnet ([`verification/v216_marks_gauss_bonnet.py`], [`verification/v217_marks_emergence_scan.py`]). The K1 lever ("exactly four marks") is the one that can be *executed* today. If the marks are $\mathbb{Z}_2$ branch points (cone angle π, deficit π) on the flat seam sphere ($\chi{=}2$ — the same χ that supplies the 8 in c_3), Gauss–Bonnet *forces* their number: $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\mathrm{fam}}{+}1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and *three independent* criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), rank $H^1 = \#\text{marks} - 1 = 3 = N_{\mathrm{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal $(3,3,3)$), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling v217 confirms it on the actual DtN/state: scanning the number n *and* the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap $(2/3)^6$}. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays **[O]**. The light emergence test ([`verification/v503_qgeo_emergence_light.py`]) sharpens this residual to a *convergence*: given the clock, the modulus freezes automatically ($P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ with $j = 1728$ identically, v493), and the order-4 element on H^1 is the Coxeter element of $W(A_3)$ — exactly the P2 cusp-class carrier datum of v499: the QGEO modulus rest and the P2 typing rest R1 are *one* common residual ("why does the raw seam carry the order-4 clock"), not two. And the clock-rigidity theorems (v506) reduce that common residual to *one alignment bit* — "the collar deck is the central involution of the mark-D_4" — with the clock's *order* fermionically forced ($U^2 = (-1)^F$ exact); the bit-origin theorems (v507) then derive the deck's *class* (every mark-free collar deck is a half-period translation of the seam double) and characterise the bit as an equivalence tetrad with a physical extension-class face, so only its *position* — which half-period — stays the input; the freedom theorems (v510) finally type that position half as *topology* (the collar deck is a covering deck, hence free; the edge class is excluded), so the residual is the square-modulus datum $\tau = i$ alone — restated by the flag-transitivity web (v512) as one discrete symmetry-lift bit ($V_4 \to D_4$: flag transitivity $\iff$ $\tau = i$). **[E]** (count + equality), **[O]** (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

And the four marks are a *four-interval* modular geometry ([`verification/v480_multilocal_four_interval.py`]). The seam circle minus its four marks is

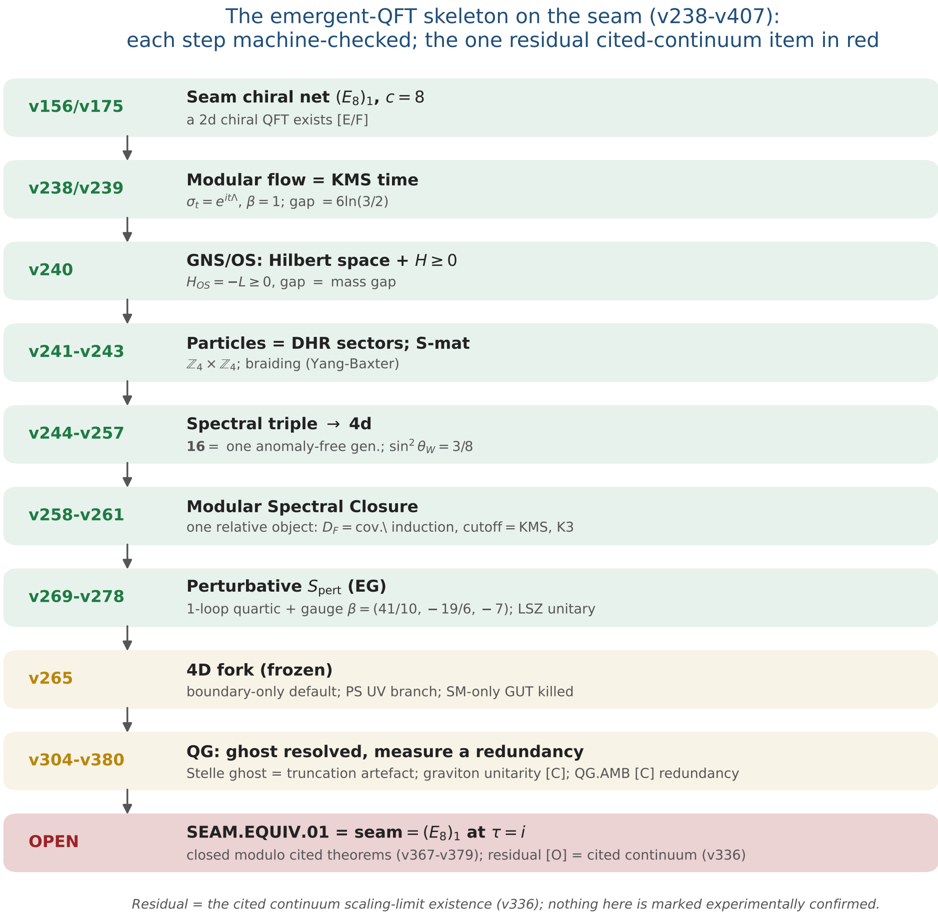

Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier's $SO(10)$ **16**. Each step is machine-checked; the one residual — `SEAM.EQUIV.01`, whose MMST route `SEAM.EQUIV.MMST.01` is closed modulo cited theorems (v367–v379) with the cited continuum scaling-limit existence (v336) as its [O] leg (parent [O]) — is in red, the frozen 4D fork and the QG certification layer in gold. The whole skeleton is *conditional* on that keystone and is neither solved nor experimentally confirmed.

the $n{=}4$ disjoint-interval region of a chiral free fermion — the one continuum class whose *multi-interval* modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; and Kawahigashi–Longo–Müger: *holomorphic* $\Leftrightarrow$ trivial multi-interval inclusion, $\mu{=}1$ — the same one bit as $\det K{=}1$). Until now the suite used only the single-ball CHM boost (v358); v480 exhibits the multilocal mechanism on the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with $\rho^4{=}{-1}$ *exactly* (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15} (the diagonal 4-fold-cover isomorphism), each sector a *single*-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [E] $\rho C \rho^{-1}{=}C$ at 10^{-14}, so $\omega{\circ}\rho{=}\omega$ holds *manifestly* at the 4-interval state level — and a one-site displacement of a single interval re-couples the sectors at 10^{-2} (the invariance is the μ_4-symmetric configuration, not an artifact); and the cross-interval modular coupling peaks on the conjugate-point diagonal, the lattice shadow of the Casini–Huerta mixing term (numerical). [C] the continuum multi-interval theory is cited, not re-derived; [O] the raw-seam premise (`QGEO.SYM.01`) is untouched — this puts the bedrock and the AP2 continuum leg (v476/v478) in one established, exactly solvable frame; it does not close them.

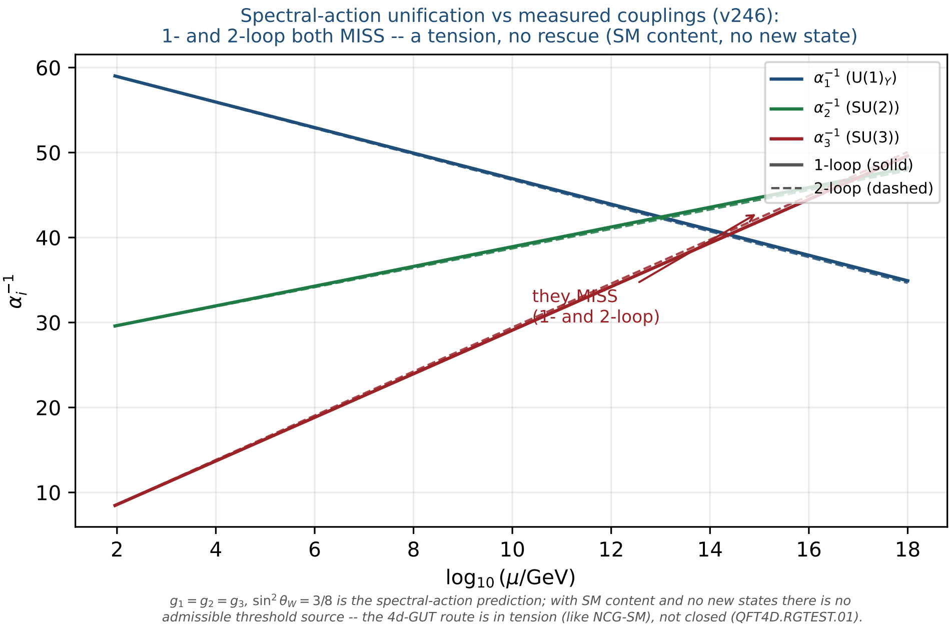

Figure 2: Honest data cross-check (v246): the spectral-action prediction $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; 2-loop minimal spread ≈ 3.2 at $1.7{\times}10^{14}$ GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

`QGEO.SYM.01` is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the μ_4-deck structure of the carrier clock — i.e. the carrier (P2) and the seam's conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam's degree-4 mark divisor on the genus-0 double has $h^0 = \deg{+1} = 5 = g_{\rm car}$ and $\operatorname{rank} H_1 = \deg{-1} = 3 = N_{\rm fam}$ (with $\Lambda^{\rm even}(\mathbb{C}^5) = 1{+}10{+}5 = 16 = \dim S^+$), so P2 is the genus-0 mode count of the four-mark seam divisor ([`verification/v228_rr_index_gate.py`]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than `QGEO.CONF.01`/`ISO.01`.** It does *not* name $\mathbb{P}^1$ or μ_4 (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* `QGEO.SYM.01`, equivariant uniformisation puts the order-4 clock in the form $z \mapsto iz$ on $\mathbb{P}^1$, the four parabolic marks are its free orbit μ_4, H^1 carries the $(1,2,3)$ grading, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$ — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus > 0 (uniformisation to $\mathbb{P}^1$ fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ($N_{\rm fam} \neq 3$ collapses $b_1 = N_{\rm fam}$ and the four-mark μ_4 structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

Honest framing. The correct external statement is *conditional*: "*Given* `QGEO.SYM.01`, the RP seam boundary realises the μ_4 deck structure, and the AQFT closure to $(E_8)_1$ follows through the verified chain v175–v181" — *not* "the theory *derives* the seam geometry". The premise stays **[O]**; everything downstream of it is **[E]**. Per v323 (Bisognano–Wichmann) `QGEO.SYM.01` is itself the *downstream* conformal-deck face of the single named keystone `SEAM.EQUIV.01` (the raw RP seam = the holomorphic $(E_8)_1$ net at $\tau{=}i$), which is now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the lattice/S_3

stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays [O]; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]).

This is not a gap to apologise for — it is TFPT's one fundamental postulate. Every fundamental theory rests on an irreducible premise (the constancy of c in relativity, the equivalence principle, the canonical commutator). `QGEO.SYM.01` is that premise for TFPT: a single geometric identification (seam conformal structure = carrier μ_4 deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

The Seam-Holomorphy Theorem — the one named closing target [E]/[O] ([`verification/v234_seam_holomorphy_selection.py`])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that "closing the theory" becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force E_8:

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* (μ-index 1, single vacuum sector). Holomorphic $+\, c = 8 \Rightarrow$ the unique even unimodular rank-8 lattice net $(E_8)_1$, and $c = 8 = g_{\text{car}} + N_{\text{fam}}$ is itself structural.
2. **Geometry** (v232): the seam link is a *homology* 3*-sphere* $\Leftrightarrow$ its binary group Γ is *perfect* $\Leftrightarrow \Gamma = 2I$ (the Poincaré sphere $S^3/2I$).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number: #(1-dim irreps) $= |\Gamma^{\text{ab}}| = |H_1(S^3/\Gamma)| =$ #(mark-1 affine-Dynkin nodes), which equals 1 *only* for E_8 ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$). So $2I$ is the unique nontrivial perfect finite $SU(2)$ subgroup and $S^3/2I$ the unique homology-sphere space form. **Consequence:** Research Contract 2 (G_{net}/Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), [O] at one step:

$$\text{RP} + \text{gap} + \text{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net}$$
$$\xrightarrow{[\text{E}]} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I,\ (2,3,5),\ \text{P2} \xrightarrow{\text{v179–v181+Kleinian genus-0}} \texttt{QGEO.SYM.01}.$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is "free Gaussian RP bulk $\Rightarrow$ holomorphic (single-sector) boundary net"** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That step is now *closed modulo cited theorems* (the MMST route `SEAM.EQUIV.MMST.01`): the lattice/S_3 stack pins it at every computable level (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`), only the cited continuum existence staying [O] (v336); v_{geo} stays the No-Unit metrology primitive and F_{transfer} stays external physics. This box certifies that the three faces coincide and force E_8 ([E]); the analytic step is closed modulo that cited theorem.

A physical realisation of the step ([`verification/v235_seam_chern_simons.py`]). The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons / K-matrix theory (even integer K), where #anyons $= |\det K|$, the edge chiral central charge is $c = \text{signature}(K)$, and the edge net is *holomorphic* $\Leftrightarrow |\det K| = 1$ — so "no nontrivial abelian sector" is precisely $\det K = 1$. The extension tower of v92 *is* the anyon-condensation tower read by determinant: $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8 with $c = \text{signature} = 8 = g_{\text{car}} + N_{\text{fam}}$, so the holomorphic end is the *Kitaev* E_8 *quantum-Hall state*. Holomorphy $\Leftrightarrow$ condensing the order-$|\mu_4| = 4$ *Lagrangian* glue (det $16{\to}16/4^2{=}1$); a partial order-2 isotropic only reaches det 4 (D_8). The residual is thereby a single integer condition — "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" — the same sheet/μ_4 selection as `GATE.QGEO`, but now with holomorphic $\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`, [E]/[O]).

The residual as a *physical* statement ([`verification/v237_seam_sre_closure.py`]). The same integer condition is the physics of topological order: a K-matrix phase has ground-state degeneracy $|\det K|^g$ on a genus-g surface, so $\det K = 1 \Leftrightarrow$ *no* topological ground-state degeneracy on any closed surface $\Leftrightarrow$ the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite K ($c = 8 = g_{\text{car}} + N_{\text{fam}}$) only E_8 is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev E_8 state, whose edge is the holomorphic $(E_8)_1$. TFPT already has a unique vacuum on the plane

(RP + gap + cluster + OS); the plane cannot see $\det K$ (no non-contractible cycles), the torus can. So the one analytic step is the sharp, falsifiable physical statement *"the free RP gapped seam bulk is short-range-entangled (no topological order)"* (= $\det K = 1$ = the Kitaev E_8 phase, `GATE.HOLO.03`, [E]/[C]) — and the lattice/S_3 stack now answers it at every computable level (torus $\det K{=}1$, v378), so the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is *closed modulo cited theorems*, only the cited continuum existence (v336) staying [O] (the parent stays [O]).

Lemma 2 (U1 — wall landscape, *corrected*)**.** *Parabolic degree zero forces the line weight-sums onto the wall* $(w_1, w_2, w_3) = (2, 1, 1)$*. An explicit balanced representative (weights in* $\frac{1}{3}$*) is*

$$W_{\text{wall}} = \tfrac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of $(0, 1, 2)$*, row sums* $(2, 1, 1)$*.* ***Honest result:*** *the* $6^4{=}1296$ *column-permutation matrices give* 144 *walls; under the* full *group* $D_4 \times S_3(\textit{lines}) \times S_3(\textit{values})$ *they form* five *orbits,* not one*. So* W_{wall} *is* not *singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;* W_{wall} *is one of the five. Machine: fully certified* $C_U^{(1)}$ *(*[`verification/v29_research_contract_certs.py`]*).*

The splitting type is now algebraic (RH route, [`verification/v32_rh_splitting.py`]***).*** *Passing from the monodromy to the Fuchsian* residues A_k *(eigenvalues = weights* $\{0, \frac{1}{3}, \frac{2}{3}\}$*,* $A_k = U^{k-1} A_0 U^{-(k-1)}$*,* $U = \operatorname{diag}(1, i, -i)$*) the twisted* $\mathbb{Z}_4$*-average collapses* exactly*:* $\sum_k U^k A_0 U^{-k} = 4\operatorname{diag}(A_0)$*, so the exponents at infinity are* $4\operatorname{diag}(A_0)$*. A flat bundle with regular* ∞ *needs integer exponents summing to* $4{=}-\deg E$*; with the cusp weights the only options are perms of* $(2, 1, 1)$ *and* $(2, 2, 0)$*, i.e.*

$$\boxed{\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \operatorname{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})}$$

(the sibling $(2, 2, 0) \to \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$*). Such a Hermitian* A_0 *(spectrum* $\{0, \frac{1}{3}, \frac{2}{3}\}$*, that diagonal) exists by Schur–Horn. So the* splitting *is an algebraic condition on* $\operatorname{diag}(A_0)$*.* Honest wall: *the global* $\prod M_k = \mathbf{1}$ *is the path-ordered monodromy (*not $\exp(2\pi i \sum A_k)$*), constraining* $\operatorname{off-diag}(A_0)$ *via the genuine RH solve; and R-extraction still needs the* $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ *bridge. The RH route fixes the* splitting *algebraically but not* c_u/c_d*.*

Lemma 3 (U2 — the kill switch, *run and hardened* [E])**.** *The selected polystable wall representative must* not *collapse to the diagonal abelian* $U(1)^3$ *point if it is to give non-trivial* $SU(3)_F$ *transport mixing:* ***(A)*** $\nabla_F^\star$ *polystable but* not *simultaneously diagonal (the desired breakthrough), or* ***(B)*** *diagonal collapse, in which case* c_u/c_d *stays honestly [C].* ***Result: the cheap test is run robustly and lands on A.*** *The kill-switch fires iff the* D_4*-symmetric cusp representation is* reducible *(commutant non-scalar). Across* 400 *random cusp-class generators of the* $\mathbb{Z}_4$ *family* $M_k = U^{k-1} M U^{-(k-1)}$ *the commutant is scalar at* every *point (irreducible* $400/400$*), the reducible locus is hit with probability* 0 *(measure-zero, codimension* ≥ 1*), and the mixing strength stays bounded away from zero (*$\min$ *off-diag* $= 0.35 > 0.05$*). So case A is the* generic *behaviour, not an accident of the one explicit point of* [`verification/v33_explicit_flat_bundle.py`]*:* (U_{wall}) survives *its own cheapest falsification test.* [`verification/v38_uwall_killswitch.py`]

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, *positive-dimensional*)**.** *With* $M_k = U^{k-1} A U^{-(k-1)}$ *(*$U = \operatorname{diag}(1, i, -i)$*, the* $\mathbb{Z}_4$ *rotation) and the reflection* $M_1 = V M_1^{-1} V^{-1}$*,* $M_3 = V M_3^{-1} V^{-1}$*,* $M_2 = V M_4^{-1} V^{-1}$ *(with* $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$*,* $V^2{=}1$*,* $VUV^{-1}{=}U^{-1}$*), the relations become polynomials in the entries of* $A \in C_{\text{cusp}}$*;* $\mathcal{X}_{\text{cusp}}^{D_4}$ *is cut out as* $\bigcup_k C_k$*, parametrised by trace coordinates* $x = \operatorname{tr}(M_1 M_2)$ *(note* $\operatorname{tr} A = 1{+}\omega{+}\omega^2 = 0$*).* ***Result*** *(*$C_U^{(2)}$*,* [`verification/v30_d4_character_variety.py`]*)***:** *adding the reflection to the* $\mathbb{Z}_4$ *locus of* v19 *does* not *isolate the point — the* full *D_4-fixed product locus is still* positive-dimensional *(*$|\operatorname{tr}(M_1 M_2)|$ *varies continuously over* $\approx [0, 3]$*). So even the full* D_4 *symmetry cannot select* $\nabla_F^\star$*.* Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations)**.** *The readouts* $\rho \mapsto \det R(\rho)$*,* $\rho \mapsto \mathrm{Spec}(Q_+(\rho))$*,* $\rho \mapsto \Lambda_{f,j}(\rho)$ *are well-defined (cyclotomic/algebraic) on* $\mathcal{X}^{D_4}_{\mathrm{cusp}}$*.* ***Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}^{D_4}_{\mathrm{cusp}} : \ \det R = 8,\ \mathrm{Spec}(Q_+) = \{1,2,3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not *closed.* ***The bottleneck (made precise by U3):*** *since the* D_4*-fixed variety is positive-dimensional, the* selector *must do all the cutting — and evaluating* $\det R(\rho)$*,* $\mathrm{Spec}(Q_+(\rho))$ *as functions on* $\mathcal{X}^{D_4}_{\mathrm{cusp}}$ *requires the* parabolic-degree ↔ residue dictionary $R(\rho)$*, i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the* (U_{wall}) *gate.* Status update: since v118–v122 the dictionary itself is exact — *hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the* $W(A_3)$ *monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller* $R4'$*, the* establishment of the selectors *(*$n = (5,-9,6)$ *+ the diamond determinants; since [`verification/v134_dual_anchor.py`]: the pair* (d,n)*).*

What $R(\rho)$ ***is (*** [`verification/v31_R_dictionary.py`]***).*** *Two facts pin its nature.* (i) Case A is alive: *every* D_4*-fixed cusp monodromy found is* irreducible*, so the wall point carries non-trivial* $SU(3)_F$ *mixing — the U2 kill switch does* not *fire to the diagonal case B.* (ii) $R(\rho)$ is not algebraic: *R is integer-valued, but the algebraic invariants of* ρ *(e.g.* $\mathrm{tr}(M_1M_2)$*) vary* continuously *on the locus; an integer readout cannot be a continuous algebraic function. Hence* $R(\rho)$ *is the discrete invariant of the* non-abelian Hodge / Mehta–Seshadri *map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle* E_ρ*): integer output, but its* computation *is the transcendental harmonic-metric (Hitchin) solve* on the generic Higgs locus*.* ***Refinement (U4″,*** [`verification/v40_harmonic_metric.py`]***):*** *at the* polystable *point TFPT actually selects this pessimism does not apply — see the box below.*

Progress: the *selector side* of U4 is closed on the explicit point ([**E**]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 *selectors* are *algebraically* accessible on the explicit flat bundle of [`verification/v33_explicit_flat_bundle.py`] — no PDE. Computed exactly from A_0: the exponents at infinity $\mathrm{eig}(\sum_k A_k) = \{2,1,1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1,1,2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\boxed{\begin{array}{c} \det R = 8 = n\cdot a \quad \text{(anchor pairing on the splitting type)}, \\ \mathrm{Spec}(Q_+) = \{1,2,3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{array}}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\mathrm{Spec}(Q_+)$ is the affine image (d=3 the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data.* **Residual (sharpened below by Readout Rigidity):** the quark *ratio* c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [`verification/v39_uwall_selectors.py`]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([**E**]). The U4 "transcendental Hitchin solve" is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [`verification/v33_explicit_flat_bundle.py`] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim\ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So** $C_U^{(3)}$ **reduces from a transcendental PDE to: this finite unitarisation**

(done) + the combinatorial word-length R (closed: H1 + $\det R = 8$). *Residual:* the feared PDE is gone; the quark *ratio* c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [`verification/v40_harmonic_metric.py`]

The final leg-assignment test (honest result) ([**E**]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1, so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\mathrm{diag}\,\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [`verification/v41_leg_assignment.py`]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([**E**]). The v41 "genericity is future work" item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for *any* M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$-family flatness constraint *is* the torsion statement "$T = MU$ is a fourth root of unity": the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1} v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\boxed{\mathrm{diag}\,M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \qquad \mathrm{spec}\,M = \{1, \omega, \omega^2\}\ \text{automatic}}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + det 1). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a *μ_4-torsion identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\mathrm{diag}\,M = (r, z, \bar{z})$ with r real and $\mathrm{Re}\,z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4-fixed flat cusp locus realises only $\mathrm{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [`verification/v114_torsion_delta.py`]

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle ([**E**]). The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i) μ_4-average lemma [E]:* $\sum_k U^k X U^{-k} = 4\,\mathrm{diag}\,X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4\,\mathrm{diag}\,A_0$ and the anchor splitting holds *iff* $\mathrm{diag}\,A_0 = (2, 1, 1)/4 = a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33's diagonal was forced, not chosen). *(ii) The $(8, 0, 5)/144$ lemma [E]:* the anchor diagonal and the cusp spectrum $\{0, 1, 2\}/N_{\mathrm{fam}}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$ the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2,\ |a_{13}|^2,\ |a_{23}|^2) = \frac{(8,\ 0,\ 5)}{144}} \quad \begin{array}{l} 8{=}\mathrm{rank}\,E_8,\ 5{=}g_{\mathrm{car}}, \\ 144{=}(|\mu_4| N_{\mathrm{fam}})^2,\ 8{+}5{=}13{=}\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$. *(iii) Exact normal form [E]:* $A_0^\star = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$ has $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ *exactly*. *(iv) $A_0^\star$ is the RH solution [E]:* its big-circle monodromy is trivial to 10^{-10}, $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\operatorname{diag}\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, $\operatorname{tr}(\widetilde{M}_0 U) = 1$ — the v33/v40 point is gauge-equivalent to $A_0^\star$ (the a_{12}, a_{23} phases are diagonal gauge). *(v) The anchor forces the $d_1 = 0$ slice [E]:* a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\operatorname{diag}\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue [C]:* $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining U_{point} step. [`verification/v115_anchor_residue.py`]

The resonance theorem: $M_\infty = \mathbf{1} \iff a_{13} = 0$ — one gauge orbit [E] ([`verification/v116_resonance_uniqueness.py`])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw = -(1/w)(B_0 + B_1 w + \cdots)Y$ with $B_m = \sum_k i^{km} U^k A_0 U^{-k}$, the *twisted* μ_4 averages collapse [E]:

$$B_0 = 4 \operatorname{diag} A_0, \qquad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents $\operatorname{diag}(2,1,1)$ have resonance gap 1; the level-1 formal-gauge equation $(1 - \operatorname{ad} R_0)H_1 = R_1$ is singular exactly on the cells $\{(2,1),(3,1)\}$, where B_1 has entries $(0,\ 4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i \operatorname{diag}(2,1,1)} = \mathbf{1}$. Hence [E]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation. Uniqueness corollary [E]:* $a_{13} = 0$ + the $(8,0,5)/144$ moduli + phases = diagonal gauge ⇒ the flat anchor locus is **exactly one gauge orbit**, the exact $A_0^\star$: within the μ_4-equivariant anchor class the (U_{wall}) bundle datum is *algebraically unique. Sharpness [E]:* $\|M_\infty - \mathbf{1}\| \sim 8.5\,|a_{13}|$ (0.17 at $a_{13} = 0.02$) vs 10^{-10} at $A_0^\star$. *Honest scope [C]:* the equivariant ansatz is the established D_4 selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal $(0, \frac{1}{2}, \frac{1}{2})$) remain [E] — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 — the holonomy is exact [E] ([`verification/v117_monodromy_weyl_a3.py`])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact $A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\boxed{\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}}$$

unitary, $\det = 1$, $\operatorname{tr} = 0$, $\chi = \lambda^3 - 1$ (the cusp class *exactly*), $\widetilde{M}_0^3 = \mathbf{1}$, and $\operatorname{diag}\widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$ — so $d_1 = 0$ and $\delta = \frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0 U)^4 = \mathbf{1}$ with $\operatorname{tr}(\widetilde{M}_0 U) = 1$ (the $\{1, i, -i\}$ branch, $d_1 = 0$ slice — exactly where v114/v115 located the physical point). *The group* [E]: exact enumeration of $\langle U, \widetilde{M}_0 \rangle$ gives order 24 with order statistics $(1,9,8,6)$ and character values $(3,-1,0,1)$ — uniquely S_4 in its standard ⊗ sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice** A_3:

$$\boxed{\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \qquad |W(A_3)| = 4! = 24}$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]:* the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10}, and the invariant form H is U-invariant to 10^{-11}. *Status:* holonomy values [E]; only the analytic identification stays [E]. The δ thread (v20 → v40/v41 → v114 → v115 → v117) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E] ([`verification/v118_hexagon_family_dictionary.py`])

The missing "$C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary" (H2) acquires its first exact entry. *Sign-twist lemma [E]:* $-\widetilde{M}_0$ has order 6 and

$$\boxed{\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6}$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle \widetilde{M}_0 \rangle$); the v20 denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]:* $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1-t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\mathrm{fam}}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4|\det_-/N_{\mathrm{fam}} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the matrix itself. *Sheet-extended group [E] (audit):* $\langle U, \widetilde{M}_0, -\mathbf{1}\rangle$ has order $48 = \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+$ — the seed's quartic coefficient. *Residue [C]:* the dictionary's *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20's established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C] ([`verification/v120_address_table.py`])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today's readings (the red-team firewall). *(i) The lepton words are the atoms [E]:*

$$\boxed{(L_e, L_\mu, L_\tau) = (8,5,3) = (\mathrm{rank}\,E_8,\ g_{\mathrm{car}},\ N_{\mathrm{fam}}) = (p_0{+}e_2,\ e_2,\ p_0)}$$

in mass order (longest word = lightest lepton). *(ii) Addresses = division by the hexagon [E]:* $(r,w) = L$ divmod p_2 with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \to (2,1)$, $\mu \to (5,0)$, $\tau \to (3,0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. *(iii) Sheet parity [E]:* by the v118 dictionary, e sits on the *untwisted* sheet ($\omega \in \mathrm{spec}\,\widetilde{M}_0$), μ, τ on the *twisted* sheet $(-\omega, -1)$; the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where v20's product rule replaces the resolvent. *(iv) Quark sum rules [E] (frozen v18 words $7,7,5,3,2,0$):* up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1{+}p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, v117); **all nine charged fermions** $= 40 = p_1 p_3 = a^T Ra$ (v119); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0,0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]:* five independent closures $(16, 10, 14, 24, 40)$ constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E] ([`verification/v121_address_pinning.py`])

The per-fermion assignment closes at the *information* level. The established identity $L = R{+}2U$ (v95) says the word-length table *is* L[sector][generation] — rows (up; down; lepton) $= ((7,3,0); (7,5,2); (8,5,3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R's first column is the **anchor** $a = (1,1,2)$. The margins are atoms [E]: row sums $(4,8,10) = (e_1, \mathrm{rank}\,E_8, p_3)$, column sums $(4,13,5) = (e_1, \Delta_Q, g_{\mathrm{car}})$. And the census closes it [E]:

$$\boxed{\begin{array}{c}\text{among all } 3\times 3 \text{ matrices with entries in the hexagon } \{0..5\},\\ \text{exactly 17 have the atom margins — and exactly ONE has } \det = 8\end{array}}$$

— and it is R (also the unique one with SNF $= (1,1,8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]:* the open H2 item shrinks from "derive 9 per-fermion integers" to "derive the six atom margins + det = rank". *Firewall:* R, L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems ([`verification/v122_margin_theorem.py`]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — *(S1)* the

D_4 annihilator $n = (5, -9, 6)$ with $n^T R = (8, 0, 0)$ (it kills the unchanged (c_2, c_3) plane; $n \cdot a = 8$), *(S2)* $\det R = 8$, *(S3)* $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices [E]: the $(S1)$ census has $4 \times 4 \times 4 = 64$ candidates, $(S2)$ leaves 12, $(S3)$ leaves **exactly one**, and it is R. Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]:* on the $(S1) + (S2)$ locus $\det(M + 2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]:* H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class.*

The dual-normal selector: (d, n) pins R columnwise ([verification/v136_dual_normal_selector.py]). The selector list shrinks once more. With the dual anchor $d = a^\top R^{-1}$ ([verification/v134_dual_anchor.py]) and the torsion normal n, each *column* of R obeys $d \cdot c_j = a_j$ and $n \cdot c_j = 8\delta_{1j}$; the common kernel of (d, n) is the primitive vector $(6, 8, 7)$ (audit: p_2, rank E_8, scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d, n) \Rightarrow R$, and the 6^9 census above collapses to three column problems [E]. *Honesty:* K is *not* pinned directly ($d^\top K = (1, \frac{1}{2}, 1)$); $\det K = 4$ comes via the μ_4 wall ([verification/v135_det_surface.py]). *The mechanism is bounded:* the zero mode of $A_0^\star$ carries the exact spectral normal $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\operatorname{adj} A_0^\star = \frac{2}{9} P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$, $\sigma \cdot a = N_{\text{fam}}$, $n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \to n$ is a genuine mechanism beyond the monodromy group. *R4′ restated [C]:* derive the *pair* (d, n); then R is forced columnwise.

The selector triangle ([verification/v139_selector_triangle.py]). The pair itself now collapses further. First, d is *pure anchor data*: $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ exactly — the closed form never mentions R, so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\operatorname{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$\boxed{n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \qquad n \cdot a = 8 = \operatorname{rank} E_8, \qquad n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2}$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1}, a)$-constrained line the σ-pairing is $11(11+t)$: divisible by 11 for *every* t, a perfect square exactly at the true n (t=0). The review's predicted lift is exact as an audit identity: $n = \operatorname{reverse}(\sigma) + |\mu_4| e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings → two ([verification/v142_frame_integrality.py]). The third pairing is *not* independent. The pairing triples $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$\boxed{x - 3y + z \equiv 0 \pmod{11}} \qquad (\text{index } 11 = \|\operatorname{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x, y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8) = (2, 8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11+t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and t=0 is the unique even t with a square σ-pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$, v134) — restructuring, not establishment.

The pairing values are atom identities ([verification/v145_pairing_atoms.py]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally — $n = w_0(\sigma) + |\mu_4| e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h - m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n \cdot \mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \qquad n \cdot a = |\mu_4|\, a_3 = 8 = \operatorname{rank} E_8 \ \text{ via } \boxed{|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9,\ \sigma \perp w_0(a))},$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4| g_{\text{car}} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \|\operatorname{Pl}(K)\|_1$. *R4′ residue [C] (final form):* derive the *one* map $n = w_0(\sigma) + |\mu_4| e_3$ — no independent number remains; everything else in the chain anchor → (d, n) → R is exact.

Both dual normals are anchor data ([verification/v149_cusp_normal.py]). The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$\boxed{n \cdot e_1 = 6 = p_2(a), \qquad n \cdot e_2 = 3 = p_0(a), \qquad n \cdot e_3 = 5 = e_2(a)}$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data $(6,3,5)$, the frame pairings $(2,8,121)$ and the w_0-lift are three presentations of one covector, all atomic (equivalences exact). *Audit:* $\sigma \to (5,1,2)$; alternate reading $(6,3,5) = (|R^+(A_3)|, N_{\rm fam}, g_{\rm car})$; sum $14 = \dim G_2$. *R4′ residue [C] (cusp form):* derive the *assignment* — why the Q_+ degrees $(1,2,3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r,w) = (1,1)$ ($L = 7$ for both), and the C_6 resolvent $c(r,w) = |\mu_4|^w/(\frac{5}{4} - \cos\frac{r\pi}{3})$ reads *only* (r,w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\mathrm{Pl}_{\mathbf{1},a}(K)$. The "11" is then *generated*, not read:

$$(K, \mathbf{1}, a) \to B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix}\right) \to \mathrm{Pl}(K) = (-1,6,4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \to \|\,\mathrm{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\rm car}\,\|\,\mathrm{Pl}_{\mathbf{1},a}(K)\|_1}{N_{\rm fam}^2\,\Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}} \quad \left(\text{anchor microcode } = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}\right).$$

Typing: [E] for the Plücker / microcode identities ($\mathrm{Pl}(K) = (-1,6,4)$, $\|\cdot\|_1 = 11$, $5{\cdot}11/(9{\cdot}13) = 55/117$); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio $\mathcal{C}_{ud}(\rho^\star)$ = this $\Lambda^2 F$ readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from "find the right scalar y" to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale ($U_{\rm point}$) stays as an anchor, no new scalar, no fishing.

Audit (the 121 *lemma, [`verification/v119_review_validation_2.py`]):* the "11" has a second, independent appearance [E]: with $h = \mathbf{1} + a = (2,2,3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\,\mathrm{Pl}(K)\|_1^2 = (p_3 + 1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [`verification/v42_exterior_leg.py`]

Bridge test ([`verification/v43_exterior_bridge.py`]). The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \overline{\widetilde{M}}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\mathrm{Pl}(K) = (-1,6,4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^\star \to \mathrm{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R; $\det R = 8$ and $\mathrm{Spec}(Q_+) = \{1,2,3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([`verification/v44_carrier_exterior.py`]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\rm even}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The "Λ^2" of the exterior leg is therefore not exotic Hodge data — it is the carrier's own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from "discrete non-abelian-Hodge invariant" to "the carrier exterior degree" — a standard [E] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family $\mathrm{Pl}(K)$ [E], and the carrier-Λ^2 $\leftrightarrow$ family-$\Lambda^2 F$ link (via family $\otimes$ carrier $= 15 = 16-1$) plus the normalisation remain [C].

The "11" is Pascal — a third, simplest origin ([`verification/v45_family_exterior.py`]). Following that link, $\mathrm{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3 = SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row $4 = (1,4,6,4,1)$ with full sum $2^4 = 16 = \dim S^+$, and

$|\,\mathrm{Pl}(K)| = (1,4,6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\boxed{\|\,\mathrm{Pl}(K)\|_1 = \textstyle\sum_{k=0}^{2} \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\mathrm{car}}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree $2 = \deg(u^c)$), and family $\otimes$ carrier $= 15 = \dim \mathfrak{su}(4) = \Lambda^2(5)\oplus\Lambda^4(5) = 10{+}5 = N_{\mathrm{fam}} g_{\mathrm{car}}$ read three ways. So the "11" is generated a *third* way — the cumulative family exterior algebra up to the up-quark's degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the "11" worry, *discharged* by Readout Rigidity)**.** *The original worry was that $c_u/c_d = \mathcal{C}_{ud}(\rho^\star)$ must be evaluated at the uniquely selected $\rho^\star$* without *using* 11 *as input, so that until then* 55/117 *would be a mere table-level rational shadow.* ***This is now resolved (v49):*** *$\mathcal{C}_{ud}$ is* constant *on the discrete selector stratum $\mathcal{S}_{8,Q}$, so the "11" is* earned *as the rigid Plücker readout $\|\,\mathrm{Pl}(K)\|_1$ (v42) —* independently *of point-uniqueness. Point-uniqueness of $\rho^\star$ enters only the* absolute *amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates)**.** *$C_U^{(1)}$ wall enumeration (done); $C_U^{(2)}$ D_4-fixed character solver; $C_U^{(3)}$ selector uniqueness (one S-equivalence class); $C_U^{(4)}$ amplitude extraction — the* ratios *$R, Q, c_u/c_d$ are read off the (derived) selector stratum (closed, v49/v71); only the* absolute *$U_f^\star, c_q$ normalisation is the anchor.*

Progress: an explicit valid flat bundle is realised (RH solve) ([E]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, diag $= (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)}/(z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \mathrm{diag}(2,1,1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1 M_2 M_3 M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope:* this is one realising point (existence + case A), likely one of a residual family; the unique $\nabla_F^\star$ still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [`verification/v33_explicit_flat_bundle.py`]

H2 bridge probed ([`verification/v34_h2_bridge_attempt.py`]). The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\mathrm{diag}\, M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes $(0.475, 1.107, 0.917)$. So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family $\rho^\star$) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the $W(A_3)$ monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4-fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\mathrm{SNF}(R) = (1,1,8)$, $\mathrm{Spec}(Q_+) = \{1,2,3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [E]: parabolic degree 0 + polystable $\Rightarrow$ unitary (Mehta–Seshadri) $\Rightarrow$ $\Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([`verification/v40_harmonic_metric.py`]).
- U_{H2} [E]: R, Q, K are read off the bundle (splitting type $= a$, $\det R = 8$, $\mathrm{Spec}(Q_+)$) ([`verification/v39_uwall_selectors.py`]).
- U_{Λ^2} [E]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \mathrm{SNF}(R) = (1,1,8), \mathrm{Spec}(Q_+) = \{1,2,3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\mathrm{car}} \|\,\mathrm{Pl}_{\mathbf{1},a}(K)\|_1 / (N_{\mathrm{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([`verification/v49_readout_rigidity.py`],

[verification/v42_exterior_leg.py]).

- U_{point} [O]: full uniqueness of $\rho^\star$ — needed only for the *complete* $U_f^\star$ amplitude matrix, *not* for c_u/c_d.

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; the only residual, U_{point}, is not an independent gate — it reduces to the single dimensionful anchor v_{geo} (Gate 1 below).

The simple closure ([verification/v71_simple_r_bridge.py]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \operatorname{rank} E_8$ with $\mathrm{SNF}(R) = (1,1,8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\text{fam}}$, v70), and $\mathrm{Spec}(Q_+) = \{1,2,3\}$ is the D_4-equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3,4,8,20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\text{car}}\cdot 11/(N_{\text{fam}}^2\cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts — no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point}, is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

Gate 1 complete: $U_{\text{point}} \to v_{\text{geo}}$ ([verification/v75_upoint_to_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c-ratio is closed (Plücker), and each *sector product* is a clean φ_0^{ret}-power (Grand Mass Volume, v46: $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$; the lepton product is $\frac{16}{7}\cdot\frac{4}{3}\cdot\frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}}/N_{\text{fam}}^2$). Since (pairwise ratios) + (product) $\Leftrightarrow$ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale* v_{geo}. **So U_{point} is not an independent gate: it reduces to v_{geo}, the *same* one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — (G_{metric})

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi\to\infty} \int_{\mathcal{G}_\chi/\mathrm{Diff}} \exp\!\big(-S_{\text{rel},\chi}[g,A,\psi] + J\cdot O\big)\, d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \operatorname{Tr} f\Big(\frac{D_{\text{rel}}+A_\Sigma}{\chi}\Big) - \operatorname{Tr} f\Big(\frac{D_{\text{ref}}}{\chi}\Big) + \frac{i\pi}{2}\Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm)**.** *For $s > 5/2$ a Sobolev space of Θ-symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\text{gh}} = K^*K$ and $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}}\Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (*not *rhetorically).* Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* [E])**.** *$S_{\text{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion* $\operatorname{Tr} f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \cdots$*. For the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ (so $E = -\frac{R}{4}$, spinor trace $\operatorname{tr}\mathbf{1} = 4$) the Gilkey coefficients give, per $(4\pi)^{-2}$,*

$$a_2 = \tfrac{1}{6}\operatorname{tr}(6E+R\mathbf{1}) = -\tfrac{R}{3} \quad (\textit{Einstein–Hilbert } \textstyle\int R), \qquad a_4\big|_{R^2} = \tfrac{1}{360}\operatorname{tr}(180E^2+60RE+5R^2\mathbf{1}) = \tfrac{R^2}{72}$$

(the R^2/Starobinsky term), so the spectral action structurally *produces $S_{\text{eff}} = \int\sqrt{g}\,\frac{\bar M_{\text{Pl}}^2}{2}(R+\frac{R^2}{6M^2})+\cdots$ with the scalaron mass ratio fixed by the cutoff moment, $M^2/\bar M_{\text{Pl}}^2 = 6(4\pi)^2/f_0$. The TFPT closure $M^2/\bar M_{\text{Pl}}^2 = c_3^7$ fixes $f_0 = 6(4\pi)^2/c_3^7$, i.e. $M = c_3^{7/2}\bar M_{\text{Pl}} = 3.06\times 10^{13}$ GeV — the boundary normalisation c_3* is *the gravitational scale — and hence the Starobinsky slow-roll forms $n_s \simeq 1 - \frac{2}{N}$, $r \simeq \frac{12}{N^2}$. The $R+R^2$* structure *is convention-independent (standard Gilkey); the precise rational $\frac{1}{72}$ and factor 6 are the standard Dirac conventions used here ([verification/v36_spectral_action_g2.py], [verification/v28_gravity_fR.py]).*

The simpler resolution: "full QG" for physics is the gap-decoupled admissible sector ([**E**]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\text{eff}} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

$$\boxed{\begin{array}{c}\text{admissible IR sector} = \text{closed under stated RP;}\\ \text{ambient metric measure (G6)} = G_{\text{metric}}\text{ gate}\end{array}}$$

G2 supplies the local action ($R+R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce (G_{metric}) to the ambient projective limit (G6). **Precise status (Alessandro's boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure is now discharged as a [C] *redundancy* (v369, the holographic-route keybox below), a certification object rather than a missing piece. Its absence does *not* affect the bounded IR claim. [`verification/v36_spectral_action_g2.py`]

The glue Q-system: the index-4 object of the G_{net} statement exists explicitly ([**E**]/[C]). The G_{net} target — *"the seam–Calderón inclusion has Jones index* $4 = |\mu_4|$*"* — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$, unit δ_0) *all* Longo Q-system axioms hold exactly [**E**]: associativity, unit law, the Frobenius identity $(m\otimes 1)(1\otimes m^*) = m^*m = (1\otimes m)(m^*\otimes 1)$, and specialness $mm^* = |\mathbb{Z}_4|\,\mathbf{1}$ — a special C* Frobenius algebra with

$$\boxed{\dim\theta = 4 = |\mu_4| = [B:A]}$$

exactly the Carrier Index Lemma value (KLM: $\mu_{\text{carrier}} = 16 = 4^2\mu_{E_8}$ re-verified). *Locality = isotropy [E]:* on the Lagrangian glue $\langle(1,1)\rangle$ the discriminant form gives $q(k(1,1)) = k^2 \in \mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step [E]:* $\{0,2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2\times 2$. *R2 sharpened [C] (recorded, not claimed):* what remains is the operator-algebraic *identification* "the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system" — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

The hull is the graded module of the Q-system ([`verification/v128_graded_hull.py`]**).** The Q-system is not merely an abstract companion of E_8 — E_8 *realises* it [**E**]: the 240 roots, built as explicit norm-2 vectors in $D_5 \oplus A_3$+glue coordinates, decompose as $240 = 52 + 64 + 60 + 64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\text{coset}(r_1+r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$:

$$\boxed{E_8 \text{ is a } \mathbb{Z}_4\text{-graded Lie algebra over the carrier; the grading group is the glue} = \text{the Q-system}}$$

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries*. [`verification/v125_glue_qsystem.py`]

Graded Frobenius ([`verification/v143_graded_frobenius.py`]**).** The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 [**E**]: (i) *coset duality* — $\text{coset}(-r) = -\text{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k \times g_{-k} \to \mathbb{C}$; (ii) the glue character average $\frac{1}{4}\sum_k i^{k\,\deg}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5)\times W(A_3)$ orbit with multiplicity-one dimensions $64 = 16\cdot 4 = (\mathbf{16},\bar{\mathbf{4}})$, $60 = 10\cdot 6 = (\mathbf{10},\mathbf{6})$ — simple currents, sector ring $= \mathbb{Z}_4 =$ the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0,2\} = SO(16)_1$. *R2 residue [C] (final form):* exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census (`[verification/v148_fock_census.py]`**).** The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* $(D_5, SO(6))$ weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0,0),(2,0),(0,2),(2,2)\}$: **the odd glue classes $k(1,1)$ — the actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128+128$ into the odd sectors of the *two* Lagrangian glues $\langle(1,1)\rangle$ and $\langle(1,3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* ($128 =$ one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$. *Consequence [C]:* the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated)**.** *For each Θ-invariant g in the Calderón ball the physical boundary kernel $\mathcal{K}_g = P_{\mathrm{adm}}\Pi_{\mathrm{phys}}\mathcal{N}_g^{TT}\Pi_{\mathrm{phys}}P_{\mathrm{adm}}$ is reflection positive, $\langle\Theta F, F\rangle_{\mathcal{K}_g} \ge 0$ (G3); and $d\mu_{\mathrm{Cal},\chi}$ is Θ-invariant, supported where G3 holds uniformly, so $\int\langle\Theta F, F\rangle_{\mathcal{K}_g}\, d\mu_{\mathrm{Cal},\chi} \ge 0$ (G4).* ***G4 is the decisive step:*** *RP after integrating over dynamical metrics, not merely at fixed background.*

Lemma 11 (G5 — gap dominance [E])**.** $2\|V_{\mathrm{metric}}\|_{\mathrm{rel}} < \Delta = 6\log\frac{3}{2}$*, with* $\|V_{\mathrm{metric}}\|_{\mathrm{rel}} \le 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$*, hence* $2\|V\| = 0.7852 < 2.4328$ *and* $\Delta_{\mathrm{eff}} \ge \Delta - 2\|V\| > 1.647 > 0$*. The numerical inequality is trivial (* `[verification/v29_research_contract_certs.py]`*); the work is the operator-norm bound (hard).*

Sugawara reading + atomic OS moment (`[verification/v171_os_moment_cluster.py]`**).** The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)|\log(N_{\mathrm{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$ the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length $\xi = 1/(6\log\frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full G_{metric})**.** If *the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,* $2\|V_{\mathrm{metric}}\| < \Delta = 6\log\frac{3}{2}$ *(G5),* then *the admissible IR measure is stable under the ambient completion: a strictly positive effective gap* $\Delta_{\mathrm{eff}} = \Delta - 2\|V\| > 0$ *protects the admissible sector from the (un-built) ambient/deep-UV, so every* physical low-energy *read-out (masses, mixings, α^{-1}, the $R{+}R^2$ regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\mathrm{RP} \;+\; 2\|V\| < \Delta \;\Longrightarrow\; \textit{admissible IR measure stable under ambient completion}\,.}$$

Thus (G_{metric}) *is* not *a diffuse "full quantum gravity missing" block: it is the sharply delimited* ambient projective limit *(G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on.* ***Typing:*** *the inequality is machine-checked [E]; the operator-norm bound on $\|V_{\mathrm{metric}}\|$ is the conditional input.*

Lemma 12 (G6–G7 — projective limit & Ward identities)**.** *A projective system μ_{χ_n} with $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$, uniform moment bounds $\sup_n\int\|g\|^p_{H^s}\,d\mu_{\chi_n} < \infty$, tightness and regulator-independence yields $\mu_{\mathrm{QG}} = \lim\mu_\chi$ on $\mathcal{G}/\mathrm{Diff}$ (G6, the hardest step); the limit obeys the BRST Ward identities $\langle s\mathcal{O}\rangle = 0$, whence $\nabla^\mu T_{\mu\nu} = 0$ and the η boundary phase cancels the APS anomalies (G7).*

Gate 2 reduction: G6 is a *boundary* projective limit, not a bulk one ([`verification/v76_gmetric_reduction.py`])

The exact decoupling margin is $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6\log\frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\text{metric}}\|_{\text{rel}} \leq 248\, c_3^2 = \dim(E_8)\, c_3^2 = \frac{31}{8\pi^2}$ $(31 = 2^{g_{\text{car}}} - 1)$. **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so $G6$ is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\text{RP(seam-Calderón boundary kernel)} + \text{tightness} \implies G_{\text{metric}} \text{ closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — now discharged as a [C] redundancy (v369, keybox below). So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) a [C] redundancy, no longer a diffuse "full QG missing". [C]

The measure as one-loop fluctuations around the *parameter-free* saddle ([`verification/v365_qg_oneloop_saddle.py`], `QGAMB.SADDLE.01`)

With the saddle now parameter-free (v358/v359), the boundary measure of Tier B organises as a Gaussian one-loop fluctuation determinant whose ingredients are atom-fixed. **Fixed quadratic form [E]:** the fluctuation inverse-propagator stiffness is the saddle coefficient $1/c_3 = 8\pi$ — no free dimensionless dial, so the one-loop problem is *well-posed* (it was not before v358). **Gap-controlled convergence [E]:** the fluctuation operator obeys $M \geq \Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337), so the Gaussian integral converges mode-by-mode with no zero/negative modes besides the saddle directions; on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$ the regularised one-loop log-det $\text{Tr}'\log M = 6\log\frac{9}{2}$ is finite. **Conformal direction [C]:** the one wrong-sign mode ($c_{\text{conf}}(4) = -\frac{3}{2}$) is made convergent by the GHP contour ($\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$) with IDG dressing (v334). **Tightness [E]:** $\chi = 1/(1-(2/3)^6) = 729/665 < \infty$ (v330), so the one-loop measure has a projective limit on the admissible sector, with *zero* new dimensionless dials (v364). **Residual:** the full *non-perturbative* projective limit (G6 on the ambient sector) is now discharged as a [C] redundancy (v369, keybox below) — this sharpens C7 from "diffuse full QG" to a one-loop-controlled Gaussian problem around a parameter-free saddle and then to a certification object. [E]/[C]

The holographic route: the ambient measure is boundary-*redundant*, not missing dynamics ([`verification/v369_qgamb_redundancy.py`], `QGAMB.REDUNDANCY.01`)

There is a second, strategically stronger way to dispatch Gate 2 than building the measure: prove that the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the measure `QG.AMB.01` is a non-fundamental *certification* object — the role a coordinate choice plays in general relativity — rather than a missing piece of dynamics. The discriminators that make this *ambient-redundancy* statement well-posed are all in hand. **Trivial superselection [E]:** the number of DHR sectors of a chiral net is $|\det \text{Cartan}|$, which is 1 for E_8 (one primary, the vacuum; $\text{DHR}((E_8)_1) = \text{Vec}$) against 4 for the same-c rival $SO(16)_1$ — a holomorphic boundary net has *no* nontrivial superselection charge that a bulk could carry invisibly. **No torus degeneracy [E]:** the ground-state degeneracy on T^2 equals the number of anyons $= |\det K| = 1$ for E_8, so there is no topologically protected bulk d.o.f. hidden from the boundary (the genus-1 obstruction of v344, in its redundancy reading). **Bounded reconstruction [E]:** the seam transfer deviation subspace contracts at the finite gapped Petz rate $(2/3)^6$ (v221), and the gap-decoupling margin $\Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) separates the un-built ambient sector from the admissible one. **The statement [C]:** with the Bisognano–Wichmann modular reconstruction map (v258/v309/v323), under `SEAM.EQUIV.01` every gauge-invariant bulk observable is boundary-reconstructible, $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\text{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$, and two ambient measures sharing the same boundary state are physically equivalent. **Residual [O]:** the construction consumes exactly two premises — `SEAM.EQUIV.01` (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct `QG.AMB.01` but supplies the *alternative* route, so that Gravity-complete = Boundary-complete + Ambient-redundancy. [E]/[C]/[O]

The classical field equation, parameter-free: entanglement equilibrium gives the linearised Einstein equation with G from c_3 ([`verification/v358_grav_entropy_equilibrium.py`])

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient [E].** Equilibrium gives $G_{ab} = c_3^{-1}T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed*: all four c_3-roles are mutually consistent — the Unruh factor $1/(2\pi) = 4c_3$, the EH action $1/(16\pi) = c_3/2$, the Einstein coefficient $8\pi = 1/c_3$, the Bekenstein density $1/4 = 1/|\mu_4|$ — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence [E] (new).** The first law requires the coefficient $2\pi/\eta$ (η the entropy density); the geometry gives $1/c_3 = |\mathbb{Z}_2|\, 2\pi\, \chi(S^2)$ (one-sided Gauss–Bonnet, v58); these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\, \chi(S^2) = 4$ — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of c_3 are *one*. **The matter side, assembled [C] (closing J3).** The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323), $K_B = 2\pi\int_B w\, T_{00}$, $w = (R^2 - r^2)/(2R)$; the 3-ball weight integral is $\int_B w\, d^3x = 4\pi R^4/15$, so $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\,\delta\langle T_{00}\rangle$ — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed [E].** The Bekenstein numerator $1/4 = 1/|\mu_4|$ (v57) and the entanglement normalisation = the central charge $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$ pin the dimensionless coefficient; the area-law form is established (v59). **Residual:** the full covariant extension is now *closed* (v359, next keybox); what remains is only (i) the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) **[O]** and (ii) the absolute scale v_{geo} (the Planck-area unit) **[O]**, while the global ambient measure (Gate 2/C7) is now a [C] redundancy (v369) — *no free dimensionless parameter survives in the gravity coupling.*

And that boundary measure is a *rigorously constructed* net ([`verification/v77_e8_conformal_net.py`]). The level-1 central charges $c = \dim\mathfrak{g}/(1 + h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \operatorname{rank} E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5)+c(A_3) = 5+3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ *E_8-lattice* VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the $(E_8)_1$ lattice net* (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\mathrm{eff}} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The full covariant field equation, parameter-free: fixed-volume equilibrium gives the Einstein tensor with conservation as an output ([`verification/v359_grav_nonlinear_einstein.py`])

The linearised result above (v358) extends to the *full* covariant equation by Jacobson's *fixed-volume* (maximal-vacuum-entanglement) equilibrium — no linearisation assumed. **Einstein tensor, not Ricci [E].** Stationarity of the entanglement entropy at fixed volume forces the divergence-free combination: the 4D trace identity $g^{ab}G_{ab} = (1 - \frac{d}{2})R = -R$ (for $d = 4$) selects $G_{ab} = R_{ab} - \frac{1}{2}R\, g_{ab}$, so

$$\boxed{G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}, \qquad c_3^{-1} = 8\pi}$$

with *both* coefficients TFPT-fixed: the Newton coupling $8\pi = 1/c_3$ (v358) and the cosmological constant Λ from α, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60). **Conservation is an output [E].** By Lovelock's theorem the only divergence-free, second-order, local symmetric tensor built from the metric is $a\, G_{ab} + b\, g_{ab}$, so $\nabla^a G_{ab} = 0$ forces $\nabla^a T_{ab} = 0$ — matter conservation is *derived*, not assumed. **So Gate B6 — the classical metric-sector field equation — is closed parameter-free at the *local* level. Residual:** the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) and the single unit v_{geo} stay **[O]**, while the global ambient measure (Gate 2/C7, the constructive boundary projective limit) is now a [C] redundancy (v369) — no free dimensionless gravity dial remains. An *external candidate* for the missing action level is quantified in the next keybox (v473); it does not change this typing. [E]/[C]

The entropic-action bridge: an external candidate for the missing action level, quantified — nothing closes ([`verification/v473_entropic_action_bridge.py`])

The one honest **[O]** left in the classical sector — "equation of state, not a from-action quantisation" (v359) — now has an explicit *external* candidate: Bianconi's entropic action $S_\mathrm{B} = -\operatorname{Tr}_F \ln\big(\widetilde{G}\widetilde{g}^{-1}\big)$, the quantum relative entropy between the spacetime metric and the matter-induced metric (G. Bianconi, *Gravity from entropy*, Phys. Rev. D **111**, 066001 (2025); `arXiv:2408.14391`; her Appendix C identifies $\Delta^{1/2}_{\widetilde{G},\widetilde{g}} = \widetilde{G}\widetilde{g}^{-1}$ with an Araki-type modular operator) — a local covariant action whose variation yields the Einstein equations plus an emergent nonnegative Λ at low coupling: exactly the missing variational layer, *if* the bridge identities hold. What is machine-checked today (`GRAV.ENTROPIC.ACTION.01`): **carrier Hodge count [E].** Her $\Omega^{0,1,2}$ matter fiber, placed on the five-slot carrier $\mathbb{C}^5$ (*not* on 4d spacetime, where the fiber is 1+4+6=11), counts $1{+}5{+}10{=}16{=}2^{g_\mathrm{car}-1}{=}\dim S^+_{D_5}$ and Hodge-folds $(\star : \Lambda^1{\cong}\Lambda^4)$ onto $\Lambda^\mathrm{even}\mathbb{C}^5$ — the carrier half-spinor's own Pascal row (v2/v44); a vector-space identity *only*. **Coefficient match [E] (the one quantitative pin).** Her weak-coupling Lagrangian $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_\mathrm{m} + \cdots$ (the 3 = the three form blocks) matched to the TFPT EH normalisation $1/(16\pi G) = c_3/(2G)$ fixes her free constant *exactly*,

$$\boxed{\beta'_\mathrm{B} = \tfrac{c_3}{6} = \tfrac{1}{48\pi}}$$

(the $6 = 3 \times 2 = p_2 = |R^+(A_3)|$ is recorded as a structural echo, not a proof). **Entropy potential = quadratic Λ [E].** Her $\Lambda_\mathcal{G} = \frac{1}{2\beta}\operatorname{Tr}_F(\mathcal{G} - I - \ln\mathcal{G})$ is nonnegative and quadratic at the fixed point; with the TFPT displacement $\varepsilon_\star = e^{-\alpha^{-1}}Q$ it reproduces the closed Λ branch $\rho_\Lambda/\bar{M}^4_\mathrm{Pl} = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), and the prefactor match yields the *exact* normalisation target $\operatorname{Tr}_F Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\,\delta_\mathrm{top}$ — a closed algebraic target for the two-form (pair) block ($\binom{5}{2}{=}10$), **[C]** until an explicit Q is constructed. **R^2 kill test [E] (pre-registered).** The raw log expansion carries a curvature-squared coefficient of order $(c_3/6)^2$; the TFPT Starobinsky sector needs $\bar{M}^2_\mathrm{Pl}/(12M^2_\mathrm{scal}) = 1/(12c_3^7)$ (v36) — the exact mismatch is $3c_3^{-9} = 3(8\pi)^9 \approx 1.2\times10^{13}$: a direct identification of the raw entropic action with the TFPT R^2 sector is *false* without a mechanism (the scalaron as a light trace mode of her $\mathcal{G}$-field generating c_3^{-9}, or a KMS-spectral renormalisation of the log action). **The bridge proper [C] (recorded, not proven).** The compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma = \Delta^{1/2}_\Sigma$ — the physical Calderón compression of her relative metric operator equals the TFPT seam modular operator; if proven, S_B is a local volume representation of the TFPT relative spectral action $S_{\mathrm{rel},\chi}$ (her log action is the all-scale-integrated heat kernel, by Frullani $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$, verified). Open with it: the operator-level Hodge items (Clifford action, Dirac intertwining, leaf involution, Calderón polarisation), and Lorentzian positivity of $\widetilde{G}\widetilde{g}^{-1}$ is *not* automatic (timelike gradients) — TFPT's OS/reflection-positive route is the sturdier construction. **Residual [O] (unchanged).** Nothing closes: the v359 equation-of-state typing stays **[O]**; adopting S_B would close it *only* after the compression conjecture *and* an R^2 mechanism; zero TFPT knobs are added.

The operator level, executed ([`verification/v474_entropic_hodge_carrier.py`]). Work packages 1 and 4-algebra of the bridge are now machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ ($\dim 2^{g_\mathrm{car}}{=}32$). **Spinor structure exhibited [E]:** the ten CAR gammas satisfy $\{\Gamma_a,\Gamma_b\}{=}2\delta_{ab}$ exactly, the 45 bilinears span $\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — S^+ *is* the even exterior algebra with its spinor structure, and Bianconi's Hodge–Dirac operator is Clifford multiplication at symbol level ($(c(\xi)^2{=}-|\xi|^2$, symbolic). **The fold is the $5{\to}\bar{5}$ conjugation [E]:** the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber 1+5+10 becomes the GUT half-spinor $16{=}1{+}\bar{5}{+}10$ — the SM carrier decomposition, not a dimension match; honest obstruction: $\Omega^{\le 2}$ is *not* Clifford-invariant, so the identification must go through the fold, never through degree truncation. **The Q-target, decided [E]:** the uniform-block ansatz $\operatorname{Tr}Q^2{=}d\,k^2c_3^4{=}32c_3^4$ with integer k admits *exactly* the supports $(d,k){=}(2,4),(8,2),(32,1) = \{|\mathbb{Z}_2|, \operatorname{rank}E_8, 2^{g_\mathrm{car}}\}$, with the minimal uniform solution $q{=}c_3^2$ per Fock mode and the identity $\operatorname{Tr}Q^2/\delta_\mathrm{top}{=}32/48{=}2/3$; the two-form block 10 (the naive pair-sector reading) and the fiber 16 require irrational multipliers and are *excluded*. The physical choice among the three supports stays **[C]**; AP2 and the R^2 mechanism stay **[O]**.

The R^2 kill test, executed ([`verification/v475_entropic_scalaron_gate.py`]). On a maximally symmetric background the relative curvature operator has eigenvalues *exactly* $\{R;\ R/4\times4;\ R/6\times6\}$ (her Appendix-B flattening verified, incl. the 6×6 two-form block), so the exact vacuum action is $\mathcal{L}_\mathrm{B}(R) = -[\ln(1{-}\beta R) + 4\ln(1{-}\beta R/4) + 6\ln(1{-}\beta R/6)] = 3\beta R + \frac{17}{24}\beta^2R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) and $\frac{17}{24}$ is the exact tensorial factor behind the pre-registered gap. With the pinned $\beta'{=}c_3/6$

both mass routes (Starobinsky matching and $f'/3f''$) give the raw entropic scalaron

$$\boxed{m_{\rm raw}^2 = \tfrac{4608\pi^2}{17}\,\bar M_{\rm Pl}^2 \;\Rightarrow\; m_{\rm raw} \approx 51.7\,\bar M_{\rm Pl}}$$

— *trans-Planckian*, not light: the naive reading "the TFPT scalaron is the light trace mode of her $\mathcal{G}$-field" is *dead* [E]; a viable mechanism must supply exactly $m_{\rm raw}^2/m_{\rm TFPT}^2 = \frac{72}{17}(8\pi)^9 \approx 1.69\times10^{13}$ in mass^2. The domain is not the obstruction (first log pole at $R{=}384\pi^2\bar M_{\rm Pl}^2$, trans-Planckian), and the Lorentzian caveat is now an explicit witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2 \le 0$ for $v^2 \ge 1/\alpha$ (ln undefined), while spacelike/Euclidean stay positive — the OS route is the sturdier construction. **Net verdict [C]:** of the three bridge interpretations, the entropic action covers the Einstein + Λ level (survives); the light-mode reading is dead; KMS-spectral renormalisation is the *only* surviving route to the R^2 sector. Nothing closes.

AP2, made well-posed ([`verification/v476_entropic_compression_toy.py`]). The compression conjecture itself is sharpened on an exactly solvable gapped chain. **The reading is decided:** on a *pure* bulk (the seam situation: pure bulk, thermal boundary) the covariance is a projection, so $f(x){=}x/(1{-}x)$ is singular on its spectrum — the literal operator-side reading $P_\Sigma f(C) P_\Sigma$ is *ill-posed*, while the state-side $f(P_\Sigma C P_\Sigma)$ (Peschel) is finite: the identity can only mean *build* Δ_Σ *from the compressed relative metric*, which is exactly Bianconi's own local construction — AP2 retyped, not weakened. **The mismatch is exactly second order [E]:** symbolically, $[C^n]_{AA} - (C_{AA})^n$ carries the cross-cut block twice (first order vanishes identically), so for any analytic f the two readings agree up to $O(\|C_{AB}\|^2)$ — the modular (entanglement) content *is* that quadratic correction. **Gap control:** at the bounded (covariance) level the mismatch obeys $d \le x^2$ and falls with the gap, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\rm eff}{>}0$, v76/v337). **Residual [O] (unchanged):** the continuum seam statement itself. No marker moves.

The surviving R^2 route, typed as one moment ([`verification/v477_entropic_scaleflow.py`]). The Frullani identity upgrades to a *scale-flow representation*: $-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}{-}e^{-t}) = 2\int_0^\infty \frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ — Bianconi's log action *is* the flat scale-integral of relative heat-kernel actions, while TFPT's $S_{{\rm rel},\chi}$ evaluates *one* KMS scale with a weight f: the bridge between the two action principles is a choice of *scale measure*. In the moment dictionary (weight $w(t)\,dt/t$, μ_k its e^{-t}-moments) the coefficients read $3\beta R\,\mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2$; the flat weight has $\mu_1{=}\mu_2{=}1$ (= the raw v475 coefficients), and demanding $m^2{=}c_3^7 \bar M_{\rm Pl}^2$ forces *exactly*

$$\boxed{\mu_2/\mu_1^2 = \tfrac{72}{17}(8\pi)^9} \qquad \text{with} \qquad \frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7 \text{ (identically)}$$

— the v475 kill-test number *is* a moment ratio, and the "13 orders of magnitude" are not a bridge mismatch but a scale-measure datum: the flat measure fixes it wrongly, the TFPT KMS measure ($f_0{=}6(4\pi)^2/c_3^7$, v36) fixes it correctly — one KMS moment, two parametrisations, zero new dials. [C] (consistency, not derivation: the measure is TFPT input); the R^2 gate and the v359 typing stay [O].

The two remaining legs, first computable steps ([`verification/v478_entropic_continuum_legs.py`]). *Leg A (AP2 continuum):* on the exactly solvable critical chain (infinite-volume kernel, 50-digit precision) the compressed interval state carries the conformal/BW structure quantitatively — the entanglement entropy follows Calabrese–Cardy with $c_{\rm est}$ = 1.00089/1.00038/1.00021 (monotone → the Dirac $c{=}1$; gapped control $\sim 6{\times}10^{-8}$), and its Peschel modular Hamiltonian organises on the CHM weight $\beta(x){=}x(L{-}x)/L$ (corr 0.87→0.99, even-range bands vanishing identically, odd tail 1.8%): the state-side Δ_Σ (the reading *forced* above) flows to the CHM/Bisognano–Wichmann geometric form — *exactly* the object the entanglement-equilibrium derivation consumes (v323/v358); the bridge's modular side and the gravity side meet in the continuum limit (a lattice exhibit, not a continuum proof — AP2 stays [O]; the natural continuum frame for the seam version is the *four-interval* multilocal modular geometry of the four marks, v480). *Leg B (the measure):* the single-scale corollary is exact — $d\mu{=}\delta(t{-}t_0)dt$ gives $\mu_2/\mu_1^2{=}e^{t_0}$, so the one-moment condition forces $t_0{=}\ln\frac{72}{17}{+}9\ln(8\pi){=}30.461$, one dimensionless KMS time in closed form; the near-miss to $h(E_8){=}30$ (difference 0.461) is explicitly *declined* per the no-free-pattern rule (v354/v355) — deriving the measure from the entropic structure stays [O]. [E]/[C]/[O]

The Simple-Current Extension Theorem — the central $G_{\rm net}$ closing statement [E]/[C] ([verification/v154_simple_current_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension.* Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle(1,1)\rangle \subset \mathrm{Disc}(D_5) \oplus \mathrm{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1,1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$\boxed{[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5{+}3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1}$$

so B is *holomorphic* (KLM μ-additivity), and a holomorphic c=8 chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $\boxed{B \cong (E_8)_1}$ (the same-c rival $SO(16)_1$, μ=4, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([verification/v125_glue_qsystem.py]), the $\mathbb{Z}_4$-graded Frobenius E_8 hull ([verification/v143_graded_frobenius.py]), the Ramond glue sectors ([verification/v148_fock_census.py]). **Consequence:** $G_{\rm net}$ is internally *closed* if the seam–Calderón boundary net is *defined* to be A; it stays the *one* external identification "the RP seam–Calderón net is A" if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not "find a quantum gravity".

And that one identification reduces to a single shared premise ([verification/v155_quasifree_boundary.py]). The identification splits (v110) into (a) "the seam Calderón involution is sheet-odd" — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2\cdot 4\pi}$ (structural) — and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1,1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of Γ_d). Hence (b) *follows* from "the seam bulk is a reflection-positive free field" — the *same* premise already load-bearing in the area law ([verification/v59_area_law_evidence.py]) and the EH mechanism ([verification/v150_replica_eh_model.py]/[verification/v151_bfk_split.py]; since exercised on the discretized collar with the seam's own kernel and real replica sheets, v471 — the R3 kernel premise is discharged at the finite level, its residual retyping to the continuum scaling limit shared with SEAM.EQUIV.01, v336, plus the v152 anchor). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by $G_{\rm net}$, the area law and R3, and given it the chain Gaussian bulk $\Rightarrow$ quasi-free boundary $\Rightarrow$ one kernel = net (v113) $\Rightarrow$ carrier $A \Rightarrow (E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([verification/v156_seam_net_construction.py]). The "$B \cong (E_8)_1$" step is no longer an assertion — the chiral net is built explicitly and the identification *checked.* The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|\,u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1. Sixteen Majoranas give $c = 8 = 5{+}3$; the weight-1 currents are $248 = 120_{\rm NS} + 128_{\rm R}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\boxed{\chi = \frac{E_4}{\eta^8} = q^{-1/3}\big(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \cdots\big) = j^{1/3}}$$

(verified by q-series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate ([verification/v157_rigid_fixed_point.py]). Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 ΨDO with principal symbol $|\xi|_g$ (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic c=8 theory is *rigid*: every field has weight $(h, \bar h{=}0)$, an exactly-marginal deformation needs $(1,1)$, and with $\bar h$=0 *no* $(1,1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss–Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2|\,2\pi\,\chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=6 log $\frac{3}{2}$>0) boundary flow reaches the holomorphic fixed point" — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a *proven* stable fixed point ([`verification/v158_fixed_point_stable.py`]**).** An exact operator-dimension count settles it: the chiral Majorana has h=$\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h, lowest 1) — nothing relevant flows the free theory away; the h=1 currents are chiral (v157), not $(1,1)$-marginal; and the lowest genuinely-interacting operator (the quartic, h=2>1) is *irrelevant*, as are all higher $2k$-fermi operators (h=k). So the free chiral c=8 fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped (Δ=6 log $\frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise ([`verification/v160_seam_gaussianity_from_pf.py`]**).** Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \mathrm{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 "one kernel = the net" property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56's gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — the Schwinger cone the transport must contract has dimension $\gg 3$, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement ([`verification/v161_one_particle_reduction.py`]**).** The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ, and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree-m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\mathrm{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap *equals* the one-particle gap**: the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation ⇔ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim \mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport's one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A_2), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items ([`verification/v162_seam_transport_identification.py`]**).** Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4-equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^{3} U^k X U^{-k} = |\mu_4|\,\mathrm{diag}(X)$ for $U = \mathrm{diag}(1,i,-i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights spec$(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the ω_k=$|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4-equivariant and admissible — which is the R1 "seam clock" identification (the transfer operator as the seam's gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The "free-bulk cloud" is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) **[E]**/**[O]**.

And both residual identifications are pinned to their floor, so (A) has no independent open gate left (`[verification/v165_premise_a_closure.py]`). A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types **[O]** → **[C]** (closed by assembly). R_1 (the seam clock) reduces to `GATE.QGEO`: a μ_4-equivariant flat gapped generator on the carrier is *unique* (= $A_0^\star$, v115/v116), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established (v137, b_1=3=$N_{\rm fam}$). That is `GATE.QGEO`, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (v153) the irreducibles stay exactly $\{\pi, v_{\rm geo}\}$ — closing (A) adds none. **Final accounting:** premise (A) = (A_2 assembly, **[C]**) + (R_1=`GATE.QGEO`, already open) + the $\{\pi, v_{\rm geo}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate **[E]**/**[O]**.

And the two structural residuals — net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the one irreducible open premise (`[verification/v175_net_existence_full_cone.py]`). The CAR second-quantisation functor $\Gamma(t)$=$\bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ (dim 2^{16}=65536); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of spec(t), so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity 2^8=256 (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every m* (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem.* And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character E_4/η^8=1+$248q$+$4124q^2$+$34752q^3$+$213126q^4$+$1057504q^5$ (level-1=248), the embedding 248=120+128, and the E_8 Cartan matrix even with det 1 (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the *same* input**: that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. `QGEO.REALIZE.01`, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly **[O]**— the single irreducible structural premise; net existence and full-cone RP are **[E]**.

The No-Unit Theorem — $v_{\rm geo}$ is an irreducible metrology primitive [E]/[O] (`[verification/v153_no_unit_theorem.py]`)

$v_{\rm geo}$ is not an open gap. Every load-bearing compiler datum — $g_{\rm car}, |\mu_4|, N_{\rm fam}, {\rm rank}\, E_8, \alpha^{-1}$, the determinant ladder $(3, 4, 8, 20)$, $c_3 = \frac{1}{8\pi}$, $\varphi_0^{\rm ret}$ — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension-d≠0 output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — $v_{\rm geo}$ is a *forced input.* The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\rm point} \sim v_{\rm geo}, \qquad \frac{1}{G} \sim v_{\rm geo}^2, \qquad m/\mu = e^{3/4} \text{ (a pure ratio)},$$

so the irreducibles stay exactly {one dimensionful unit, π}. *Re-typing:* $v_{\rm geo}$ is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Sharpened across gravity, and the final tally (`[verification/v364_vgeo_sharpen.py]`). After the parameter-free gravity (v358/v359/v361) the *gravity* sector adds no new scale either: the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim$ (dimensionless) v_{geo}^4 both reduce to v_{geo} times atoms. Hence v_{geo} is the *single* dimensionful input of the *complete* theory (matter and gravity), and the final tally is $\{v_{\text{geo}}, \pi\}$ with *zero* dimensionless dials — a ∼26 → 1 reduction against the Standard Model. ($1/G$ stays UV-sensitive (Sakharov, v68), so it is a metrology readout, not a clean prediction — consistent with v_{geo} being the anchor.)

Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure μ_{QG}. Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \tfrac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: the local covariant field equation is parameter-free (v358/v359); Theorem G remains open only for the projective-limit ambient* measure μ_{QG} *(G6, the deepest item). G5 certified; the regime equation closed [`verification/v28_gravity_fR.py`].*

Research Contract — `CELEST.SEAM.01` (the celestial and twistor continuum route)

Reader's note. A compact synthesis of this contract's executed work packages (WP1–WP5d, both WP5d stages, + the WP5e α, β, γ, δ_2, ε_2 and ε_1 stages + the first back-reaction milestone M1) — the narrative arc from the μ_4 clock to the $(E_8)_1$ boundary shadow — is given as a dedicated section in `tfpt_3_e8_audit_bootstrap` ("The celestial and twistor continuum route"); this contract remains the full technical reference (typing fence, kill tests, work-package statements). The bridge from this contract's compiler-side constructions to reconstructed Minkowski physics is *not* part of this contract: it is the separate central contract `WOIT.OS.TWISTOR.01` below — everything in `CELEST.SEAM.01` is preparation for it.

The object diagram (read this first). The route is one chain of objects; each arrow is a separately typed claim. Where mathematics ends and physics begins is visible at a glance: the last two arrows are [O].

Arrow	Status	Claim / modules	Kill test
$(\Sigma, \mu_4, \rho, \Theta) \to \mathbb{P}^1\backslash\mu_4$	[E]	`SEAM.MARKS` chain: four marks, clock, cross-ratio 2, $\tau{=}i$ pillowcase (v168/v214/v216/v180; bit reduction v506/v507/v510/v512)	a fifth mark, a non-order-4 clock, or a marks-preserving root without flag transitivity
$\mathbb{P}^1\backslash\mu_4 \to \mathbb{C}^2/\mathbb{Z}_4$	[E]	`CELEST.WP1.01`/`CELEST.WP2.01`: the glue is the flat $\mathbb{Z}_4$ monodromy on the A_3 ALE, clock = Kähler $U(2)$ phase, clock^2 = deck (v492/v493)	equivariant-sector closure failure (3112/6720 control) or a surviving shape modulus
$\mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma$	[E]/[C]	`CELEST.WP5E.*`: equivariant twistor uplift — anomaly ledger, level-from-flux, axion slot, back-reacted Ω_N (v505/v509/v511/v513/v514/v515)	the preregistered level kill ("inflow demands $k \neq 1$"); the δ_1/M2–M3 kills (the M1 kill was evaluated by v515 and did not fire)
$\mathbb{PT}/\Gamma \to \mathcal{A}_{\mathrm{hol}}$	[O]	the *interacting* holomorphic algebra (global BCOV+SDYM quantisation; WP5e proper; target of `WOIT.OS.TWISTOR.01`; CFT-side anchors v497–v504)	anomaly mismatch at the Costello–Li grade
$\mathcal{A}_{\mathrm{hol}} \xrightarrow{\mathrm{OS}} \mathcal{A}_{\mathrm{Mink}}$	[O]	OS reconstruction with the real structure Θ — the `WOIT.OS.TWISTOR.01` target (the free-collar RP/OS witnesses v379/`FORM.SEAM.MMST.01` are the *free* anchor only)	the seven kill tests of `WOIT.OS.TWISTOR.01` below

Definition (the group extension, exactly). The shorthand "$\mathbb{PT}/\mathbb{Z}_4$" used in this contract is precise but deserves unpacking, because *two different* order-4 structures are in play and only one of them is quotiented:

- $\Gamma_{\mathrm{ALE}} = \langle\mathrm{Deck}\rangle \cong \mathbb{Z}_4 \subset SU(2)$, the deck group $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$: *triholomorphic*, preserves the holomorphic symplectic form Ω; this is the $\mathbb{Z}_4$ that is *quotiented* ($\mathbb{C}^2/\Gamma_{\mathrm{ALE}}$ is the A_3 singularity $XY = Z^4$, and $\mathbb{PT}/\Gamma$ means the quotient by Γ_{ALE} acting on the twistor space of the ALE) [E] ([`verification/v492_celestial_z4_orbifold.py`]).
- $\rho = \mathrm{Clock} = \mathrm{diag}(i, 1) \in U(2)$: a Kähler, *not* triholomorphic isometry ($\det\rho = i$ rotates Ω by the μ_4 generator — the WP1 S5 computation); it is *not* in Γ_{ALE} and is *not* quotiented — it *normalises* Γ_{ALE} and survives the quotient as the residual clock symmetry [E] (v492).
- The global object is the extension generated by both: since $\rho^2 = \mathrm{Deck}$ exactly (v492/v506), $\langle\Gamma_{\mathrm{ALE}}, \rho\rangle = \langle\rho\rangle$ is *cyclic* — projectively $\mathbb{Z}_4$ (the clock on the sphere) with ρ^2 generating the deck $\mathbb{Z}_2$ there, and on the spin/fermionic level the canonical $\mathbb{Z}_8$ tower ($V^2 \propto U$, $V^4 \propto (-1)^F$, nonsplit; $8 = 2|\mu_4|$) [E] (v506/v507).

The action table, entry by entry ([E] = arithmetically established at the current stratum, [O] = a quantisation question):

Object	Γ_{ALE} (Deck)	ρ (Clock)	Status / witness
$K_{\mathbb{PT}}$ (canonical class)	trivial	weight via $\det\rho = i$	[E] (incidence-forced column weights $(+1, +1, -1, -1)$, v514)
Ω (hol. symplectic)	preserved	rotated by i	[E] (v492 S5; back-reacted Ω_N closed-form with $(2\pi i)^2$-integral periods and forced charge 4, v515)
BCOV fields ($O(-2)$ tower)	orbifold sectors	character series $P_0..P_3$	[E] ledger / [O] quantisation (v514)
open-string fields (SDYM E_8)	glue-equivariant sector	sector rotation $j \mapsto j{+}1$	[E] ($D[d] = 60(d{+}1)/64(d{+}1)$, v492) / [O] interacting
boundary states ($(E_8)_1$ shadow)	μ_4 sector split	clock phase on $\lvert s\rangle$ etc.	[E] character/GNS level (v497–v500) / [O] as an actual net

The A_3 role table (no silent identifications). Several rank-3 A_3's and external gauge factors circulate near this contract; the following table states, for each pair, either the connecting functor (with its witness) or an explicit *not identical*:

Object	Role and relations
A_3^{family}	the family factor in the lattice $D_5 \oplus A_3$ ($\mathfrak{su}(4)$-flavour; three families from its exponents, v128/v137). Related to A_3^{ALE} by the McKay correspondence *of types* only (both are type A_3); **not identical** as realisations — one is a sublattice of E_8, the other the singularity type of $\mathbb{C}^2/\mathbb{Z}_4$. The bridge that *does* connect them is the Kronheimer–Nakajima quiver (v479: same ADE datum, different functors).
A_3^{ALE}	the ADE type of $\mathbb{C}^2/\mathbb{Z}_4$ ($XY = Z^4$); its resolution carries the three exceptional spheres (v492/v493).
three exceptional spheres	the Coxeter/Picard–Lefschetz structure of the resolved A_3^{ALE}: the clock acts as a Coxeter element of $W(A_3)$ on H_2 (eigenvalues $\{i, -1, -i\}$, v493); they carry the three twisted axion slots (v505) and the lockstep fluxes (v509).
D_5^{carrier}	the carrier factor ($g_{\text{car}} = 5$); glued to A_3^{family} by the μ_4 Lagrangian glue into E_8 (v92/v125); **no** role on the ALE side beyond the zero-mode algebra $\mathfrak{g}_0 = D_5 \oplus A_3$ (v492).
Woit's $SU(3)_{\text{color}}$	*external programme reference only* (Euclidean Twistor Unification: colour as the rank-three quotient bundle on $\mathbb{PT}$). **Not identical — no identification claimed**; in particular $A_3^{\text{family}} \not\simeq SU(3)_{\text{color}}$ (an $\mathfrak{su}(4)$ flavour factor is not a colour gauge bundle), and TFPT's colour lives in the SM readout of the carrier, not on $\mathbb{PT}$.
Woit's $SU(2)_{\text{weak}}$	*external programme reference only* (the internal spin factor of the Euclidean twistor construction). **Not identical — no identification claimed**; whether an internal $SU(2)$ of that kind can be realised on $\mathbb{PT}/\Gamma$ is exactly kill test (6) of `WOIT.OS.TWISTOR.01`.

Hypothesis (as corrected by WP1). *The seam is the glue-equivariant $\mathbb{Z}_4$-orbifold sector of the celestial chiral algebra on the A_3 ALE space $\mathbb{C}^2/\mathbb{Z}_4$; the μ_4 clock is the Kähler $U(2)$ phase* $\operatorname{diag}(i, 1)$ *whose square is the deck group (spin bridge $\mathbb{Z}_8$, $8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ seam winding integer); and the associated twistorial bulk is the* self-dual *sector of TFPT gravity.* This is the fourth research contract, alongside (U_{wall}), (G_{metric}) and `CONTRACT.F.01`: a numbered chain of work packages with pre-registered kill tests, *not* a claim. The method is the celestial-chiral-algebra (CCA) construction of Bittleston–Homans–Sharma on Eguchi–Hanson space (`arXiv:2305.09451`, JHEP 09 (2023) 008 — the $\mathbb{Z}_2$ case, refined here to $\mathbb{Z}_4$ with a flat connection at infinity in the Kronheimer–Nakajima / heterotic-on-ALE sense), inside the celestial-holography programme (Strominger, PRL **127**, 221601: $w_{1+\infty}$; Costello–Paquette–Sharma, PRL **130**, 061602: Burns holography, the $SO(8)$ WZW$_4$ model; Bittleston–Sharma–Skinner, CMP **403** (2023): quantizing the non-linear graviton). The gauge-algebra admissibility gate is already passed in the literature: Costello's one-loop axionic-consistency analysis (`arXiv:2111.08879`) admits the exceptional algebras *including* E_8 for SDYM on twistor space (Okubo: no independent quartic Casimir). The spin bridge in the hypothesis is no longer a bookkeeping convention: on the 16-Majorana seam the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$) and the quarter-shift lifts to a canonical $\mathbb{Z}_8$ tower with $V^2 \propto U$ — "$8 = 2|\mu_4|$" fermionically explained ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`); and the nonsplit class is arrangement-*sensitive*: the edge (silver) arrangement splits ($U_e^2 = +1$, zero roots), so the fermions *measure* the alignment bit as a Fidkowski–Kitaev-type extension class ([`verification/v507_seam_bit_origin.py`], `SEAM.BIT.ORIGIN.01`) — and that class is nonsplit *iff* the deck acts freely (all 17 seam-circle involutions, v510), which the covering deck does by topology.

WP1, executed and verified: the E_8 μ_4-glue *is* the flat $\mathbb{Z}_4$ monodromy of the equivariant CCA on the A_3 ALE space — verdict B ([`verification/v492_celestial_z4_orbifold.py`], `CELEST.WP1.01`)

Everything in this box is exact arithmetic (sympy, no floats). **Inner grading / flat monodromy [E]:** the v128 glue grading ($240 = 52+64+60+64$, additive on all 6720 root pairs, dims with Cartan $(60, 64, 60, 64)$, $\mathfrak{g}_0 = D_5 \oplus A_3$ carrier $= 60$) is *inner*: $h = (2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\operatorname{Ad}(\exp(2\pi i h/4))$ — an order-4 torus element, i.e. a *flat $\mathbb{Z}_4$ monodromy* in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5; 1, 1, 1, -3)$ reads the *same* class, so the glue charge is the diagonal $(1, 1)$ of the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ — the v92/v125 Lagrangian glue. **The spatial side [E]:** $\mathbb{C}^2/\mathbb{Z}_4$ under the deck $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$ *is* the A_3 singularity $XY = Z^4$ (invariant ring exact; Hilbert series $(1+t^4)/((1-t^2)(1-t^4))$; Molien table verified). **The equivariant sector closes [E]:** the glue-equivariant SDYM(E_8) modes ($j + m \equiv 0 \bmod 4$) form a Lie subalgebra with graded dimensions $D[d] = 60(d+1)$ (even d) / $64(d+1)$ (odd d) — uniform per parity *only* because $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = 60$ and $\dim \mathfrak{g}_1 = \dim \mathfrak{g}_3 = 64$: the v128 dimensions are exactly the consistency condition. The zero modes ($d{=}0$) are $\mathfrak{g}_0$ alone — *the unbroken algebra on the orbifold is the carrier* $D_5 \oplus A_3$+Cartan $= 60$; the asymptotic density is $62/248 = 1/4 = 1/|\mathbb{Z}_4|$ (the orbifold index), and the $\mathbb{Z}_2$ halfway sector reproduces the Eguchi–Hanson towers $120/128 = SO(16)$ adjoint/spinor — μ_4 refines the sheet $\mathbb{Z}_2$ exactly as $4 = 2 \times 2$ (v125). **Character level [E]:** Θ_{E_8} splits into the four glue-coset thetas (shell $1 = (52, 64, 60, 64)$ re-derived from the *lattice*); the four sector characters Θ_{C_j}/η^8 sum *exactly* to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (the v377 series); the sector conformal weights are $h_{D_5} = (0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$, $h_{A_3} = (0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$ — the v92/v125 discriminant form $(5x^2{+}3y^2)/8 \bmod 1$ — and the glue diagonal has *integer* $h = (0, 1, 1, 1)$: the v125 isotropy *is* the locality of the extension to $(E_8)_1$. **The critical correction [E] (why verdict B, not A):** the A_3 deck $\mathrm{diag}(i, i^{-1})$ induces on the celestial sphere $z = z_1/z_2$ only $z \mapsto -z$ (projective order 2, the sheet flip) — it is *not* the order-4 clock $z \mapsto iz$. The exact bridge: any $SU(2)$ lift of the clock has order 8 (spin group $\mathbb{Z}_8$, which would quotient to A_7), and (spin clock)^2 = the deck generator *exactly*, with $|\mathbb{Z}_8| = 8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ winding integer. The order-4 clock *is* realised, as the $U(2)$ phase $\mathrm{diag}(i, 1)$: projective order 4, fixed points $\{0, \infty\}$, free μ_4 orbit (all v180 clock properties), normalising the deck (so preserving the glue flat connection), with $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator (a Kähler, *not* triholomorphic isometry — it rotates the twistor sphere by $\pi/2$). **Gravity side [E]:** clock-invariance selects from the 3-parameter versal A_3 family $XY = Z^4{+}a_2Z^2{+}a_1Z{+}a_0$ *exactly* the 1-parameter family $XY = Z^4{+}a_0$, whose four branch points are *one* μ_4 orbit with cross-ratio 2 — the seam marks (v179 clock orbit; v168/v214 cross-ratio 2 $\Rightarrow$ $j{=}1728$ pillowcase). **Negative controls [E]:** the false spatial action $\mathrm{diag}(i, i)$ is killed three ways ($\det = -1 \notin SL(2, \mathbb{C})$, no CY/ALE resolution; Poisson closure fails in 14/14 nonzero cases; conjugation symmetry broken), the false glue (60/64 blocks swapped) breaks additivity on 3112/6720 root pairs, and of all 24 relabelings exactly $\{\mathrm{id}, j \mapsto -j\} = \mathrm{Aut}(\mathbb{Z}_4)$ are additive — the grading is rigid. **Verdict [C]:** the celestial and glue $\mathbb{Z}_4$ structures match under *one* precisely named extra identification — clock^2 = deck on the sphere (the sheet-$\mathbb{Z}_2$ spin bookkeeping) — so WP1 lands at verdict B (match modulo one named identification), close to but honestly short of A. [E]/[C]

Typing and non-circularity. Admissible *inputs* are the finite seam/glue data already in the suite — the μ_4 Q-system glue (v92/v125), the graded hull (v128), the seam rigidity and pillowcase geometry (v168/v214), the Möbius clock (v180) — plus the WP1 [E] arithmetic ([`verification/v492_celestial_z4_orbifold.py`]). *Not* admissible as an input: the *continuum existence* of the holomorphic $(E_8)_1$ net on the seam — that is precisely the `SEAM.EQUIV.01` target, and assuming it would make the contract circular. *Scope fence:* the twistorial bulk of this contract is the *self-dual* sector only, never full Einstein gravity — the classical field-equation route (entanglement equilibrium, v358/v359) and the entropic-action bridge (`GRAV.ENTROPIC.ACTION.01`, v473) are separate, non-competing statements about the full equation. `SEAM.EQUIV.01` stays [O] throughout; nothing in this contract moves it.

Work packages.

- **WP1 (*done*, verdict B) [E]/[C]:** the structural $\mathbb{Z}_4$ arithmetic of the keybox above, executed and machine-verified as `CELEST.WP1.01` [`verification/v492_celestial_z4_orbifold.py`].
- **WP2 (*done*, verdict B) [E]/[C]:** the clock-invariant deformation theory, executed and machine-verified as `CELEST.WP2.01` [`verification/v493_celestial_wp2_clock_deformation.py`] (sympy exact, 47 checks). Both targets landed, *sharpened*: (i) a_0 is the pure *scale* of the seam configuration — the binary-quartic invariants of $w^2 = Z^4{+}a_0$ are $I = 12a_0$, $J = 0$ *identically*, so $j = 1728$ and the $\tau{=}i$ pillowcase shape (v168/v214) is *frozen* for every $a_0 \neq 0$ (no shape modulus survives clock-invariance; the negative controls $a_1Z \Rightarrow j{=}0$ and $a_2 \neq 0 \Rightarrow j{=}1556068/81$ show the test has teeth); (ii) the three resolved spheres ($3 = \mathrm{rank}\, A_3 = N_{\mathrm{fam}}$) carry *exactly* the three nontrivial μ_4 characters $\{i, -1, -i\}$ under the clock, which acts on H_2 as a *Coxeter element* of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, order

$h(A_3)$=4=$|\mu_4|$=N_{fam}+1) and *is* the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); the surviving deformation direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$ — *not* the naive invariant diagonal ($\det(A{-}1) = -4$: no invariant cycle) — and all three sphere volumes move in lockstep ($|\Pi_j| = \sqrt{2}\,t$). The BHS deformed-algebra pattern transfers from $\mathbb{Z}_2$ to $\mathbb{Z}_4$: the fibre bracket $-4\,\text{Nambu}(XY{-}Z^4{-}a_0)$, anchored at the a_0=0 orbifold, closes with corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade (no sector leak — the WP1 equivariant sector deforms consistently), with a_0 in the weight-0 slot of BHS's c^2. Verdict B via three named [C] identifications ($-4\,$Nambu as *the* k=4 CCA bracket; period = root difference; the seam-scale reading via clock^2 =deck); the full deformed twisted-sector OPEs and the $W(\mu)$-type scaling limit for k=4 stay [O]. The v216 [O]-residual is *typed*, not moved: given the order-4 clock, the deformation freezes the $(2, 2, 2, 2)$ modulus to the square automatically — a relocation of the same order-4 carrier input, not a new derivation (since v510 that input reads: the square-modulus datum $\tau = i$; its position half is topology; since v512 it reads equivalently: one discrete symmetry-lift bit, flag transitivity $\iff \tau = i$).

- **WP3 (*done*, verdict B) [E]/[C]:** the Green–Schwarz alignment test, executed and machine-verified as `CELEST.WP3.01` [`verification/v495_celestial_wp3_gs_alignment.py`] (exact Fraction/sympy, 25 checks). The arithmetic side is sharp: the Okubo coefficient $\text{Tr}_{\text{adj}}X^4/(\text{Tr}_{\text{adj}}X^2)^2 = 5/(2(\dim\mathfrak{g}{+}2))$ is *derived* as a polynomial identity for all 8 algebras on Costello's list (with a sharp $\mathfrak{sl}_4$ negative control: 5/32 vs 3/32), the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim{+}2) = h^\vee{+}6$ holds exactly across the Deligne series, and for E_8 the axion coefficient is *exactly* $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\text{fund}} = 1/10$ (adjoint-trace), so the GS coupling $\kappa = \lambda_{\mathfrak{g}}/(8\pi\sqrt{3})$ satisfies $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$ resp. 1/300 — exact anchor rationals, with the anchor decomposition $h^\vee$=30=$|\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}}$, $\varphi(30)$=8=rank, $\tilde{\lambda}$=$|\mathbb{Z}_2|N_{\text{fam}}$, λ_{fund}=$1/(|\mathbb{Z}_2|g_{\text{car}})$ (a byproduct: the printed $\lambda^2_{\mathfrak{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip; his own weight content gives 3, Okubo-consistent). κ/c_3 itself is *irrational* ($2\sqrt{3}$): only the squared alignment is exact. **The look-elsewhere caveat is part of the result [C]**: the same squared-rational alignment holds for *all eight* algebras on the list (8/8 — the test as such has zero selective power), $\tilde{\lambda}$-integrality passes 2/8 (shared with $\mathfrak{sl}_3$), and the only $\mathfrak{e}_8$-selective single test is g_{car}=5 | $h^\vee$ (1/8); the conjunction that isolates $\mathfrak{e}_8$ is post hoc, not preregistered. *Alignment survives; selectivity does not* — the c_3-connection of the celestial route is convention-level compatibility plus genuine λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence, and never a derivation of c_3 from celestial data.
- **WP4 (*done*, verdict B(ii)) [E]/[C]:** the type mismatch, executed head-on and machine-verified as `CELEST.WP4.01` [`verification/v496_celestial_wp4_character_shadow.py`] (exact integer/Fraction/sympy, 25 checks). The $(E_8)_1$ character is *not* a conformal block of the celestial $E_8[\mathbb{C}^2]$ S-algebra in its own (jet) grading — the obstruction is localised *three* independent ways: (a) the CP grading gives spin $1{-}d/2$, unbounded below, and 248 is never a jet-tower dimension ($60(d{+}1)/64(d{+}1)$, $d \le 100$); (b) the cumulative generator count is quadratic ($31s^2{+}92s{+}60$, $31 = k{+}h^\vee$), so the jet Fock grows as $n^{2/3}$ against the character's $n^{1/2}$ (Meinardus poles s=2 vs s=1; the ratio f_n/χ_n increases strictly for $n = 1..12$); (c) the level-2 null ideal of $(E_8)_1$ deletes *exactly* $27000 = 30^3 = (h^\vee)^3$ out of $\text{Sym}^2(248) = 1{+}3875{+}27000$ (the character keeps $4124 = 1{+}248{+}3875$) — and has no jet analogue. **But the boundary/period reading holds exactly at the current stratum**: the zero-mode slice is the 60 vacuum-sector currents, the single-line jet slice cycles the four sector counts $(60, 64, 60, 64)$, one full μ_4 period of loop energies sums exactly to 248 (the single-particle level of the $248q$ stage), and the glue-diagonal weights $(0, 1, 1, 1)$ are integers — while the free loop Fock at level 1 counts $897266 \gg 248$, i.e. the rational truncation must be *imposed in a limit.* That is precisely the MMST scaling-limit shape of `SEAM.EQUIV.01` (v336/v449: the rational net exists in the *limit* of the lattice collar, not in the pre-limit algebra): WP4 is hereby *retyped* from “character as conformal block” to “character as boundary/limit shadow”, and the constructive limit

question (which limit, which topology, convergence) passes to WP5 — where WP5a has since answered the "which limit" part at the counting level (the coefficientwise χ_w limit below), WP5b has constructed the deleting operator, WP5c the limit state whose GNS kernel contains the ideal, WP5d the local-net legs (the α-stage two-interval index chain and the β-stage split + strong-additivity witnesses — all three KLM ingredients of complete rationality on the lattice), and WP5e-$\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$ have anchored *both* faces of the twistor uplift, closed its exchange sub-branch, executed the full-tensor ledger, executed the level-from-flux dial and built the bulk-axion slot (the CFT-side prefactor + level pinning; the equivariant twistor skeleton with the index bridge and the rigid $32\,T_3$ residual; the exchange no-go with certificate; the full-tensor confirmation of the collapse with the one cubic d-symbol channel that opens; the lockstep theorem making "one level" a theorem of clock invariance, with the sector-counter dial pinning $k = 1$; the equivariant $O(-2)$ slot construction with $\tilde{\lambda} = 6$ triply pinned and the first GH/A_3 back-reaction step), and the first quantised back-reaction milestone M1 has since been executed (the closed-form Ω_N with $(2\pi i)^2$-integral lockstep periods and forced source charge $4 = |\mu_4|$, v515), leaving the global BCOV quantisation (WP5e proper) as the remaining open piece — narrowed to the δ_1 τ-integral (twisted contact terms, *undecided* at exploration level), the physical justification of the cubic GS term and the remaining $\mathbb{PT}/\mathbb{Z}_4$ back-reaction milestones (throughout, $\mathbb{PT}/\mathbb{Z}_4$ is the quotient by Γ_{ALE}, normalised by the clock $\langle\rho\rangle$ — the definition block above; the ε_1 slot itself is executed; of the quantised milestones M1–M3 preregistered in v514, M1 is executed via v515, M2–M3 stay preregistered [O]).

- **WP5 (the summit) — subdivided; WP5a–WP5d (both WP5d stages) plus the WP5e α, β, γ, δ_2, ε_2 and ε_1 stages and the back-reaction milestone M1 are landed.** The twistorial construction as a *second continuum route* for `SEAM.EQUIV.01` (Costello–Li grade), carrying the WP4 constructive-boundary-limit question. Subdivision WP5a–WP5e:
  - **WP5a (*done*) [E]/[C]:** "the null ideal from the limit", executed and machine-verified as `CELEST.WP5A.01` [`verification/v497_celestial_wp5a_null_ideal.py`] (exact integer/Fraction arithmetic, 34 checks). Three results make the WP4 limit reading precise. (i) *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) *contains* the chiral jet grading as its w=2 member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n$+1 — strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$): the WP4 "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit stabilisation threshold $w = n$+1, not a slice. (ii) *The null ideal is derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with peeling residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; level-1 integrability ($\langle 2\theta, \theta^\vee\rangle = 4$, $\langle \omega, \theta^\vee\rangle = 2$) leaves the vacuum as the only level-1 primary ([C] Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. (iii) *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: 34752). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components (5304+1820+135+1), $5304 \neq 14^3$ (the $h^{\vee\,3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), and block weights $(0, \frac{1}{2}, 1, 1)$ that can*not* fuse into one local character; only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor. *Honest limit* [O]: the limit does *not* generate the truncation (the loop Fock counts $897266 \gg 248$ at integer level 1) — WP5a fixes the ideal's *size and location* quantitatively and gives the celestial route the same two-step shape as MMST (limit + maximal ideal); the operator content is exactly WP5b–e, of which WP5b–WP5d and the WP5e α, β, γ, δ_2 and ε_2 stages are

executed below.

- **WP5b (*done*) [E]/[C]:** "the singular vector as an operator", executed and machine-verified as CELEST.WP5B.01 [verification/v498_celestial_wp5b_singular_vector.py] (exact integer/Fraction arithmetic, 53 checks, deterministic) — *success on the preregistered criterion*: the deleting object exists and is explicit. $|s\rangle = (E^{\theta}_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) is constructed in a machine-built Chevalley/Frenkel–Kac basis (cocycle asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign *forced* by Jacobi, SGN $= -1$; κ derived, $\kappa(\theta^\vee, \theta^\vee) = 2$), and $J^a_1|s\rangle = 0$ is machine-verified for *all* 248 generators with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$, the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight and Shapovalov norm 0. *Level dial*: the F^θ_1 coefficient is $2(1-k)$ — nonzero at k=0 and k=2, zero exactly at k=1: without the central extension the deletion operator does not exist. μ_4 compatibility: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle\theta, h\rangle = 5$, height $29 = h^\vee-1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even (WP5a-consistent) with the deleting θ-$\mathfrak{sl}_2$ crossing the sheet-odd classes $(1, 3)$; 8 quarters $= q^2$ via the per-period dictionary. The module it generates is *the* ideal: weight 2θ has multiplicity 1 in the level-2 Fock, and the direct $\mathfrak{g}_0$-lowering orbit BFS from $|s\rangle$ reproduces the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ($27000 = (h^\vee)^3$, quotient 4124; [C] Weyl complete reducibility beyond depth 4). Negative controls separate honestly: the level-1 current state and generic level-2 states are *not* singular; at k=2 the *cube* is (the $(E^\theta)^{k+1}$ pattern is generic Kac level mechanics); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion) — the singular-vector mechanism is level-1 *generic*, the *one-block closure* is the E_8/μ_4-specific part. *Handover to WP5c* (answered below): in the twisted quarter-*slot* moding of the χ_w limit Fock, two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-*period* dictionary (v496), not the per-slot identification, carries $|s\rangle$ to q^2; exactly the GNS/limit-state question (kernel $\supseteq$ ideal) that WP5c answers.
- **WP5c (*done*) [E]/[C]:** "the GNS limit state", executed and machine-verified as CELEST.WP5C.01 [verification/v500_celestial_wp5c_gns_limit_state.py] (exact integer/Fraction arithmetic, 35 checks, deterministic) — *success on the preregistered criterion*: the family ω_w exists and its limit has the null ideal in its GNS kernel. ω_w is the quasi-free state with loop sector = the affine k=1 vacuum n-point functions (via the machine-determined compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$, the unique anti-automorphism sign on all 61504 basis pairs) and radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic); *the WP5a threshold holds at the state level*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N+1$, sharp at $w = N$. The core is fully machine: the *complete* exact level-2 Gram — 31124 monomials in 9361 weight blocks — has rank 4124 exactly (the preregistered target), kernel $27000 = V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights), every block PSD, rank table per Weyl orbit $(0, 0, 1, 8, 44)$, level-1 rank 248 positive definite (the current layer survives). μ_4: the clock descends to GNS with level-2 rank split $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the WP5a two-routes identity at the *state* level — and $|s\rangle$ *is* the zero vector of GNS(ω_∞), resolving the WP5b twisted-slot tension. The KILL branch is probed and not triggered: a CCR obstruction shows *no* w-uniform state can damp the radial modes (the family formulation is *necessary*), and the wrong constructions are caught (family $E'_w = mw+r$ erases the 248 layer, $710955 \neq 248$; no damping keeps $897266 \neq 248$); k-dial and D_8 controls separate honestly (k=2: $\langle s|s\rangle = +4$, no ideal in the kernel; D_8: one block of four, $h = \frac{1}{2}$). [C] quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2.
- **WP5d — both stages (*done*): α [E]/[C], β [E]/[C] (continuum uplift [O]):** "the local net", approached from the lattice. The α stage — the KLM

two-interval index as the operational Haag-duality discriminator ($(E_8)_1 \Rightarrow \mu{=}1$, one sector; $SO(16)_1 \Rightarrow \mu{=}4$) — is executed and machine-verified as `CELEST.WP5D.01` [`verification/v501_celestial_wp5d_two_interval_index.py`] (39 checks; Gaussian lattice machinery ED-validated to 10^{-15}, exact Fractions for the arithmetic). Measured on the 16-layer seam carrier (critical Majorana ring, $c_- = 8$): the fermionic two-interval mutual information is *extensive* (the $\mu{=}1$ reference; c fit 0.5000, residual $\to 0$ with N), while the sector-summed (orbifold) prescription pays exactly one classical bit — the $\ln 2$ plateau at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at $N{=}512$), so $[F : F_{\text{even}}] = 2$ and $\mu_{\text{gauged}} = 4$, the entropic twin of the v490 parity census (two independent lattice witnesses); the orbifold also breaks two-interval complementarity ($S(E) \neq S(E')$, the direct duality-failure witness), with the complementary-pair budget bounded by $\ln 4 = \ln \mu(SO(16)_1)$ as a *double-limit* statement. The condensation arithmetic is anchored at both measured ends: $\det \mathrm{Cartan}(D_5) \cdot \det \mathrm{Cartan}(A_3) = 16$, KLM/Longo–Rehren $16/4^2 = 4/2^2 = 1$, $\Sigma d_i^2 = (4, 4, 1)$, $\theta_v = 1$ exactly at $\nu = 2c_- = 16$ (rivals $\neq 1$) — so $\mu = 1$ after condensation and the preregistered KILL ("μ-offset $\neq 0$ after condensation") is *not* triggered; the $\nu{=}1$ control shows a non-removable offset ($\theta_v \neq 1$: the discriminator has teeth), the trivial phase shows none, and the wrong sector sum loses the full $\ln 2$. [C] the continuum dictionary "offset = ln index" (KLM; Longo–Xu; Casini–Huerta–Magan–Pontello). **The β stage (*done*) [E]/[C] — split + strong additivity, the two remaining KLM legs** — is executed and machine-verified as `CELEST.WP5DB.01` [`verification/v504_celestial_wp5d_klm_completeness.py`] (37 checks; Gaussian lattice machinery ED-validated, exact GF2/integer algebra for the exact legs; the $U(1)$/Dirac number-conserving subnet as the preregistered negative control — the current-net analogue where strong additivity famously *fails* in the continuum). *Strong additivity is algebraically exact on the lattice*: with the shared boundary Majorana ("touching = share the point"), $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ *exactly* (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32), while disjoint intervals give exactly *half* (rank 16/32, index 2 — the missing sector is odd $\otimes$ odd, the single $\ln 2$ bit of the α stage, localised at the split point); the $U(1)$/neutral algebras do *not* generate the union even with the shared site (gaps 2/10/52, growing). The entropic witness uses the honest discriminator *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold touching deficit stays $< \ln 2$ with the Ising approach $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (the naive "defect $\to 0$" expectation is *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control *bursts* $\ln 2$ from $L = 128$ on and grows as $\frac{1}{2} \ln \mathrm{Var}\, Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$: infinite index). The split property is witnessed quantitatively: the coupling-block singular values fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% over $d = 16..512$, $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$), trace norm summable — and the orbifold *inherits* the same ladder exactly at machine level (P_A flips $C \to -C$, σ_k identical to 8.3×10^{-17}; the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ with singular values $\{\sigma_a \sigma_b\}$ — Longo heredity, [C]). The Pimsner–Popa bound is a one-line *identity*, $E(a) - a/2 = PaP/2$, so $\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]$ exactly, attained in exact integers ($16384 + 2048$ monomial sweeps, 0 violations; random pencil $\min \lambda = 0.5000000000$); the two-interval group gives $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly, and the index is pinned by *two independent routes* ($e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$; $e^{2\Delta_\infty} = 4 = 1/\lambda_{E_4} = \mu$); the $U(1)$ control has $\lambda = 1/(m{+}1) \to 0$. *With the α stage, all three KLM ingredients of complete rationality — split, strong additivity, finite μ — are now witnessed on the lattice.* The continuum uplift stays **[O]** and is honestly fenced: the theorem "finite-group orbifolds of completely rational nets are completely rational (hence split + strongly additive)" is *Xu's theorem* (algebraic orbifold CFTs) — cited, *not* claimed; the continuum proof for the concrete seam quotient net on the collar (v336/v449) and for the interacting condensed $(E_8)_1$ net remains WP5e / Costello–Li territory.

– **WP5e (subdivided): α stage (*done*) [E]/[C] — the CFT-side prefactor + level pinning; β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space, including the honest a_0 refutation; γ stage (*done*) [E]/[C] — the sphere-axion pairing check, a rigid *negative* result with certificate; δ_2 stage (*done*, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse confirmed, one cubic door opens; ε_2 stage (*done*, verdict B) [E]/[C] — the CPS level-from-flux dial, with "one level" lifted to a theorem; the global BCOV quantisation stays [O].** The α stage is executed and machine-verified as `CELEST.WP5E.ALPHA.01` [`verification/v502_celestial_wp5e_prefactor_level.py`] (exact sympy/Fraction arithmetic, 33 checks). Both exactly computable consistency anchors of the uplift hold. (i) *The $q^{-1/3}$ prefactor is exact μ_4 vacuum-energy bookkeeping*: the clock is *inner* ($h = (2,2,2,2,2;0^4)$ and the A_3 detector $h' = (0^5;1,1,1,-3)$ both read the glue class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32), so the induced twist on the 8 torus bosons is $\theta = 0^8$ in *every* sector — a *shift* orbifold, not a rotation orbifold — and each sector carries the same $8 \times (-\frac{1}{24}) = -\frac{1}{3} = -c/24$ (Hurwitz-ζ exact, $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$); the sector ground exponents are $-\frac{1}{3} + (0,1,1,1)$ with $h_j =$ the discriminant form, an identity holding via spectral flow ($j^2(h,h)/32$, mod 1) *and exactly* via the $10 + 6 = 16 = 2^{g_{\rm car}-1}$ Majorana free fermions of the WP5d seam carrier (R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); the four sector characters sum to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$; the rotation reading *fails* (the geometric A_3 deck gives $\frac{3}{16} \neq \frac{3}{8}$ — the WP1 "clock $\neq$ deck" lesson at the Casimir level). (ii) *$k = 1$ is forced three independent ways*: the current condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes onto the E_8 adjoint $60+64+60+64 = 248$ only at spin 1); the conformal embedding ($D_5+A_3 \subset E_8$: $47k^2+219k-266 = 47(k-1)(k+266/47) = 0$; $D_8 \subset E_8$: $128k(1-k) = 0$); and the central charge $248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor itself); plus the WP5b singular-vector dial ($\|(E^{\theta}_{-1})^m 0\|^2 = m!\,k(k-1)\cdots(k-m+1)$, singular vector exactly at grade $k+1$; $k = 1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). *Honest sharpening*: glue-h *integrality* $h(J^j;k) = k\,(0,1,1,1)$ holds for *all* $k = 1..8$ and fixes nothing — the originally conjectured integrality route is dead; the three other dials carry. Controls: $D_8/SO(16)_1$ has the *same* $q^{-1/3}$ prefactor but $h = (0, \frac{1}{2}, 1, 1)$ (the discriminator is glue-h integrality, not the prefactor); $\mathbb{Z}_2$ or μ_4 *rotation* twists break the common prefactor; wrong $k \in \{2,3,4\}$ fails all five dials coherently. **The β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space** — is executed and machine-verified as `CELEST.WP5E.BETA.01` [`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`] (exact sympy/Fraction arithmetic, 47 checks). The twistor side is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on $\mathbb{C}^2/\mathbb{Z}_4$ with the μ_4-glue monodromy. (i) *The AB ledger is exact*: denominators $\det(1 - g_j^{-1}) = (2,4,2)$ with Dedekind sum $5/4 = (|\mathbb{Z}_4|^2-1)/12$; equivariant characters $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248, 0, -8, 0)$ by two independent routes; invariant average $60 =$ the carrier; Frobenius 61568. (ii) *Okubo per sector, the honest sharpening*: only the *invariant* sector is Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{\mathfrak{e}_8}$ — the v495 identity re-derived through the glue grading); the twisted sectors carry irreducible T_5/T_3 content, and the AB-weighted sum cancels the D_5 quartic *exactly* while leaving the *rigid* residual $32\,T_3$ (no admissible reweighting fixes it; the graded GS exchange is rank-obstructed in sectors 1–3) — twisted-sector closed strings must carry the rest [O]. (iii) *The index bridge* (the centrepiece): $f(m) = \frac{1}{4}\sum_j (i^{jm}-1)/\det_j = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ *exactly* — the fixed-point ledger and the McKay/Kronheimer intersection ledger are the same arithmetic; the glue-bundle ch_2 defect is the integer -78 by *both* routes (classes $(-24,-30,-24)$). (iv) *The level dials say $k = 1$ geometrically*: the lattice current count is 240 at $k{=}1$ and exactly 0 at $k = 2,3,4$; the embedding residual $47k^2+219k-266 = (0, 360, 814, 1362)$; $h(J;k) = k$ is integer for *all* k (integrality alone fixes nothing — honest, exactly as on the CFT side); one scale $\Rightarrow$ one level (v493 periods;

since the ε_2 stage a *theorem* of clock invariance, v509). (v) *The honest refutation*: the clock-invariant modulus $a_0 \in O(8)$ fills the BSS *graviton* slot $O(2)$, *not* the axion slot $O(-2)$ (Hamiltonian weights $X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$, typed [C] as a coincidence) — "the theory brings its own GS axion as a_0" is *refuted*; what does match: the three $H^2(\mathrm{ALE})$ classes carry exactly the three twisted-sector Coxeter characters $\{i, -1, -i\}$ (bijection, no invariant class), so the bulk axion must come from the $O(-2)$ tower field itself (construction [O]). Controls: $\mathrm{diag}(i, i)$ breaks the ledger four independent ways; $SO(16)$ glue gives characters $(120, 0, +120, 0)$ and defect $-30 \neq -78$ with a *failing* bulk Okubo (independent quartic Casimir); $k = 2$ dies on the closure dial ($-31/48 \neq -\frac{1}{3}$) while naive integrality dials stay integer. The preregistered kill does *not* fire on the equivariant skeleton. **The γ stage (*done*) [E]/[C] — the sphere-axion pairing check, an honest rigid *negative* result** — is executed and machine-verified as `CELEST.WP5E.GAMMA.01` [`verification/v508_celestial_wp5e_gamma_pairing.py`] (exact sympy/Fraction arithmetic, 27 checks). Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on $H^2(\mathrm{ALE})$, the β-stage slot bijection) cancel the rigid $32\,T_3$ residual ($A_{\mathrm{fix}} = 9P_1 - 30P_2 - 15P_3 + 32\,T_3$) by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? *No, with a certificate.* (i) *Vertex-space theorem*: the $W(D_5)\times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are *exactly* $\mathrm{span}\{s_5, s_3\}$ (dim 2) and the invariant quartics *exactly* $\mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5) — complete by Weyl nullspace arithmetic over all bidegrees; the *product theorem* is the master kill: the product of any two invariant quadratics has identically zero T_5/T_3 content, so every exchange image — any axion count, any selection rule, any couplings — is annihilated by Φ_{T_3} while $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$. (ii) *Selection enumeration*: the strict two-index $\mathbb{Z}_4$ rule collapses (the class bilinears are nonzero only for $j' = -j$, forcing $m = 0$: the sphere axions have *no* invariant bilinear Cartan vertex at all); the generous twist-insertion channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$ give a rank-2 exchange matrix with $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$ and annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$ — two independent obstructions, the second driven by the side discovery $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are parallel); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are rigidly *silent*), and the sector-0 Okubo mechanism is undisturbed ($\Phi_P(Q^{(0)}) = 0$). (iii) *Naturalness dissolves*: ch_2-natural and Atiyah–Bott-weight couplings both carry certificate $(0, 0)$ against the required $(32, 72)$ — the failure is scale- and coupling-independent (the ansatz space misses the target identically, not by an unnatural factor). Controls: the flipped rule changes the enumeration but not the T_3 kill (Φ_P is rule-dependent, Φ_{T_3} is not); $SO(16)$ has *no* sphere partners (odd McKay classes empty) *and* uncancelled T_5 ($(15, -18, -15, -4, 40)$ — E_8 is doubly special); D_8 has no T_3 structure at all ($16\,T_8 + 12\,s_8^2$). Number fences typed [C] only: $32 = (h, h) + (h', h') = 20 + 12$, $72 = 2\tilde{\lambda}_{e_8}^2$. The β-stage slot bijection is untouched and the preregistered level-kill still does *not* fire — what dies is the exchange *realisation* of the pairing. **The ε_2 stage (*done*, verdict B) [E]/[C] — the CPS level-from-flux dial** — is executed and machine-verified as `CELEST.WP5E.EPS2.01` [`verification/v509_celestial_wp5e_eps2_level_from_flux.py`] (exact sympy/Fraction arithmetic, 28 checks). Target: $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). (i) *The CPS skeleton is replicated exactly*: the S^3 period of the brane back-reaction 3-form is $(2\pi i)^2 N$ (pullback density $-\sin 2\theta$, their eq. 3.33 — large-gauge invariance quantises N); the exceptional-sphere Kähler flux is $2\pi N$ exactly; the boundary level magnitude is $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$), Dynkin normalisation $T(\text{vector}) = 1$ — "level = flux quantum × Dynkin index", *linear* in the flux. (ii) *The pairing matrix is pinned, not postulated*: the β-stage ch_2 ledger + integrality + unimodularity + effectivity force $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by complete enumeration (row candidates 8/6/8; 48 unimodular solutions; with $P \geq 0$

exactly 2: identity + the A_3 diagram flip), so the glue fluxes are $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ (total $188 = 248 - 60$, flip-invariant) and every charged root current carries *exactly one* flux quantum through exactly one sphere. (iii) *The level formula, honest*: "level = total open-string flux" is *killed by the lockstep test itself* ($k_i = (64, 60, 64)$ unequal — F_i is the flavour multiplicity, the brane matter content); what works is $k =$ flux per adjoint current $= F_i/\dim \mathfrak{g}_i = (1, 1, 1)$, anchored non-circularly by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ and the embedding index $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$ ($60 = 2h^\vee$). The ord-4-vs-level-1 tension is *resolved*: the clock order counts *fractional flux sectors*, not the level ($\mu = 16 = \mathrm{ord}^2$; the μ_4 glue diagonal is isotropic and Lagrangian; KLM $16 \to 16/4^2 = 1$, zero residual fractional flux), and a *new dial* pins the value: #primaries$((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly *one* sector iff $k = 1$, independent of the CPS transplantation. (iv) *The lockstep theorem* (the lifting of the β-stage "one scale $\Rightarrow$ one level" to a *theorem*): clock invariance forces $a_3 = a_2 = a_1 = 0$ sharply, the four branch points are one μ_4 orbit, $|\Pi_j| = \sqrt{2}\,t$ is *equal* on all three spheres, the monodromy acts as the pure phase i and $\det(A - 1) = -4 \neq 0$ (no invariant flux vector) — with the CPS flux-level dictionary, $k_1 = k_2 = k_3$ *is a theorem* of clock invariance; the new falsifier: the clock-forbidden family $Z^4 - Z$ (smooth, disc -27, branch points not a μ_4 orbit) gives *zero* lockstep chains over all 24 orderings. Honest limit: uniform flux doubling keeps the lockstep — the theorem proves *equality*, not the value; the value 1 comes from the current/sector counters and the CPS minimal quantum $N = 1$. Controls separate honestly: $k = 2$ dies on the closure dials (count 0, residual 360, prefactor $-31/48$, #prim 3) while the lockstep dial honestly does *not* fire; $SO(16)$ glue leaves spheres $1/3$ flux-dark with condensation stuck at 4; the A_2 form has $\mu = 12$ (no Lagrangian subgroup exists at all) and Coxeter order $3 \neq 4$. Typed honestly: the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ stays [C] (their geometry is the $\mathbb{C}^2$ blow-up with one (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE); the type-I B-model back-reaction on $\mathbb{PT}/\mathbb{Z}_4$ stays [O] (its first milestone M1 has since been executed, v515; M2–M3 remain). **The δ_2 stage (*done*, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse is real, and one cubic door opens** — is executed and machine-verified as `CELEST.WP5E.DELTA2.01` [`verification/v511_celestial_wp5e_delta2_tensor_ledger.py`] (exact Kostant/Weyl weight arithmetic + sympy/Fraction, 41 checks). The γ-stage collapse was a *Cartan-restricted* statement; δ_2 recomputes the ledger on the full adjoint tensor structure. (i) *Structure*: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is *semisimple* with no $\mathfrak{u}(1)$ factor (class-0 roots split 40+12, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, each irreducible; and the *innerness theorem* decides the collapse wholesale: the grading element h lies in the Cartan *of* $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge 0 mod 4 — the γ-stage collapse is a *full-tensor* statement *and* holds across arities (all 15 non-neutral trilinear sector triples have Hom = 0: the sphere axions have no invariant partner operator at any arity ≤ 3). (ii) *The Hom tables*: bilinear invariants exist only at $j' = -j$ (dims 2/1/1/1, the Killing blocks); the neutral trilinear multisets carry dims $(3, 2, 2, 1, 1)$, all cross-sector trilinears are pure brackets (antisymmetric in their repeated legs) — the *only* totally symmetric trilinear on all of $\mathfrak{e}_8$ is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir: $\mathrm{Inv}(S^3\,45) = 0$ protects T_5). (iii) *The new channel*: the quartic projection of the d-symbol exchange is $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$ by two independent routes (Killing propagator $2h^\vee = 60$ from the roots; hand route $|x^2 - (\mathrm{tr}\,x^2/4)\mathbf{1}|^2/60$), and $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the γ-stage master kill (the product theorem) covers *quadratic* vertices only and does not extend to cubic ones: the T_3 direction *is* reachable through the d-symbol channel, while brackets, tadpoles and mixed $d \times f$ die exactly and no T_5 analogue exists. (iv) *The new certificate*: $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3, augmented rank 4 — the pairing in the charge reading remains *obstructed*, but with the genuinely *weaker* certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\rm fix}) = 72 - 8 = 64$; and the relaxed reading (charge bookkeeping dropped on the bilinear P-block) becomes *exactly solvable*: $A_{\rm fix} = -u + 8v + 2w + 1920\,Q_{dd}$ (residual 0) — where the γ stage found $T_5 = T_3 = 0$ persisting even relaxed, the d-channel closes the T_3 gap with small integer coefficients. Number fences typed [C] only: $64 = \dim \mathfrak{g}_1 = 2^6$; $1920 = |W(D_5)| = 8 \cdot 240$ — and the 1920 fence has now been stress-tested and *fails as a derivation claim* ([`verification/v513_celestial_dterm_nonderivation.py`], `CELEST.DTERM.NONDERIV.01`): the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 pre-declared catalog expressions hit 1920, vs. 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920: Gell-Mann gives 120, $\mathfrak{su}(4)$-Killing 256, plain trace 32, ch_3-normalised 69120); the charge-legal flux pairings give 2048/1800 — they bracket and miss — while the only flux hit is the charge-forbidden $(1, 2)$ pairing; quantisation delivers only lattices ($60\,\mathbb{Z}$, with 1920 the 32nd multiple; strict ch_3 units exclude it); and the twisted cubic index $32i$ clashes in class provenance with the true quartic 32 (classes 1/3 vs. 0/2: same number, disjoint mechanism). The convention-stable [E] core is the factorisation $c_d = \Phi_{T_3}(A_{\rm fix}) \times 2h^\vee(E_8) = 32 \times 60$; the fence stays [C], the physical generation of the 32 stays [O] — no marker moves. Controls: the Killing control replicates the γ-stage channels and $A_{\rm fix}$ exactly; $SO(16)/D_8$ has *no* symmetric cubic ($\mathrm{Inv}(S^3\,\mathrm{adj}_{D_8}) = 0$), no odd sectors and no A_3 block — the δ_2 channel cannot even be posed (E_8/μ_4 doubly special); the false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$ inflates every Hom dimension ($(5, 2, 2) \neq (2, 1, 1)$, $\mathrm{Inv}(S^3\mathfrak{g}_0) = 6 \neq 1$) and the false sector content $(16_s, \bar{4})$ kills every pairing. What stays open after δ_2 [O]: the arity ≥ 4 ledger, the *physical* justification of the cubic GS term ($c_d = 32 \times 60$, the convention-stable core; the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded and convention-contingent, v513; ch_2-naturalness for cubic vertices *undecided* — no canonical geometric datum exists for a cubic vertex), and the burden on δ_1 is *sharpened*: it must supply either the $\psi = 64$ slice or the selection-rule relaxation. **The δ_1 stage — executed at exploration level, *undecided*: a method boundary proven at the $\mathbb{Z}_2$ anchor.** The twisted BCOV contact-term ledger on $\mathbb{C}^2/\mathbb{Z}_4$ (the binary test of the γ stage) has been run as a preregistered exploration probe (exploration-level, v-numbered on demand); its verdict is *undecided*, and its four exact sharpenings now fence the target: (i) the strict holomorphic q^0 reading is *refuted at the $\mathbb{Z}_2$/Eguchi–Hanson anchor* (the published EH celestial chiral algebra has a nonzero one-loop deformation, so a reading that returns an identically empty twisted one-loop term at $\mathbb{Z}_2$ is a method boundary, *not* a kill of the contract); (ii) the forced leading (T_5, T_3) ratio is 4:3 (sector 2 : sectors 1+3) — every successful measure must carry it; (iii) the $SL(2, \mathbb{Z})$ orbit decomposition of the 16 sector pairs is $15 = 12 + 3$: the contact term is the *modular completion* of the Atiyah–Bott data (a Harvey–Moore-type τ-integral), not a free number; (iv) massive coset levels $n \geq 2$ are mandatory. The remaining δ_1 target is therefore the τ-integral itself, evaluated against the δ_2-sharpened criterion ($\psi = 64$ slice or selection-rule relaxation). **The ε_1 stage (*done*) [E]/[C] — the $O(-2)$ bulk-axion slot** — is executed and machine-verified as `CELEST.WP5E.EPS1.01` [`verification/v514_celestial_wp5e_eps1_axion_slot.py`] (exact sympy/Fraction arithmetic, 34 checks, verdict B). Three results, each exact. (i) *The slot is a construction*: on $\mathbb{PT}' = \mathrm{Tot}(O(1) \oplus O(1) \to \mathbb{P}^1)$ the pushforward ledger $\pi_* O(-2)$ gives per spacetime degree d exactly $(d+1)^2$ classes = the exact wave-operator nullspaces, and the *equivariant* bookkeeping under the v492 action $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$ (incidence-forced column weights $(+1, +1, -1, -1)$) closes *block by block* for all $d \leq 6$ and all four characters; the character series are $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \ldots$, $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \ldots$, $P_2 = 6t^2 + 10t^4 + 28t^6 + \ldots$, the $d = 0$ slot has multiplicity $(1, 0, 0, 0)$ — *the bulk axion survives the projection* — and the Molien invariant ring is $(1 - t^8)/((1 - t^4)^2(1 - t^2))$: $Z \in O(2)$, $X, Y \in O(4)$, one relation in degree $8 =$ the a_0 weight (the v492/v493 bridge); the E3 bijection is sharpened per character (twisted minimal content $(2t, 6t^2, 2t)$, no degree-0 mode, = the Coxeter eigenvalues, $\det(A-1) = -4$), and the graviton control separates a_0 (the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$). (ii) $\tilde{\lambda} = 6$ *is triply pinned*: Okubo $\mathrm{Tr}_{\mathrm{adj}}\, X^4 = 36\langle x, x\rangle^2 = (6\langle x, x\rangle)^2$ on the 240 glue roots ($36 = h^\vee + 6$, $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\mathrm{fam}}$); the measure chain on $\mathbb{PT}/\mathbb{Z}_4$ cancels the quotient factor $\mu = 1/4$ *exactly* (vertex^2 μ^2, propagator $1/\mu$, anomaly μ: $\tilde{\lambda}^2_{\mathrm{eff}} = 36$ for arbitrary μ), with the wrong bookings quantified and *excluded* (anomaly-only $\Rightarrow \tilde{\lambda} = 3$, missing propagator renormalisation $\Rightarrow \tilde{\lambda} = 12$); and the flux side (minimal CPS quantum $N = 1$, a single exchange channel exactly at $k = 1$, $(\kappa/c_3)^2 = 12$ exact). (iii) *The first honest back-reaction step (Gibbons–Hawking A_3)*: clock-invariant four-centre configurations form exactly two branches (axis^4 = pure resolution or *one* free μ_4 orbit = pure deformation); the free orbit gives $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, i.e. $XY = Z^4 + a_0$ with $a_0 = -z_0^4$ — the v493 family *re-derived from the geometry*, with the a_0 monodromy = one clock step; the charge-k GH point is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k = 4$: the seam boundary $S^3/\mathbb{Z}_4$); the periods are $\Pi_j = 4\pi t_0(i-1)(1, i, i^2)$ with equal moduli (lockstep, the v509 counter-check from geometry) and the Coxeter clock re-derived ($A^4 = 1$, $\det(A-1) = -4$, $\Pi \cdot A = i\Pi$); the CPS "$N \log$" analogue carries source charge $4 = |\mu_4|$, with the honest sharpening that the Ricci-flat ALE has asymptotic log coefficient exactly 0 (the CPS log is an exceptional-locus statement; Burns contrast $\det g \neq 1$), and the multipole selection rule $m \equiv 0 \bmod 4$ stores $-a_0$ in the first symmetry-breaking $(4, \pm 4)$ harmonic. Controls: $\mathfrak{so}_8$ ($\tilde{\lambda}^2 = 12$ irrational; perfect squares in the Deligne series only $\{9, 36\}$), $\mathrm{diag}(i, i)$ (Veronese cone, no hypersurface), $k = 2$ (three channels). [C]: conditional on Costello's flat-$\mathbb{PT}$ matching $\lambda_g^2 E = A$ (the measure chain is classical bookkeeping). [O] (the honest fence): the *quantised* BCOV one-loop coefficient on $\mathbb{PT}/\mathbb{Z}_4$ and the twisted channels ($32\, T_3$) — preregistered as M1–M3 below; M1 has since been executed (v515). **WP5e proper [O] (the global BCOV quantisation):** the BCOV/Kodaira–Spencer theory on $\mathbb{PT}/\mathbb{Z}_4$ itself (quantisation, non-fixed-point contributions, the twisted closed-string measure). The exchange sub-branch is *closed* by the γ stage, the full-tensor ledger (δ_2), the level-from-flux dial (ε_2) and the bulk-axion slot (ε_1) are *executed*, and δ_1 stands *undecided* at exploration level; the named remaining targets, with preregistered success/kill criteria: **(δ_1, sharpened)** the Harvey–Moore τ-integral behind the twisted contact terms — success = the one-loop coefficient comes out exactly $(9, -30, -15, 0, 32)$, computed, not fitted (equivalently: it supplies the $\psi = 64$ slice or the selection-rule relaxation), carrying the forced 4:3 leading ratio; kill = any other exact value; **(the cubic GS term)** the physical justification of the d-symbol vertex with $c_d = 32 \times 60$ (the honest target after v513: derive the factor $32 = \Phi_{T_3}(A_{\mathrm{fix}})$ physically — the $1920 = |W(D_5)|$ fingerprint is look-elsewhere-loaded and convention-contingent, not a target; a ch_2-type naturalness datum for *cubic* vertices does not yet exist); **(the back-reaction milestones, preregistered as M1–M3 in v514; M1 executed)** the type-I B-model *back-reaction* on $\mathbb{PT}/\mathbb{Z}_4$, which would upgrade the ε_2 CPS dictionary from [C] to derived: **M1 (*done*, SUCCESS)** [E]/[C] the A_3 Ω_N (the Beltrami/Kodaira–Spencer deformation sourced by the glue currents on the three lockstep spheres; preregistered success = a closed-form Ω-deformation whose $S^3/\mathbb{Z}_4$ periods are $(2\pi i)^2\times$ an integer flux vector with the lockstep phases $(i-1)(1, i, i^2)$; kill = unequal sphere fluxes or a non-integer period — then the CPS transplant *and* the ε_2 level dial die) — executed and machine-verified as `CELEST.WP5E.M1.01` [`verification/v515_celestial_wp5e_m1_omega_n.py`] (exact sympy, 30 checks): the residue form $\omega_2 = dX \wedge dZ/X$ is *derived* (three ambient routes; orbifold-cover factor $4 = |\mu_4|$; clock phase i), the twistor family closes as $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; seam fibre $\lambda = 0$ = the v493 family) with the CY-compatible clock lift $(Z, \lambda) \mapsto (iZ, -i\lambda)$, $\Omega \to +\Omega$, and $\Omega_N = \Omega_0 + \sum_p N_p K_p$ is *closed form* (CPS/Bochner–Martinelli kernels on the four centre twistor lines) with every $S^3/\mathbb{Z}_4$ period $(2\pi i)^2$-integral, the seam-fibre phase vector $2\pi i\, t_0(i-1)(1, i, i^2)$ with clock covariance $\Pi \to i\Pi$, the flux vector $N(1,1,1,1)$ forced uniform (clock orbit + K_4 connectivity) and the lens geometry *forcing* the source charge $4 = |\mu_4|$ ($N \in 4\mathbb{Z}$); the 12 conifold nodes sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$); honest fence: on the clock-forbidden $Z^4 - Z$ family the $(2\pi i)^2$ quantisation *also* holds — integrality alone is *not* the discriminator, the discriminator is the lockstep phase structure (0/24 vs 8/24) plus the clock forcing of equal fluxes; neither kill fired; **M2** the twisted KS measure (quantise the $O(-2)$ tower on $\mathbb{PT}/\mathbb{Z}_4$ with the ε_1 mode ledger and compute the one-loop box coefficient per sector; success = the bulk sector reproduces $\tilde{\lambda}^2 = 36$ *and* the twisted sectors pair with the three sphere axions cancelling the rigid $32\,T_3$ residual; kill = a leftover independent T_3-type quartic in any sector); **M3** the a_0 uplift (lift the $(4, \pm 4)$ GH multipole to the Beltrami differential and compute the induced Kähler-potential correction; success = a log-type correction with coefficient tied to the source charge $4 = |\mu_4|$; kill = decoupling from the centre count). Success (unchanged): partition function = E_4/η^8 *including* the $q^{-1/3}$ prefactor *derived from the twistor side*; kill: an anomaly mismatch at the Costello–Li grade — neither the CFT-side dials of v502, nor the equivariant skeleton of v505, nor the exchange no-go of v508, nor the flux/sector dials of v509, nor the full-tensor ledger of v511, nor the axion-slot construction of v514, nor the M1 back-reaction of v515 trigger it (all computable dials say $k = 1$, "one level" is now a theorem, the bulk axion survives the projection, and the back-reacted Ω_N carries the forced integral lockstep flux), but the global quantisation is untested.

Honest status line: WP5a–WP5d (both WP5d stages), the WP5e α, β, γ, δ_2, ε_2 and ε_1 stages and the M1 back-reaction milestone are exact-arithmetic [E] constructions (plus ED-validated lattice witnesses in WP5d) with [C] named identifications ("radial decoupling = collar thickness $\to 0$"; Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the quasi-free extension; the KLM/Longo–Xu/CHMP index dictionary; Longo split heredity; Kac generation of the maximal submodule for $k \geq 3$; the WP5e-β dictionaries — $\mathrm{CS}(\rho)$ mod 1 = discriminant weight, $32 = (h,h)+(h',h')$, $-30 = -h^\vee$, the Ω-contraction; the WP5e-γ number fences $32 = 20+12$, $72 = 2\tilde{\lambda}^2_{e_8}$; the WP5e-$\delta_2$ number fences $64 = \dim \mathfrak{g}_1$, $1920 = |W(D_5)|$ — the latter typed look-elsewhere-loaded and convention-contingent by v513; the WP5e-ε_2 CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$; the WP5e-ε_1 conditionality on Costello's flat-$\mathbb{PT}$ matching; the M1 global kernel patching, Hitchin small resolution and CPS brane dictionary, v515); *WP5a–WP5d plus WP5e-$\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$ plus M1 are landed — all three KLM ingredients of complete rationality carry lattice witnesses, both faces of the inflow are anchored, the exchange sub-branch is closed with a certificate, the collapse is confirmed full-tensorially with one cubic door open, "one level" is now a theorem of clock invariance with the sector-counter dial pinning $k = 1$, the bulk-axion slot is a construction with $\tilde{\lambda} = 6$ triply pinned, and the back-reacted Ω_N is closed-form with $(2\pi i)^2$-integral lockstep periods and the source charge $4 = |\mu_4|$ forced* — WP5e proper (the global BCOV quantisation) is the single remaining open piece, with the continuum uplift of the WP5d lattice witnesses honestly fenced as Xu's theorem (cited, not claimed); the exchange sub-branch is closed by v508, the full-tensor ledger executed by v511, the level-from-flux dial executed by v509, the axion slot executed by v514, the M1 back-reaction milestone executed by v515, δ_1 *undecided* at exploration level (method boundary proven at the $\mathbb{Z}_2$ anchor), and the named remaining targets are the δ_1 Harvey–Moore τ-integral (sharpened to the $\psi = 64$ slice or the selection-rule relaxation), the physical justification of the cubic GS term ($c_d = 32 \times 60$; the $1920 = |W(D_5)|$ reading look-elsewhere-loaded and convention-contingent, v513) and the remaining $\mathbb{PT}/\mathbb{Z}_4$ back-reaction milestones M2–M3 (preregistered in v514; M1 executed via v515) — `SEAM.EQUIV.01` stays [O] and nothing in WP5a–WP5e-ε_1/M1 moves it.

Kill tests (pre-registered). **(K1)** the equivariant sector could have *contradicted* the 52+64+60+64 split (uniformity of $D[d]$ needs exactly $\dim \mathfrak{g}_j = (60, 64, 60, 64)$) — *survived by WP1 and WP2*: the false-glue control shows the test has teeth (3112/6720 violations), and WP2 adds the deformation side (a shape modulus surviving clock-invariance, or a_0-corrections leaking between μ_4 sectors, would have fired — neither occurred; the clock-variant controls fail in two independent ways each). **(K2)** if one-loop associativity of the celestial E_8 theory demanded extra fields that TFPT does not possess, the contract dies (Costello's axionic mechanism is the first pass: E_8 is admissible; the TFPT-specific field content is the open half). **(K3)** — *evaluated by WP3* ([`verification/v495_celestial_wp3_gs_alignment.py`]): the trigger ("no exact alignment between the GS coefficient and the c_3 normalisation") does *not* fire — exact alignment exists ($(\kappa/c_3)^2 = 12$ resp. $1/300$). But the look-elsewhere battery *demotes the scope of K3 itself*: the alignment format passes 8/8 across Costello's whole list, so surviving K3 certifies compatibility plus λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence — alignment survives, selectivity does not. **(K4)** — *evaluated by WP4* ([`verification/v496_celestial_wp4_character_shadow.py`]): K4 fires only against the *exact*-block reading (the character is provably not a conformal block in the jet grading); it does *not* degrade the contract to "E_8 is an admissible gauge algebra for celestial SDYM", because the sector arithmetic holds exactly — WP4 lands at the boundary-limit reading (verdict B(ii), the `SEAM.EQUIV.01` scaling-limit shape), with the constructive limit handed to WP5 — and made precise there by WP5a ([`verification/v497_celestial_wp5a_null_ideal.py`]: the coefficientwise χ_w limit with threshold $w = n{+}1$ and the derived ideal), with the deleting operator itself constructed by WP5b ([`verification/v498_celestial_wp5b_singular_vector.py`]: $|s\rangle$ explicit, level dial $2(1{-}k)$), the limit state by WP5c ([`verification/v500_celestial_wp5c_gns_limit_state.py`]: the ideal in the GNS kernel, level-2 rank 4124 exactly), the two-interval index chain by WP5d-α ([`verification/v501_celestial_wp5d_two_interval_index.py`]: μ-chain $16 \to 4 \to 1$ anchored at both measured ends), the split + strong-additivity legs by WP5d-β ([`verification/v504_celestial_wp5d_klm_completeness.py`]: all three KLM ingredients of complete rationality witnessed on the lattice; the continuum uplift is Xu's theorem, cited not claimed), the CFT-side anchors of the uplift itself by WP5e-α ([`verification/v502_celestial_wp5e_prefactor_level.py`]: the $q^{-1/3}$ prefactor as exact μ_4 vacuum-energy bookkeeping, $k = 1$ forced three ways), the equivariant twistor skeleton by WP5e-β ([`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`]: the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact, glue defect -78 two routes, geometric $k = 1$ dials, the rigid $32\,T_3$ residual, and the honest a_0 refutation — the kill branch does not fire on the skeleton), the exchange no-go by WP5e-γ ([`verification/v508_celestial_wp5e_gamma_pairing.py`]: the sphere-axion pairing by quadratic-vertex exchange refuted with the three-functional certificate $(0, 32, 72)$ — the $32\,T_3$ burden moves to the twisted contact terms, δ_1 binary), the full-tensor ledger by WP5e-δ_2 ([`verification/v511_celestial_wp5e_delta2_tensor_ledger.py`]: the collapse confirmed full-tensorially by the innerness theorem, the cubic d-symbol channel opening T_3 with the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$ and the relaxed solution $c_d = 1920\,Q_{dd}$-coefficient — typed by the v513 negative certificate as $c_d = 32 \times 60$ with the $|W(D_5)|$ reading look-elsewhere-loaded), and the level-from-flux dial by WP5e-ε_2 ([`verification/v509_celestial_wp5e_eps2_level_from_flux.py`]: the lockstep theorem — "one level" as a theorem of clock invariance, falsifier $Z^4{-}Z$ at $0/24$ — plus the sector-counter dial $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$), the bulk-axion slot by WP5e-ε_1 (v514: the equivariant $O(-2)$ slot construction, $\tilde\lambda = 6$ triply pinned, the first GH/A_3 back-reaction step), and the back-reacted Ω_N by the M1 milestone (v515: closed form, $(2\pi i)^2$-integral lockstep periods, forced source charge $4 = |\mu_4|$), leaving the global BCOV quantisation (WP5e proper) as the single remaining open piece.

Honest success estimate. The probability of *full* success (WP5e: a second continuum route to `SEAM.EQUIV.01`) remains low — it is constructive holography at the Costello–Li grade. The *partial*-findings expectation is now realised and *exceeded*: WP1–WP4, all five WP5 constructive milestones (both WP5d stages), the WP5e α, β, γ, δ_2, ε_2 and ε_1 stages and the M1 back-reaction milestone are executed with sharp structural results (verdicts B, B, B, B(ii); WP5a–WP5d passed; WP5e-α pins the CFT side, WP5e-β the equivariant twistor skeleton, WP5e-γ closes the exchange sub-branch with a certificate, WP5e-δ_2 confirms the collapse full-tensorially and opens the one cubic door, WP5e-ε_2 makes "one level" a theorem and pins its value by the sector counter, WP5e-ε_1 builds the bulk-axion slot with $\tilde\lambda = 6$ triply pinned, and M1 delivers the closed-form back-reacted Ω_N with forced charge $4 = |\mu_4|$; *WP5e proper — the global BCOV quantisation — is the single* *remaining open piece*) — the WP1 critical correction (clock $\neq$ deck; clock^2 = deck), the WP2 sharpenings (the clock *is* the Picard–Lefschetz/Coxeter monodromy of the seam deformation; a_0 pure scale with the $\tau{=}i$ shape frozen), the WP3 non-selective alignment (exact λ-arithmetic, honestly typed by the 8/8 look-elsewhere table), the WP4 boundary-limit localisation of the character mismatch (three exactly-located obstructions, one surviving limit structure), the WP5a null-ideal derivation (stabilisation threshold $w = n{+}1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), the WP5b singular-vector construction (the deletion operator explicit, case tally 190/57/1, level dial $2(1{-}k)$, one-block closure typed E_8/μ_4-specific), the WP5c limit state (the ideal in the GNS kernel, the full 9361-block Gram at rank 4124, the family proved *necessary* by a CCR obstruction), the WP5d-α two-interval index (the $\ln 2$ plateau at 10^{-15}, the μ-chain $16 \to 4 \to 1$ measured at both ends), the WP5d-β KLM completion (strong additivity exact on the lattice with the shared boundary point; the bounded-vs-divergent entropic discriminator with the Ising $\frac{1}{4}$-exponent approach against the diverging $U(1)$ control; the elliptic-nome split ladder with exact orbifold inheritance; Pimsner–Popa $\lambda = \frac{1}{2}$ and $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ with integer attainment — all three KLM ingredients witnessed, Xu's continuum theorem cited not claimed), the WP5e-α prefactor/level pinning (the $q^{-1/3}$ prefactor as exact μ_4 vacuum bookkeeping; $k = 1$ forced three ways; the glue-integrality route honestly retired), the WP5e-β equivariant ledger (the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with the -78 defect; the rigid $32\,T_3$ twisted residual; the honest a_0 refutation with the three sphere classes filling the three twisted axion slots), the WP5e-γ exchange no-go (the invariant vertex spaces exactly dim 2/5; the product theorem; the certificate $(0, 32, 72)$; naturalness dissolved scale-independently — an honest negative result that converts the pairing question into the binary δ_1 contact-term test), the WP5e-δ_2 full-tensor ledger (the innerness theorem confirming the collapse across arities; the unique symmetric survivor = the $\mathfrak{su}(4)$ d-symbol with $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ opening the T_3 channel; the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$ and the exactly solvable relaxed reading with $c_d = 32 \times 60$, the $1920 = |W(D_5)|$ fingerprint honestly retired as look-elsewhere-loaded by the v513 negative certificate), and the WP5e-ε_2 level-from-flux dial (the CPS skeleton exact; the pairing matrix pinned by complete enumeration with fluxes $(64, 60, 64)$ and one quantum per current; the naive "level = total flux" killed by the lockstep test itself; the lockstep theorem with its $Z^4{-}Z$ falsifier; the sector counter $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$), the WP5e-$\varepsilon_1$ axion slot (the equivariant Penrose ledger closing block by block; the $d = 0$ slot surviving the projection; $\tilde{\lambda} = 6$ triply pinned with the wrong bookings 3 and 12 excluded; the GH/A_3 step re-deriving the v493 family and the Coxeter clock from geometry) and the M1 back-reacted Ω_N (the residue form derived with cover factor $4 = |\mu_4|$; the closed twistor family with its CY-compatible clock lift; every $S^3/\mathbb{Z}_4$ period $(2\pi i)^2$-integral with the forced uniform flux vector and the lens-forced charge $4 = |\mu_4|$; the honest integrality-is-not-the-discriminator fence) — exactly the kind of sharp, falsifiable refinement the contract format is for. Status: `CELEST.SEAM.01` **[O]** (research contract, WP5e proper the single remaining open milestone); `CELEST.WP1.01` **[E]**/**[C]** (executed); `CELEST.WP2.01` **[E]**/**[C]** (executed); `CELEST.WP3.01` **[E]**/**[C]** (executed); `CELEST.WP4.01` **[E]**/**[C]** (executed); `CELEST.WP5A.01` **[E]**/**[C]** (executed); `CELEST.WP5B.01` **[E]**/**[C]** (executed); `CELEST.WP5C.01` **[E]**/**[C]** (executed); `CELEST.WP5D.01` **[E]**/**[C]** (executed, α stage); `CELEST.WP5DB.01` **[E]**/**[C]** (executed, β stage — the continuum uplift stays **[O]**, Xu's theorem cited); `CELEST.WP5E.ALPHA.01` **[E]**/**[C]** (executed, α stage — the CFT-side prefactor + level pinning); `CELEST.WP5E.BETA.01` **[E]**/**[C]** (executed, β stage — the equivariant anomaly ledger; the global BCOV quantisation itself stays **[O]**); `CELEST.WP5E.GAMMA.01` **[E]**/**[C]** (executed, γ stage — the exchange no-go with certificate; the δ_2 ledger since executed by v511, the twisted contact terms δ_1 stay **[O]**); `CELEST.WP5E.DELTA2.01` **[E]**/**[C]** (executed, δ_2 stage — the full-tensor ledger, intermediate verdict: collapse confirmed by innerness, the cubic d-symbol channel opens T_3 with the weaker certificate $\psi = 64$; the arity ≥ 4 ledger, the cubic GS term and δ_1 stay **[O]**); `CELEST.DTERM.NONDERIV.01` **[E]** (executed, the c_d negative certificate — the $1920 = |W(D_5)|$ fence typed look-elsewhere-loaded and convention-contingent, the convention-stable core $c_d = 32 \times 60$; the fence stays **[C]**, the generation of the 32 stays **[O]**); `CELEST.WP5E.EPS2.01` **[E]**/**[C]** (executed, ε_2 stage — the level-from-flux dial, verdict B: "one level" a theorem, the sector counter pins $k = 1$;

the CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ typed [C], of the type-I B-model back-reaction milestones M1 is since executed via v515, M2–M3 stay [O]); `CELEST.WP5E.EPS1.01` [E]/[C] (executed, ε_1 stage — the $O(-2)$ bulk-axion slot construction with $\tilde{\lambda} = 6$ triply pinned and the first GH/A_3 back-reaction step; the quantised BCOV coefficient and the twisted channels stay [O]); `CELEST.WP5E.M1.01` [E]/[C] (executed, back-reaction milestone M1, SUCCESS on the preregistered criterion — the closed-form Ω_N with $(2\pi i)^2$-integral lockstep periods and the lens-forced source charge $4 = |\mu_4|$; honest fence: integrality alone is not the discriminator; M2–M3 and the quantised BCOV coefficient stay [O]); `SEAM.EQUIV.01` untouched, stays [O].

Research Contract — `WOIT.OS.TWISTOR.01` (the Osterwalder–Schrader twistor bridge)

Typing. **This is the actual bridge from compiler to physics — everything in `CELEST.SEAM.01` is preparation for it.** It is a research contract [O] (ledger row `WOIT.OS.TWISTOR.01`: Open, research contract), *not* a claim; nothing below asserts that the construction exists. The external programme it engages — Woit's Euclidean Twistor Unification (`arXiv:2104.05099`) — is a *named reference frame* for the shape of the target, never a confirmation in either direction (the Woit-bridge paragraph in `tfpt_3` states exactly what is missing).

Input. $X_+ = (\mathbb{PT}/\Gamma)_+$ (the positive-time half of the quotiented projective twistor space; $\Gamma = \Gamma_{\mathrm{ALE}} \cong \mathbb{Z}_4$, normalised by the clock — the definition block of `CELEST.SEAM.01`); $\mathcal{A}_{\mathrm{hol}}$, the *interacting* open+closed twistorial algebra (SDYM(E_8) open strings + BCOV closed strings — the WP5e-proper object, whose free/equivariant skeleton is pinned by v492–v515); ρ, the order-4 clock $\mathrm{diag}(i, 1)$; Θ, the anti-linear real structure induced by the seam reflection (the deck/covering involution, whose free-ness is topology, v510); and $\mu_{\mathrm{BCOV+SDYM}}$, the interacting functional (the global BCOV+SDYM quantisation on $\mathbb{PT}/\Gamma$).

Target theorem. $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$; the gauge-invariant algebra is reflection-positive; and the Osterwalder–Schrader quotient produces $(\mathcal{H}, \Omega, U(\mathcal{P}_+^{\uparrow}), \mathcal{A}_{\mathrm{Mink}})$ with: a positive Hilbert metric, positive energy, local causality, *one chiral fermion generation without mirror doubling*, the TFPT charge lattice, an internal $SU(3)\times SU(2)\times U(1)$ action, and the $(E_8)_1$ seam net as an *actual* boundary net (not only as a character).

Kill tests (pre-registered, all seven live). **(1)** Θ is incompatible with the clock (no anti-linear Θ with $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$ exists on $\mathcal{A}_{\mathrm{hol}}$). **(2)** Reflection positivity fails after gauge fixing (the gauge-invariant subalgebra of $\mu_{\mathrm{BCOV+SDYM}}$ is not RP). **(3)** The reconstruction produces a vector-like mirror generation (chirality is lost in the OS quotient). **(4)** The Penrose transform reaches only free / self-dual states (the interacting extension does not exist at the Costello–Li grade). **(5)** The reconstructed net has the right character but the wrong OPE / fails net equivalence (character-level agreement without operator-algebraic agreement). **(6)** The internal $SU(2)$ remains a spacetime factor (no genuine internal weak factor separates from the Euclidean rotation group). **(7)** The four μ_4 marks are not incidence-compatibly extendable over spacetime (the mark data of the seam cannot be propagated through the incidence relation).

Scope fence (explicit non-claims of the celestial/twistor branch). Self-dual is *not* full 4D physics. Until and unless this contract closes, the celestial/twistor branch claims *none* of the following: both helicities (the anti-self-dual completion), generic scattering amplitudes, local matter fields, full Einstein dynamics (beyond the self-dual sector), electroweak symmetry breaking, confinement. The classical field-equation route (entanglement equilibrium, v358/v359) is a separate, non-competing statement about the full equation; `SEAM.EQUIV.01` and its route split (`SEAM.EQUIV.MMST.01`/`SEAM.EQUIV.TWISTOR.01`) are stated in their own rows and are not moved by anything here. Status: `WOIT.OS.TWISTOR.01` [O] (research contract).

Certifiability and recommended order

Building block	Machine	Hardness
$W_{\rm wall}$ enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (*done*)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (*done*)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

$$\boxed{\text{Selector-triangle pairings} \;\to\; v_{\rm geo}\text{ scale anchor} \;\to\; G_{\rm net}\text{ inclusion theorem}}$$

The flavor interface (historically $U_{\rm wall}$) is now reduced to the *selector-triangle pairings* (the dual normal pair (d,n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The $v_{\rm geo}$ scale anchor is the one dimensionful input (same nature as $1/G$). $G_{\rm net}$ — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. $F_{\rm transfer}$ (source→pole / relic / cosmology) is the remaining downstream interface.

$F_{\rm transfer}$ is a typed functor, not a bag of open topics. It is standard physics fed TFPT *source* data, $F_{\rm transfer} = F_{\rm observable} \circ F_{\rm threshold} \circ F_{\rm RG}$, with exactly four interfaces:

Interface	Map	Status	Source data → observable
$F_{\rm pole}$	source→pole masses	[E] / [C]	Koide $53/54 = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([`verification/v183_koide_f_corner_transfer.py`])
$F_{\rm Boltzmann}$	CP source→ η_B	[C]	$\tilde m_1{=}m_3/A_\Lambda$ anchors the washout, but $M_1{=}M_R\varphi_0^4$ only relocates the free input (M_R is the seesaw scale, not a compiler power), so η_B stays a *sharper scenario*, not a closure ([`verification/v184_etaB_anchored_boltzmann.py`])
$F_{\rm relic}$	$(f_a, m_a, \theta_i, N_{\rm DW}) \to \Omega_a$	[C]	the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([`verification/v185_axion_relic_solver.py`])
$F_{\rm QCD}$	$(\alpha_s, b_3, \Lambda_{\rm QCD}, \text{lattice}) \to m_p/m_e$	[O]	a QCD matching contract ($b_3{=}-7$ a carrier output), *not* a compiler number — the rejected 1836 near-formula is the discipline working

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([`verification/v187_ftransfer_laws.py`]). Exact *algebraic* sub-parts (the Koide factor, $b_3{=}-7$) may be [E]; the *physical* prediction never is.

The functor contract `CONTRACT.F.01`, four axioms ([`verification/v213_ftransfer_functor.py`]). Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: **(1) μ_4-deck equivariance** — the discrete kernels are carrier/deck invariants $\{|\mu_4|, N_{\rm fam}, g_{\rm car}, A_\Lambda\}$, and the Koide multiplier $\lambda_2{=}(2/3)^6{=}64/729$ *is* the μ_4-deck transfer eigenvalue (v82/v54); **(2) Plücker preservation** — the dimensionless readouts are Plücker readings ($53{=}a^T(R+Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1,22,30)$; $\|\mathrm{Pl}(K)\|_1{=}11$); **(3) positivity/stochasticity** — $\operatorname{spec} T{=}\{1,(2/3)^6,(1/3)^6\}$ is positive (RP/OS),

$\kappa_f \in (0,1)$; **(4) external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite-T $\chi(T)$ ODE, the lattice C_N) is named and external, never a compiler number. So `CONTRACT.F.01` is the third research contract alongside (U_{wall}) and (G_{metric}): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure.

The discrete→dynamic lens: `F_transfer` is the readout end of the one principle (`[verification/v303_ftransfer_dynamics.py]`, `FR.DYNAMICS.01`). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $m \leftarrow (A+2I)/4\, m \to$ the E_8 marks, gap $6\ln(3/2)$). All four `F_transfer` instances share that *same shape*: [E] **F_pole** (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F'{=}(2/3)^6{=}\lambda_2$, the seam-clock subleading eigenvalue, so $|F'| < 1 \Rightarrow$ a unique attractor at the *seam* rate (v82); [C] **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$ toward the unique balance, an H-theorem); [C] **F_relic** (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); [E] **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3{=}-7$, whose Gaussian UV fixed point is the attractor (asymptotic freedom, Λ_{QCD} generated). *Honest boundary:* only F_pole has the seam rate $(2/3)^6$ exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So `F_transfer` is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory's simplicity is preserved precisely because the external rates are honestly fenced (v187), not because they reduce to the seam. (Even F_pole's flow time is non-integer, v93: the relaxation is the *continuous* semigroup $e^{t\log F}$ generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.)

The single-flow reduction (`DYN.TRANSFER.UNIVERSAL.01`). The dynamic lens is sharpened to a precise reduction (`[verification/v425_dyn_transfer_universal.py]`): the shared "shape" is literally the *one* native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6\ln(3/2)$) — restricted to each sector, extending the static universal-gap principle (v383) to a dynamic one. [E] the v99 Koide generator $\dot{q}{=}(\Delta/N_{\mathrm{fam}})(q{-}2)(q{-}5)$ linearised at the attractor $q{=}2$ has rate $-\Delta$, so the time-1 contraction is $(2/3)^6$ (the recovery eigenvalue, $q{=}5$ unstable); [E] the Boltzmann washout is that same contraction, with the CP phase $4\pi/3$ native (v320); [E] the QCD rate is $b_3{=}-7$. So the transfer *dynamics* is native; only the *anchors* stay external — the IR pole scale and M_R are v_{geo}-class, C_p is the lattice $O(1)$, and F_{relic}'s cosmological initial condition θ_i is the one genuine wall. A reduction, *not* internalisation: no new premise, and the firewall (v187) holds — the observable stays [C]. So the visible residual is one geometric postulate (`QGEO.SYM.01`), one unit gauge (v_{geo}), and this one typed transfer functor — everything else is out-of-sample data.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.

- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
- P. Naaijkens, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, `arXiv:2605.10693`.

Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), Ann. of Math. **36** (1935) 527–606.

Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, Ann. of Math. Studies **61**, Princeton Univ. Press (1968).

Spectral geometry, gravity and thermal time.

- A. H. Chamseddine, A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, Class. Quantum Grav. **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48** (1993) R3427–R3431.
- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, `arXiv:1102.0440`.

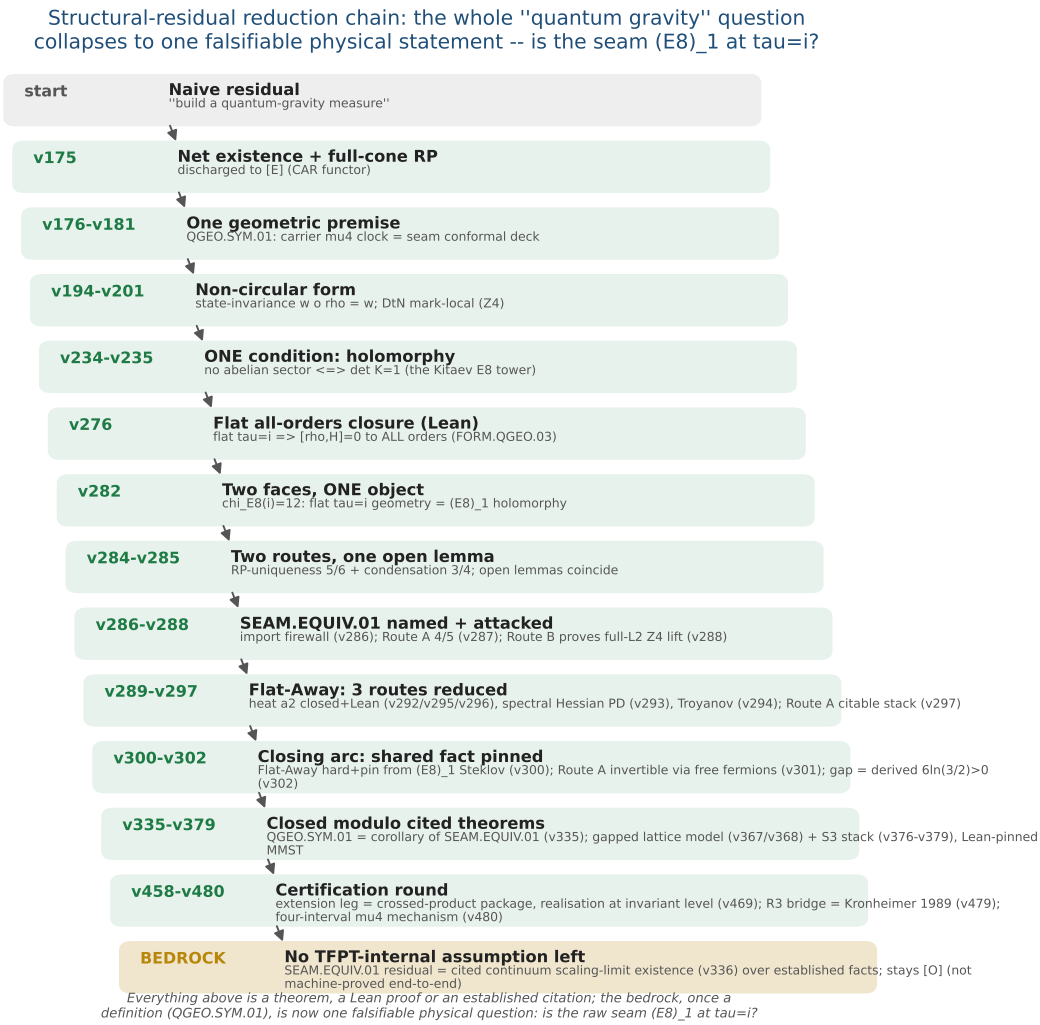

Figure 3: The structural-residual reduction chain (v175 $\to$ the certification round). The naive "build a quantum-gravity measure" shrinks, one machine-checked step at a time, from a vague measure to one falsifiable physical statement — *is the raw seam* $(E_8)_1$ *at* $\tau = i$*?* (`SEAM.EQUIV.01`; its conformal-deck face `QGEO.SYM.01` is a corollary, v335). The closing arc (v276–v302) pins the shared fact and derives the gap $6\ln(3/2) > 0$; the certification round discharges the extension leg to the crossed-product package (v469), the R3 graph$\to$geometry bridge to Kronheimer 1989 (v479) and exhibits the four-interval μ_4 mechanism (v480). Every step above the bedrock is a theorem, a Lean proof or an established citation; the residual — the cited continuum scaling-limit existence (v336) — stays honestly [O].

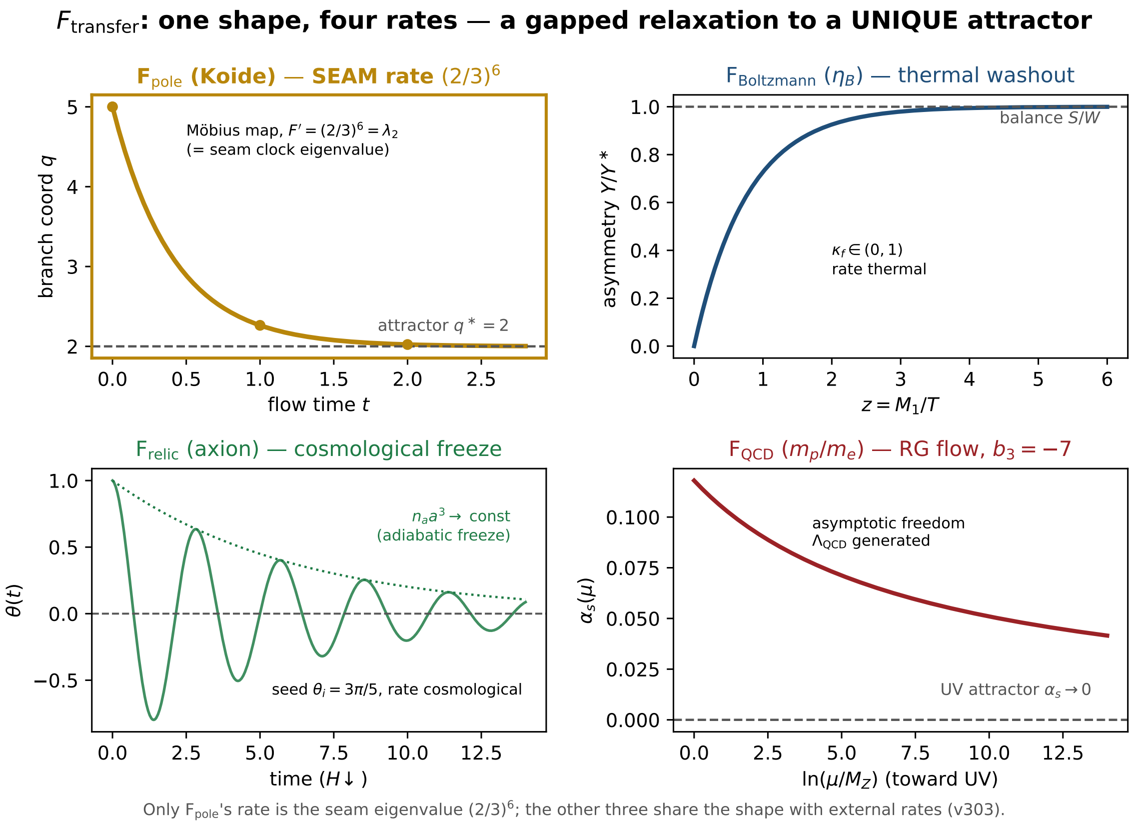

Figure 4: **The discrete→dynamic lens on `F_transfer` ([`verification/v303_ftransfer_dynamics.py`]).** All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update $m \leftarrow (A+2I)/4\,m \to$ the E_8 marks. *Only* F_{pole} (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is $F'=(2/3)^6=\lambda_2$, the seam-clock subleading eigenvalue. $F_{Boltzmann}$ (η_B, thermal washout), F_{relic} (axion, cosmological freeze) and F_{QCD} (m_p/m_e, the RG flow to the Gaussian UV fixed point with the carrier $b_3=-7$) share the *shape* with *external* rates. So `F_transfer` is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.